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Breaking the public procurement walls

Organisational changes for a collaborative approach towards sustainability

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HÖGSKOLAN I GÄVLE

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For those who believe that changes aren't permanent,
but change is

Abstract

Sustainable public procurement (SPP) aims to foster sustainable consumption and production. SPP implementation has been addressed through several disciplinary approaches, entailing efforts to understand and manage organisational changes. Previous research has proposed collaboration between stakeholders as a driver for organisational changes towards implementing SPP. However, there is a research gap in providing a more complete understanding of collaboration in SPP.

This thesis provides a more complete understanding of collaboration in SPP, proposing a framework for understanding the complexity of implementing SPP – including knowledge on managing organisational changes for collaboration. This thesis is composed of five papers, supported by archival research and grounded theory as research strategies, and includes analysis of secondary data and interviews with SPP practitioners. The first provides an understanding of the elements comprising the SPP system, including stakeholder interactions. The second explores organisational changes for implementing SPP through an analysis of drivers. The third provides a more complete understanding of the barriers hindering organisational changes for SPP and strategies to overcome such barriers. The fourth analyses the elements of collaboration in SPP, including who collaborates, how to collaborate, why and why not collaborate, and collaboration facilitators. The fifth explores organisational changes for implementing collaboration in SPP.

This thesis's findings shed light on the role of collaboration in SPP. Collaboration is a key force influencing organisational changes for SPP. However, there are limits to which aspects of SPP can be addressed through collaboration, and collaboration costs must be considered. Collaboration can break the public procurement walls but is not a silver bullet for implementing SPP.

This thesis contributes to organisational change management for sustainability and collaboration, providing a systemic approach to understanding SPP. The thesis offers practical tools to implement SPP better and promote collaboration, contributing to practitioners' efforts.

Keywords: sustainability, public procurement, organisational changes, drivers, barriers, collaboration between stakeholders

Sammanfattning

Hållbar offentlig upphandling (SPP) syftar till att främja hållbar konsumtion och produktion. SPP-implementeringen har undersökts genom flera disciplinära tillvägagångssätt, vilket innebär olika sätt att förstå och hantera organisatoriska förändringar. Tidigare forskning har föreslagit samarbete mellan intressenter som en drivkraft för förändringar mot SPP. Det finns dock en lucka i forskningen, då det saknas en helhetsförståelse för samverkan för SPP.

Denna avhandling ger en mer fullständig förståelse för samarbete i SPP, och föreslår ett ramverk för att förstå komplexiteten i att implementera SPP – inklusive kunskap om att hantera nödvändiga organisatoriska förändringar. Denna avhandling består av fem vetenskapliga artiklar, som bygger på arkivforskning och grundad teori som forskningsstrategier och inkluderar analyser av sekundärdata och intervjuer med SPP-utövare. Den artikeln första ger en holistisk förståelse för SPP-systemet, inklusive interaktioner med intressenter. Den andra och tredje artikeln utforskar krafter som driver och hindrar organisationsförändringar för SPP, såsom samverkan. Den fjärde analyserar på djupet inslagen av samverkan i SPP. Den femte utforskar organisatoriska förändringar för samverkan i SPP.

Avhandlingens resultat belyser samverkans roll i SPP. Samarbete är en nyckelkraft som påverkar organisationsförändringar för SPP. Det finns dock gränser för vilka aspekter av SPP som kan hanteras genom samarbete. Samarbete kan också innebära kostnader, vilket också bör tas i beaktande.

Samarbete kan möjliggöra offentliga upphandlingar men är inte den enda faktorn för att framgångsrikt implementera SPP. SPP-implementering kräver också organisatoriska förändringar, men en ytterligare förändring måste genomföras för att använda en drivkraft som samarbete – förändringar för SPP är flera och sammansatta istället för enstaka och monolitiska.

Avhandlingen ger ett systemiskt förhållningssätt för att förstå SPP. Denna avhandling bidrar till litteraturen kring organisatorisk förändringsledning för hållbarhet och även litteraturen om samverkan. Denna avhandling bidrar också med praktiska verktyg för att bättre implementera SPP och främja samarbete.

Nyckelord: hållbarhet, offentlig upphandling, organisationsförändringar, drivkrafter, barriärer, samverkan mellan intressenter

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A doctoral thesis is far beyond one researcher's work. It comprises personal and professional histories, a supporting network, a suitable academic structure, and a balance between scientific and "real" life. One researcher can write it, but it summarises this set of powerful influences – including those who, by example, disseminate how and why one should embark on this transforming journey, i.e. a PhD. I want to show special gratitude to those who have been examples and inspirations for my – as cold as warm – PhD journey.

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List of Papers

This thesis is based on the following papers, which are referred to in the text by Roman numerals.

Paper I
Lozano, R., Santos, F., & Barreiro-Gen, M. (2024). Developing a harmonic sustainable public procurement framework. *Environment, Development and Sustainability* 26: 2291-2306. doi: 10.1007/s10668-022-02818-1

Paper II
Santos, F., Lozano, R., & Barreiro-Gen, M. (2024). Analysing the drivers for sustainable public procurement. *Environment Systems and Decisions* 44: 966-979. doi: 10.1007/s10669-024-09985-8

Paper III
Santos, F. (submitted). Overcoming the barriers to sustainable public procurement.

Paper IV
Santos, F., Hilletofth, P., & von Haartman, R. (submitted). Understanding collaboration between stakeholders in sustainable public procurement.

Paper V
Santos, F., Hilletofth, P., & von Haartman, R. (2025). Managing organisational changes for collaboration between stakeholders in sustainable public procurement. *Corporate Social Responsibility and Environmental Management*. doi: 10.1002/csr.3117

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Author's contribution to the papers

Paper I
The author of this thesis was responsible for conceptualisation, methodology, formal analysis, investigation, visualisation, and writing the original draft. The co-authors supported through conceptualisation, methodology, formal analysis, visualisation, validation, reviewing, editing, and supervision.

Paper II
The author of this thesis was responsible for conceptualisation, methodology, formal analysis, investigation, visualisation, and writing the original draft. The co-authors supported through conceptualisation, methodology, visualisation, validation, reviewing, editing, and supervision.

Paper III
The author of this thesis was responsible for the whole paper, including conceptualisation, methodology, formal analysis, investigation, visualisation, validation, writing the original draft, reviewing, editing, and supervision.

Paper IV
The author of this thesis was responsible for conceptualisation, methodology, formal analysis, investigation, visualisation, and writing the original draft. The co-authors supported through visualisation, validation, reviewing, editing, and supervision.

Paper V
The author of this thesis was responsible for conceptualisation, methodology, formal analysis, investigation, visualisation, and writing the original draft. The co-authors supported through visualisation, validation, reviewing, editing, and supervision.

Abbreviations

DOS	Demand-offering system
FPSS	Functional product-service system
GDP	Gross domestic product
GT	Grounded theory
NGO	Non-governmental organisation
OCM	Organisational change management
PSS	Product-service system
PP	Public procurement
RQ	Research question
SAPIENS	Sustainability and Procurement in International, European, and National Systems
SBM	Sustainable business model
SD	Sustainable development
SDG	Sustainable development goal
SPP	Sustainable public procurement
ST	Systems thinking
UNEP	United Nations Environmental Programme

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1. Introduction

1.1 Research background

Sustainability is aimed at meeting the needs of the present without compromising the ability of future generations to meet their own needs (WCED, 1987). This requires balancing the economic, environmental, and social dimensions of present and future generations (WCED, 1987), including their interconnections (Lozano, 2008a), through holistic (see Hjorth & Bagheri, 2006; Hopwood et al., 2005; Mebratu, 1998) and transdisciplinary approaches (Lozano, 2008a, 2014; Shrivastava *et al.*, 2013).

In this context, the United Nations Environmental Programme (UNEP) has been promoting sustainable public procurement (SPP), through efforts aimed at linking the consumption side (i.e. governments, including cities, governmental agencies, and public organisations) to the production side (i.e. businesses), focusing on developing sustainable products and services (Brammer and Walker, 2011; Malacina *et al.*, 2022) and supporting more sustainable business models (SBMs) (van Berkel and Schotanus, 2021; Meehan and Bryde, 2011; UNEP, 2014). SPP aims to foster sustainable consumption and production (UNEP, 2021) by using the notable purchasing power of governments – around 15% of global gross domestic product (GDP) (see World Bank, 2021a).

The inclusion of sustainability issues in commercial relationships between public organisations (i.e. public buyers) and businesses (i.e. suppliers) (McCrudden, 2004; McCue and Gianakis, 2001) mobilises the sustainability-oriented use of government purchasing power, i.e. SPP (UNEP, 2021). Sustainability issues addressed by SPP initiatives include economic (e.g. developing local businesses, see Brammer & Walker, 2011), environmental (e.g. waste reduction, see Grandia & Kruyen, 2020), and social (e.g. human rights, see Sattari et al., 2022) dimensions – which must be balanced over time through a holistic approach (Lozano, 2008a).

SPP initiatives have been contributing to several countries' efforts towards sustainability (Steurer *et al.*, 2007; UNEP, 2023). For example, SPP initiatives have contributed to developing a more environmentally friendly and 30% more durable asphalt in Brazil (Betiol *et al.*, 2015), reducing by 85% the CO_2 emissions of a city's bus transportation after introducing biogas-powered vehicles in Finland (Nordic Council of Ministers, 2021), and reducing the lighting cost and energy use in schools by approximately 50% in Sweden (Lozano *et al.*, 2020). Despite successful initiatives, SPP implementation poses challenges to the involved organisations (Bryngemark *et al.*, 2023; Grandia and Kruyen, 2020).

1.1.1 Organisational changes for sustainability and procurement

SPP initiatives can be complex (Igarashi *et al.*, 2015); the already complex public procurement system (Thai, 2001) is further complicated by several elements when sustainability is considered. For example, the SPP system entails sustainability dimensions (Lozano, 2008b) and specifications of sustainable products and services (Caranta, 2023; Edquist and Zabala-Iturriagagoitia, 2021). Stakeholder interactions also influence SPP (Witjes and Lozano, 2016), which adds more complexity (Oliveira *et al.*, 2020) but also leverages the potentiality of developing sustainability through collaboration (Lozano, 2008b).

SPP complexity has been addressed with the support of several disciplinary approaches (Grandia and Kruyen, 2020; Patrucco *et al.*, 2017; Sönnichsen and Clement, 2020), including a managerial one that entails efforts to understand how organisations can change to implement SPP (Erridge and Greer, 2002; Leal Filho *et al.*, 2019; Testa *et al.*, 2016). Organisational changes are aimed at moving organisations from a *status quo* (e.g. not implementing SPP) to a more desirable state (e.g. implementing SPP) (Ragsdell, 2000; Rosenbaum *et al.*, 2018). These changes are affected by complexities, such as internal and external stakeholder influence (Freeman, 1984) and forces driving and hindering change efforts (Ford *et al.*, 2008; Lozano, 2012).

Research on organisational changes has been contributing to understanding and implementing sustainability in different contexts, such as public organisations (Cheng *et al.*, 2018; Domingues *et al.*, 2017) and businesses (Lozano *et al.*, 2019; Stoughton and Ludema, 2012). Several sustainability topics have been studied through an organisational change approach in public organisations, such as circular economy (Govindan and Hasanagic, 2018; Klein *et al.*, 2022), sustainability reporting (Domingues *et al.*, 2017; Lodhia *et al.*, 2012), and SPP (Brammer and Walker, 2011; Cheng *et al.*, 2018).

1.1.2 Driving sustainable public procurement through collaboration

Several drivers for organisational changes towards SPP have been identified in the literature (mainly focused on public organisation changes) (Amann *et al.*, 2014; Grandia *et al.*, 2015; Preuss, 2009; Shaikh and Channa, 2022), including leadership (Brammer and Walker, 2011; Leal Filho *et al.*, 2019), personnel's sustainability commitment (Eikelboom *et al.*, 2018; Grandia *et al.*, 2015), and collaboration between stakeholders (e.g. public buyers' internal stakeholders and external stakeholders) (Betiol *et al.*, 2015; Oruezabala and Rico, 2012). Collaboration between public buyers and external stakeholders (e.g. potential suppliers, policy agencies, and other public buyers) can contribute to SPP by promoting knowledge-sharing (Berg *et al.*, 2022; Johnson and Klassen, 2022) and fostering the co-development of sustainable solutions (Alhola and Nissinen, 2018; Sparrevik *et al.*, 2018), especially when supported by facilitators (Lozano *et al.*, 2020) – overcoming cultural and operational walls hindering joint actions between public organisations and other stakeholders (Erridge and Greer, 2002).

Collaboration in SPP can involve governmental agencies (Berg *et al.*, 2022; Erridge and Greer, 2002), businesses and other market organisations (Bortz *et al.*, 2024; Lingegård *et al.*, 2021; Witjes and Lozano, 2016), and societal actors (Alhola and Nissinen, 2018; Kristensen *et al.*, 2021). Collaboration between stakeholders in SPP initiatives occurs through mechanisms such as market dialogues (Alhola and Nissinen, 2018; Sparrevik *et al.*, 2018) and public consultations (Edquist and Zabala-Iturriagagoitia, 2012; Holma *et al.*, 2020) – mostly aimed at providing mutual learning (Johnson and Klassen, 2022; Kristensen *et al.*, 2021; Lingegård *et al.*, 2021) and co-creation of procurement specifications (Sonnino, 2009; Witjes and Lozano, 2016) and criteria (Bratt *et al.*, 2013; Sattari *et al.*, 2022). However, collaboration between stakeholders in SPP demands additional work (Berg *et al.*, 2022; Holma *et al.*, 2020), increases procurement time (Lozano *et al.*, 2020), and raises risks of supplier dependency (Erridge and Greer, 2002) and bias due to limited stakeholder participation (Bratt *et al.*, 2013) – legal principles of equal treatment and transparency are important in any dialogue between stakeholders in SPP (Behravesh *et al.*, 2022; Lozano *et al.*, 2020).

1.1.3 Research gaps

Research on collaboration between stakeholders in SPP has offered frameworks for understanding the role of collaboration (c.f. Holma et al., 2020; Witjes & Lozano, 2016). However, research efforts on this topic are still limited and fail to capture the influence of interactions between stakeholders in the SPP system. For example, the literature has not yet provided a systemic understanding of collaboration in SPP (Kabra *et al.*, 2023; Rejeb *et al.*, 2023), including and interconnecting elements such as collaboration channels (i.e. who collaborates with whom) (Kristensen *et al.*, 2021), benefits and drawbacks (i.e. why and why not collaborate) (Guarnieri and Gomes, 2019), and mechanisms (i.e. how to collaborate) (Sönnichsen and Clement, 2020; Uttam and Roos, 2015). There is also a research gap in exploring the facilitators' perspective on implementing collaborative SPP initiatives (Lozano *et al.*, 2020). Research on how collaboration can contribute to creating value through public procurement (including SPP) is also underexplored (see Patrucco *et al.*, 2024), including topics such as solving problems and co-creating practices, exploring better collaboration between public buyers (i.e. horizontal collaboration), and understanding how public buyers can interact with businesses (i.e. vertical collaboration) to promote sustainability.

There is also limited research on the influence of a collaborative approach on implementing SPP through a comprehensive organisational change perspective, which could contribute to understanding the role of collaboration in driving SPP initiatives and overcoming barriers to implementing SPP (Bryngemark *et al.*, 2023; Hafsa *et al.*, 2021).

Another research gap can be identified in providing an organisational change approach to implementing collaboration between stakeholders in SPP. There is little to no literature dedicated to understanding the drivers for, barriers to promoting collaboration in SPP, and strategies to overcome such barriers

– with the exception of an analysis of barriers to collaboration provided by Erridge & Green (2002) in the United Kingdom context.

Research on collaboration between stakeholders in SPP has also focused mainly on exploring developed countries' experiences, such as Finland (Alhola and Nissinen, 2018; Holma *et al.*, 2020), Sweden (Sattari *et al.*, 2022; Uttam and Roos, 2015), and the United Kingdom (Erridge and Greer, 2002), with an absence of studies on developing countries' perspectives and offering comparative approaches between different world regions. There are calls for exploring different contexts of collaboration in SPP (Hafsa *et al.*, 2021; Johnson and Klassen, 2022; Sattari *et al.*, 2022) – considering that different accountability developments and legal frameworks can foster or hinder interactions between stakeholders (Behravesh *et al.*, 2022).

1.2 Research aims and questions

SPP can contribute to governments and businesses' efforts towards sustainability, addressing economic, environmental, and social issues. However, SPP implementation can be complex and requires organisational changes. Previous research has proposed collaboration between stakeholders (e.g. public buyers' internal stakeholders interacting with external stakeholders) as a driver for organisational changes towards implementing SPP. However, there is still a research gap in understanding the role of collaboration between stakeholders in SPP, including how collaboration contributes to developing SPP initiatives, how collaborative SPP initiatives can be established, and how organisations can change to implement collaboration in SPP.

The purpose of this doctoral thesis is *to provide a more complete understanding of how organisations can contribute to sustainability efforts through SPP by exploring the role of collaboration between stakeholders and considering changes required to move such organisations towards implementing SPP and collaborative initiatives*. This thesis aims to offer a systemic understanding of collaboration between stakeholders in SPP, delivering a theoretical framework for understanding the complexity of implementing SPP initiatives and managing organisational changes in the SPP context. This thesis addresses research gaps related to the role of collaboration between stakeholders in the SPP system, the contribution of collaboration to organisational efforts aimed at implementing SPP, the uptake of collaboration between stakeholders considering the SPP process, and forces influencing the implementation of collaboration in SPP.

The research questions (RQ) proposed to achieve this thesis's aims are the following:

RQ1: What is the role of collaboration between stakeholders in SPP?
RQ2: How can collaboration contribute to developing SPP initiatives?
RQ3: What are the best practices for establishing collaboration in SPP?
RQ4: What are the forces fostering and blocking collaboration in SPP?

RQ1 was designed to explore the influence of collaboration in the elements that entail SPP, aiming to develop a systemic understanding not yet presented by previous research, and support the development of the subsequent questions. RQ2 was built to unveil the benefits and drawbacks of collaboration and analyse SPP implementation through an organisational changes approach, addressing research gaps on how collaboration can drive and improve SPP initiatives. RQ3 was planned to develop a systemic approach to collaboration in SPP, allowing the analysis of mechanisms to establish collaboration, facilitators, and other interconnected elements that comprise collaborative initiatives, which are still underexplored. RQ4 was designed to answer a research gap in understanding how organisations can change to implement collaboration in SPP. The logical flow of the RQs is: by understanding the potential of using collaboration to foster SPP (RQ1), it is possible to explore how collaboration can be useful for implementing SPP initiatives (RQ2), justifying efforts on understanding what can be done to establish collaboration in SPP (RQ3) and developing an organisational changes perspective on implementing collaboration in this context (RQ4).

1.3 Scope and delimitations

This doctoral thesis provides insights into the SPP system, organisational changes for sustainability, and collaboration between stakeholders. For instance, this thesis analyses the different elements that compose SPP initiatives (e.g. sustainability dimensions, procurement specifications, interactions between stakeholders, etc), aiming to provide a more complete perspective on SPP, contributing to SPP understanding whilst exploring the role of collaboration in the SPP system. This thesis also studies the organisational change approach to SPP, focusing on understanding the influence of collaboration in driving SPP initiatives and overcoming barriers to SPP. An analysis of the influence of collaboration between stakeholders in the SPP process is also carried out by this thesis, considering collaboration channels, benefits, drawbacks, mechanisms, and facilitators. This thesis also dives into organisational changes for implementing collaboration in SPP, including the study of drivers, barriers, and strategies to overcome such barriers. Figure 1 shows a representation of the scope of this thesis.

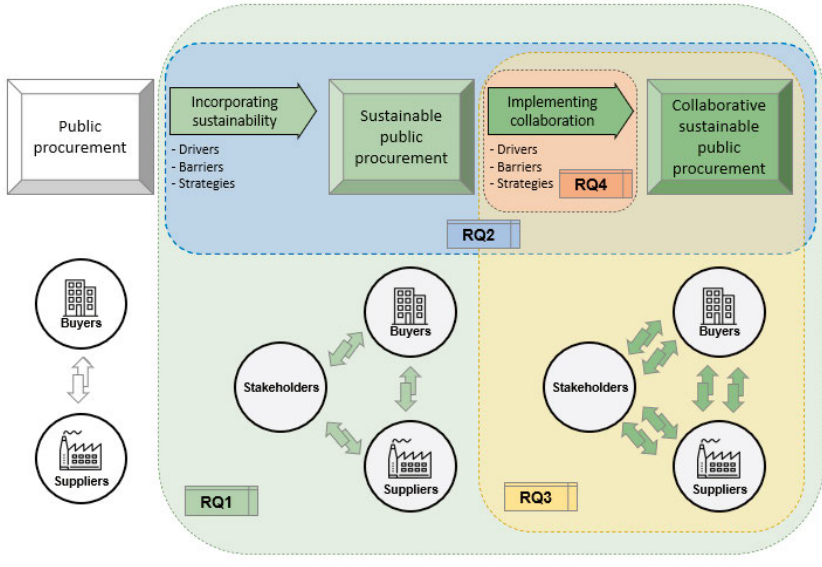


Figure 1. Scope of this doctoral thesis.

1.3.1 Contextual delimitations

This thesis aims to provide knowledge on the implementation of SPP initiatives, trying to build a more general understanding of SPP that can be used and adapted in different procurement contexts. This thesis does not investigate the effects of SPP or explore the complexities of developing monitoring efforts to analyse the outcomes of SPP initiatives. This thesis is also not aimed at discussing the different types of PP (e.g. off-the-shelf procurement, PP for innovation, see Leire and Dalhammar, 2018) or SPP (e.g. green PP, socially responsible PP, see UNEP, 2022b), allowing a focus on general aspects of implementing SPP and collaboration that can be further explored through narrowed lenses (i.e. considering each PP and SPP type, legal and bureaucratic contexts, and their peculiarities).

This thesis also proposes practical tools to implement SPP and promote collaborative SPP initiatives that can be adopted by practitioners and incorporated into SPP guidelines. The research on collaboration in SPP carried out by this thesis includes an analysis of different countries' experiences, including a developed (Sweden) and a developing (Brazil) country, offering a more complete understanding of the topic.

Despite analysing different contexts of SPP implementation and collaboration in SPP, this thesis does not aim to describe and explain Brazilian and Swedish approaches to implementing SPP (e.g. why such countries have different approaches). The analysis of different contexts used in this thesis was aimed at identifying similarities and differences in the studied social phenomena, i.e. organisational changes and collaboration in SPP.

1.3.2 Theoretical delimitations

This thesis contributes to theories related to systemic approaches, organisational changes for sustainability, and collaboration. Systems thinking (presented in Section 2.1) was used to analyse SPP and collaboration as complex phenomena, providing insights into their elements, including interconnections. An organisational change approach focused on sustainability (discussed in Section 2.2) was chosen to explore how organisations have been moving towards implementing SPP and collaboration, considering SPP as a complex process that requires organisational efforts to be implemented. Collaboration theory (presented in Section 2.3) was used to organise the elements comprising SPP collaborative initiatives, including how, why, and why not collaborate.

This thesis focuses on a theory-building approach rather than testing theories, justified by research gaps in applying the aforementioned theories to the SPP context. The theory-building journey supporting this thesis was based on research strategies such as archival research and Grounded Theory (discussed in Section 3.4). Archival research was chosen to analyse SPP discourses and provide a more complete understanding of the SPP system, including elements such as sustainability dimensions and stakeholder interactions. Grounded Theory was used to build theories on organisational changes and collaboration from empirical data collected with SPP practitioners, aiming to expand the literature and provide new insights into such phenomena.

This thesis does not delve into stakeholder theory, nor focus on understanding which stakeholders have been contributing to SPP and what their specific roles are in promoting SPP initiatives. This thesis focuses on their interactions and joint actions, addressed through collaboration theory – more related to the process rather than the subjects.

This thesis focuses on collaboration between internal stakeholders and external stakeholders, aiming to provide a better understanding of SPP implementation from an organisational perspective – mostly from public organisations, considering this thesis's research design. This thesis does not delve into collaboration between internal stakeholders (i.e. within public organisations).

The organisational changes approach developed by this thesis is more related to changes conducted by public organisations to implement SPP and collaboration, following this thesis's research design. Although SPP is a relational process and a public organisation that is moving towards implementing SPP might foster other organisations (e.g. potential suppliers) to move in a similar direction through a systemic change (as proposed in Section 5.1), this thesis does not explore forces influencing suppliers' and other stakeholders' changes.

1.4 Thesis structure

This doctoral thesis comprises six chapters and five appended papers. The chapters are structured as follows:

Chapter 1 – Introduction: Presents the thesis background, research gaps, research aims and questions, thesis scope and delimitations, and thesis structure.

Chapter 2 – Theoretical framework: Discusses concepts and theories supporting this thesis, including topics such as sustainability, systems thinking, SPP, organisational changes for sustainability, and collaboration.
Chapter 3 – Methodology: Presents the research process that organised this thesis's efforts and discusses its research philosophy, research strategies, data collection and analysis techniques, and research quality – including justification for the methodological plans and research limitations.
Chapter 4 – Summary of appended papers: Shows the summary of the five papers appended to this doctoral thesis, including their research background and contribution to this thesis.
Chapter 5 – Discussion: Discusses the research questions for this doctoral thesis, based on the contributions of the appended papers, including new understandings and applicable frameworks proposed through analyses aimed at building and improving theories.
Chapter 6 – Conclusions: Offers a summary of this thesis, including key findings, contributions to theory, and practical implications, proposing further research topics.

2. Theoretical framework

This chapter presents the theoretical framework of this doctoral thesis. The following sections will present discussions on topics such as sustainability, systems thinking, public procurement, organisational changes for sustainability, and collaboration between stakeholders. Section 2.1 will present sustainability-related concepts through a systems-thinking approach and discuss the incorporation of sustainability into public procurement. Section 2.2 will discuss organisational changes for incorporating sustainability into public procurement, based on organisational change management and literature on changes for sustainability. Section 2.3 will delve into collaboration between stakeholders – a driver for organisational changes towards incorporating sustainability into public procurement. Section 2.4 will discuss organisational changes for implementing collaboration between stakeholders in SPP, focusing on forces influencing such changes. Section 2.5 will present a summary of this thesis's theoretical framework.

2.1 Sustainability and public procurement

Humanity has realised that the triumph over nature, using capitalist economy, science, and technology to solve environmental problems, is limited and has been directing humans to an unsustainable living on Earth (Hopwood *et al.*, 2005). Achieving sustainable societies has been proposed as the common goal of human living (United Nations, 1972): societies grounded on sustainability can respect the environment´s capacity to neutralise humans´ effects on nature whilst maintaining a healthy economy, social progress, and social justice, providing equity for future generations (Elkington, 1997; Martin *et al.*, 2005).

Sustainability is aimed at meeting the needs of the present and ensuring the ability of future generations to meet their own needs (WCED, 1987) through efforts that balance economic prosperity, environmental quality, and social justice (Elkington, 1997; Mebratu, 1998) over time (Lozano, 2008a). Efforts towards sustainability have been focusing on harmonising the complex relationship between human and natural systems (Hopwood *et al.*, 2005; Princen, 2003). Systems thinking (ST) has been contributing to understanding the complexity and interconnections between human and natural systems (Mebratu, 1998).

2.1.1 A systemic approach to developing sustainable societies

The 21st century can be called the age of complexity (Dennis, 2007), considering that societies are facing new and persistent challenges from an interconnected world, such as cyber threats, climate change, demographic shifts, and a series of social problems without clear cause-and-effect relationships (OECD, 2020). Facing the complexity of today's societies demands ST, i.e. a way of

thinking that allows the identification of root causes, adaptation and management of choices, and spotting of new opportunities (Meadows, 2008).

ST aims to understand each system separately and at the same time, interconnected (Hjorth and Bagheri, 2006; Jackson, 2003), contributing to addressing complex situations (Checkland, 1985; Martin *et al.*, 2005). Systems are sets of interrelated elements (Meadows, 2008), whose purposes and outcomes are influenced by their relationships, properties, variables, and environments (Ackoff and Emery, 2017).

Systems can be formed by different elements (Meadows, 2008), such as living beings (e.g. biological), machines (e.g. railway system), organised ideas (e.g. democratic system), and other interrelated parts comprising a whole (Cabrera *et al.*, 2008). According to Ackoff (1994), systems can be classified into three types, related to their purposes: mechanical, organismic, or social. A mechanical system works by the laws of nature and has no purpose of its own, but rather that of external entities. An organismic system has a purpose of its own, such as survival. A social system not only has a purpose of its own, but its parts can also have purposes of their own, and the system itself is part of larger systems.

ST helps to identify interconnections between systemic elements and explore possibilities of changes in systems (Hjorth and Bagheri, 2006; Meadows, 2008). Systemic changes are based on holistic approaches that consider the synergies between the elements (i.e. parts) (Jackson, 2019) and understand that whole systems can be interconnected in broader systems (Hjorth and Bagheri, 2006; Jackson, 2003).

Several ST perspectives have been contributing to the social sciences (Jackson, 2019), including hard, soft, and critical ones (Ulrich, 1988). Hard ST has adopted a natural sciences approach (Flood, 1990; Midgley, 1996), proposing that human systems can be interpreted through engineering lenses, embracing a language of problems and solutions, trying to capture real-world problems, and being a model for human behaviour (Checkland, 1985). Soft ST proposes that systemic models can explore but not capture real problems (Ackoff and Emery, 2017; Checkland, 1985), incorporating humans as autonomous individuals instead of mechanical parts of engineering systems (Jackson, 2019). Critical ST aims at overcoming soft ST's neutrality (Ulrich, 1988), considered conservative regarding social structures and less able to promote changes in the real world (Jackson, 1982). Critical ST argues that ideology, theory, and practice are intertwined (Flood, 1990), motivating and justifying social changes (Flood, 1990) through an emancipatory perspective (Midgley, 2007) and a practical view (Ulrich, 2003).

ST has been building knowledge in several social sciences contexts, such as management (Ackoff, 1994; Jackson, 2003), international relations (Armenia *et al.*, 2022; Bousquet and Curtis, 2011), public health (Leischow *et al.*, 2008; Trochim *et al.*, 2006), and public policies (OECD, 2020). Sustainability has also been explored by ST approaches (Hopwood *et al.*, 2005; Mebratu, 1998; Princen, 2003), considering the complexity of developing sustainable societies and equity for future generations (Martin *et al.*, 2005), which requires a holistic approach to be understood (Lozano, 2008a; Mebratu, 1998).

The process of moving towards a sustainable society is called sustainable development (SD) (WCED, 1987), presented as a development that links environmental issues and socio-economic needs (Hopwood *et al.*, 2005; Martin *et al.*, 2005). SD embraces efforts aimed at fostering economic and social development in harmony with the environment (Mebratu, 1998), entailing four dimensions: economic, environmental, social, and time (Elkington, 2006; Lozano, 2008a).

The economic dimension includes efforts towards sustained and inclusive economic growth, full employment, resilient infrastructure, increased productivity, and technological innovation (United Nations, 2015). The environmental dimension challenges the idea that economic growth is an assured path to improve the environment and stimulates green reforms (e.g. fight climate change, forest and water management, reverse land degradation, halt biodiversity loss, see UN, 2015) to avoid a collapse in the natural system (Elkington, 2006; Hopwood *et al.*, 2005). The social dimension proposes efforts aimed at achieving social equality and justice, including food security, poverty eradication, gender equality, and decent work (United Nations, 2015). The social dimension considers that economic and environmental issues are strongly dependent on social inequality (WCED, 1987), requiring a holistic approach that balances these three dimensions to build sustainable societies (Hopwood *et al.*, 2005).

The fourth dimension of SD – time – is based on a holistic understanding of the interrelatedness of economic, environmental, and social dimensions in the short- and long-term (Lozano, 2008a). The time dimension contributes to overcoming efficiency approaches based on economy, science, and technology that have not been able to secure human and environmental long-term survival (Lozano, 2007a; Princen, 2003). The interactions between economic, environmental, and social dimensions of SD through time also reinforce the idea that sustainability is an ideal dynamic state (Lozano, 2008a; Martin *et al.*, 2005; Mebratu, 1998).

The dynamism of SD has been requiring transdisciplinary approaches to be understood (Shrivastava *et al.*, 2013). According to Sakao & Brambila-Macias (2018), transdisciplinary approaches are aimed at proposing a common system of understanding for a set of disciplines (i.e. bodies of knowledge with specific education, methods, and content). Transdisciplinarity goes beyond multidisciplinarity (i.e. parallel use of different disciplines) and interdisciplinarity (i.e. interactions between disciplines to learn from each other) – it is constituted by transcending disciplines to offer a systemic understanding. Transdisciplinarity can also be built by interactions between academics and non-academics when aimed at mixing knowledge from science and practice to build systemic solutions.

Interactions between different stakeholders (e.g. governments, businesses, and civil society) have also been proposed to strengthen efforts towards SD (Lozano, 2007a; Scheyvens *et al.*, 2016), including setting common development goals (WCED, 1987). Most recent institutional efforts to promote SD at a global level have been organised into the Sustainable Development Goals (SDGs) of the United Nations' 2030 Agenda for Sustainable Development

(United Nations, 2015). The Agenda 2030's SDGs were presented as universal objectives to be incorporated into business practices, civil society's agenda, and public policies to promote SD (Bertelsmann Stiftung & SDSN, 2018; Scheyvens *et al.*, 2016). The Agenda 2030 includes 17 goals, such as ending poverty (SDG 1), achieving gender equality (SDG 5), fighting climate change (SDG 13), and promoting peaceful and inclusive societies (SDG 16) (United Nations, 2015). Ensuring sustainable consumption and production (SDG 12) is also part of the Agenda 2030's objectives, including initiatives to promote sustainable procurement practices in businesses and governments (UNEP, 2021; United Nations, 2015).

2.1.2 Incorporating sustainability into public procurement

Public procurement (PP) is the process of buying products and services through government expenditure (Nijboer *et al.*, 2017; Thai, 2009; Thomas, 1919), aiming at achieving public policy goals (Geroski, 1990; Grandia and Volker, 2023) and fulfilling governments' needs (McCue and Gianakis, 2001; Murray, 1999). Government expenditure has long been understood as a crucial part of government: Thomas Hobbes (1651) proposed that public treasure is the government's heart and should be spent to fulfil the government's needs, especially in wartime. Niccolò Machiavelli (1532) stated that a government's power should be measured by its ability to support itself through expenditure and raise a sufficient army. Records of government expenditure (and therefore PP) are accounted for even before modern governments, such as buying army equipment in the Roman Empire (Rostovtseff, 1918) and goods in prehistorical civilisations, e.g. Syria, 2400-2800 B.C (Coe, 1989), and Mesopotamia, 2400-3000 B.C. (Sterba, 1976).

PP has gained momentum in management practice and research by being considered a strategic process that can contribute to delivering more value to organisations and societies (Malacina *et al.*, 2022; Patrucco *et al.*, 2017; Rejeb *et al.*, 2023; Trammell *et al.*, 2020). The notable share of GDP spent by countries through PP (around 15%, see UNEP, 2022) and the intersection between public policy implementation and government procurement (Assche *et al.*, 2023; Grandia *et al.*, 2023) contribute to this growing attention. PP initiatives directly support several public policies (Grandia and Meehan, 2017; Merisalo *et al.*, 2024), such as education (Falvo, 2024; Sonnino, 2009), infrastructure (Patrucco et al., 2022), innovation (Edquist and Zabala-Iturriagagoitia, 2012; Wesseling and Edquist, 2018), and healthcare (Ahsan and Rahman, 2017; Chiarini *et al.*, 2017). SD policies can be highlighted in this context (Caranta, 2022; Preuss and Walker, 2011; UNEP, 2022b), with increasing efforts being conducted to promote sustainability-friendly PP, i.e. SPP (Cheng *et al.*, 2018; Rejeb *et al.*, 2023; Sönnichsen and Clement, 2020).

SPP is aimed at connecting governments to businesses through a sustainability-linked commercial relationship (Brammer and Walker, 2011; Malacina *et al.*, 2022), fostering sustainable consumption and production (Amann *et al.*, 2014; UNEP, 2022b) and developing more SBMs (Witjes and Lozano, 2016). In SPP, the traditional PP commercial relationship between a buyer (i.e. government) and its suppliers (i.e. businesses) (McCue and Gianakis, 2001) can

be improved by incorporating sustainability-oriented practices (Brammer and Walker, 2011). This sustainability-oriented procurement uses the large government's purchasing power to acquire more sustainable products and services (Behravesh *et al.*, 2022; Sönnichsen and Clement, 2020) and support innovative and sustainability-oriented suppliers (Edquist and Zabala-Iturriagagoitia, 2021; Hafsa *et al.*, 2021).

Several countries have been using their government's purchasing power to foster SPP initiatives (UNEP, 2022b). For example, the Belgian PP, representing 23.7% of this country's GDP (around US$ 70 billion in 2021, see UNEP, 2023), has been improving the balance between economic, environmental, and social issues in its procurement activities (UNEP, 2017). The United States of America's PP also has a large purchasing power (14.7% of the USA's GDP, around US$ 637 billion in 2021, see UNEP, 2023) and has been fostering sustainability, such as promoting local businesses (UNDP, 2022) and businesses owned by women and ethnic minorities (Brammer and Walker, 2011). The Brazilian PP, which is around 12.5% of this country's GDP (approximately US$ 200 billion in 2021, see Ribeiro & Inácio Júnior, 2019), has been developing sustainable and inclusive practices (UNDP, 2022; UNEP, 2017). The Swedish PP, with a high purchasing power (18.3% of Sweden's GDP, around US$ 116 billion in 2021, see Upphandlingsmyndigheten & Konkurrensverket, 2020), is a global leader in developing innovative and sustainable solutions (UNDP, 2020; World Bank, 2021b).

SPP initiatives usually have positive results in achieving sustainability outcomes (Cheng *et al.*, 2018; European Commission, 2016; GIZ, 2021; UNEP, 2021; World Bank, 2021b), despite weaknesses in monitoring systems and a focus on products and services that might be easier to procure (e.g with sustainability labels) (UNEP, 2022a). Successful SPP initiatives have been identified in several countries. For example, Brazil has been promoting socially responsible PP, such as by establishing a mandatory requirement for suppliers to hire at least 8% of service personnel from women victims of domestic violence (AGU, 2023). Denmark has been developing an initiative to foster organic food products through PP (Daugbjerg, 2023), reaching 88% of organic content in meals offered by the city of Copenhagen in 2020 (WHO, 2022). Sweden has also been supporting sustainable public food procurement, accomplishing 70% of an organic procurement share in the city of Malmö in 2021 (Daugbjerg, 2023). However, SPP initiatives tend to be quite complex (Thai, 2009; UNDP, 2022; UNEP, 2021; WHO, 2022), especially due to the many issues that need to be addressed when implementing sustainability into procurement processes (McCue and Gianakis, 2001), such as the sustainability dimensions (economic, environmental, social, and time) (Lozano, 2008a), sustainability specifications and criteria (Sparrevik *et al.*, 2018), and stakeholder interactions (Witjes and Lozano, 2016).

According to Leire and Dalhammar (2018), PP types that can influence markets (such as in SPP) include off-the-shelf procurement, innovation-friendly procurement, and PP for innovation. Off-the-shelf procurement is related to buying existing products and services, with limited impact on innovation. Innovation-friendly procurement is an off-the-shelf procurement that

encourages potential suppliers to offer new solutions. PP for innovation is aimed at buying innovative solutions. Each procurement type may require different procedures considering regulations and bureaucratic cultures – for instance, PP for innovation can use competitive dialogue procedures, based on interaction-friendly processes, to develop innovative solutions (see Uttam and Roos, 2015), whilst off-the-shelf procurement can be more related to competitive bidding procedures such as reverse auctions, focused on simple and fast processes (see Falvo, 2024). Nonetheless, a general procurement process can be presented for a broad understanding of PP (and SPP) main phases and stages (Grandia *et al.*, 2023).

The general procurement process can be organised into four main phases (based on UNEP, 2021), although the first two preparatory phases can be combined to set a cyclical three-phase model described as preparation-purchasing-performing (3P-model) (Grandia *et al.*, 2023): 1) needs and strategies definition, 2) procurement planning, 3) supplier selection, and 4) contract execution. Each phase comprises several activities; for example, writing procurement specifications during the procurement planning phase. Figure 2 presents the cyclical four-phase procurement process, including the main activities in each phase.

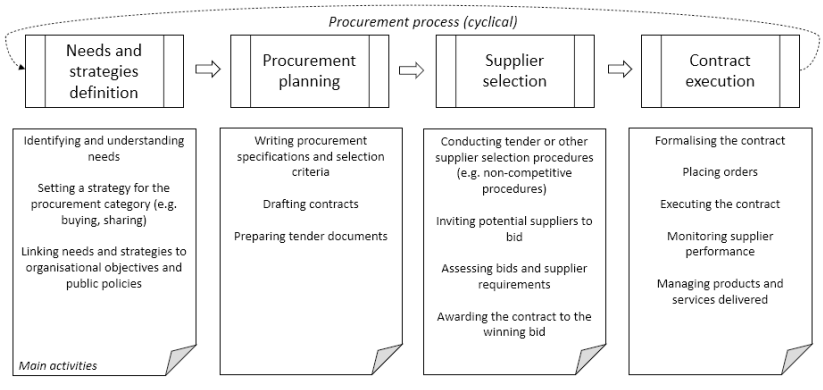


Figure 2. The cyclical procurement process organised into four main phases.

The incorporation of sustainability requirements (e.g. in procurement specifications, selection criteria, and contract performance conditions) is paramount to SPP processes (Brammer and Walker, 2011). This incorporation has indirect effects on product development and consumer demand for more sustainable products (Parikka-Alhola, 2008), as well as how these products impact societies (Preuss, 2009; Srivastava, 2007).

Implementing SPP has been posing challenges to the involved organisations (Bryngemark *et al.*, 2023; Grandia and Kruyen, 2020), considering that legal and policy efforts are not enough to achieve sustainability outcomes (Caranta, 2023). Previous research (Brammer and Walker, 2011; Grandia *et al.*, 2015; Testa *et al.*, 2016) has proposed organisational changes to overcome such challenges and understand organisational efforts towards implementing SPP.

2.2 Organisational change management

Changes have been part of organisations: to survive and prosper (Armenakis and Harris, 2009; Meyer and Stensaker, 2006), to increase organisational efficiency (Kuipers *et al.*, 2014), to contribute to sustainability efforts (Doppelt, 2010; Dunphy *et al.*, 2003), and because changes are inherent in human actions (and organisations are comprised of human actions) (Tsoukas and Chia, 2002). Organisations change by moving from unsatisfactory states (i.e. *status quos*) to more desirable (i.e. improved) states (By, 2005; Ragsdell, 2000), through change processes that can be represented by Lewin's (1947) proposal – organisations unfreeze the initial state, move, and refreeze in the improved state.

The change process – a challenging voyage (Mento *et al.*, 2002) – can be influenced by several systemic elements, such as organisational culture (Harris and Crane, 2002), resistance to change (Ford *et al.*, 2008; Waddell and Sohal, 1998), and stakeholder influence (Doppelt, 2010; Freeman, 1984). Preparatory steps can increase the chances of this process's success by developing organisational capacity for change (Dunphy *et al.*, 2003; Mento *et al.*, 2002; Supriharyanti and Sukoco, 2023). According to Meyer & Stensaker (2006), organisational capacity for change has at least three dimensions: capability to implement the change, capability to maintain the operations, and capability to implement subsequent (i.e. future) changes. Change processes can also be influenced by bundled and reinforced changes (Stouten *et al.*, 2018), considering the interconnection of multiple change processes and their contextual factors (Pettigrew *et al.*, 2001).

Organisational change management (OCM) literature has been proposing several approaches to understanding organisational changes (By, 2005; Da Ros *et al.*, 2023), including planned and emergent changes (Bamford and Forrester, 2003; Kuipers *et al.*, 2014). Planned changes include pre-planned steps (e.g. frameworks) and force field analysis (Bamford and Forrester, 2003; Rosenbaum *et al.*, 2018), following a top-down implementation (Van der Voet, 2014). Emergent changes are more adaptative, continuous, flexible, and incremental rather than planned (Burnes, 1996; Stouten *et al.*, 2018), considering bottom-up implementation processes (Van der Voet, 2014).

OCM has been contributing to understanding changes in contexts such as gender equality (Johansson and Ringblom, 2017; Meyerson and Kolb, 2000), public management (Kuipers *et al.*, 2014; Van der Voet, 2014), and sustainability (Doppelt, 2010; Dunphy *et al.*, 2003).

2.2.1 Managing organisational changes for sustainability

OCM has been proposed to understand changes towards sustainability (Doppelt, 2010), which is a dynamic state (Martin *et al.*, 2005; Mebratu, 1998) that requires continuous changes (Lozano, 2007b). Planned and orchestrated change processes can help to move organisations towards sustainability (Lozano, 2012), considering organisations as integrated whole systems (Doppelt, 2010) that need to promote successive changes to be more oriented towards sustainability (Lozano, 2007b; Martin *et al.*, 2005).

Managing organisational changes towards sustainability requires identifying drivers for, barriers to change, and strategies to overcome such barriers (Govindan and Hasanagic, 2018; Lozano, 2007b). Drivers for change are forces contributing to initiating and sustaining change processes (Harris and Crane, 2002). Barriers to change are forces that can hinder the change initiatives (Barreiro-Gen et al., 2022; Giunipero et al., 2012). Strategies are additional efforts aimed at overcoming barriers to change (Doppelt, 2010; Lozano, 2012).

Drivers for, barriers to sustainability, and strategies to overcome such barriers have been studied mainly in businesses (Lozano, 2012; Millar *et al.*, 2012; Stoughton and Ludema, 2012), with increasing interest in public organisations (Barreiro-Gen *et al.*, 2022; Domingues *et al.*, 2017; Klein *et al.*, 2022). These forces have been categorised in the OCM for sustainability literature mainly into internal and external to the organisation (Dunphy *et al.*, 2003; Engert *et al.*, 2016; Jaramillo *et al.*, 2019; Klein *et al.*, 2022; Rauter *et al.*, 2017). An internal-connecting-external categorisation (proposed by Lozano, 2015; Lozano et al., 2019) offered a more complete understanding of the topic by organising such forces considering stakeholder influence, into more dependent on internal stakeholders and proactive (internal), more dependent on external stakeholders and reactive (external), and more related to efforts dependent on both internal and external stakeholders (connecting). There has been limited research analysing the order of importance of the drivers, barriers, and strategies in the sustainability context, except for Lozano & von Haartman (2018), who provided a ranking and a classification of the drivers.

Lozano (2012) proposed a framework for understanding how drivers, barriers, and strategies can influence organisational changes towards sustainability. The framework for orchestrating the organisational change process towards sustainability presents a change process in which an organisation moves from the *status quo* towards a more sustainability-oriented state through a transition period (when forces are adjusted) with the support of an institutional framework (facilitating the institutionalisation of sustainability). This process, presented in Figure 3, is continuous due to the dynamism of sustainability.

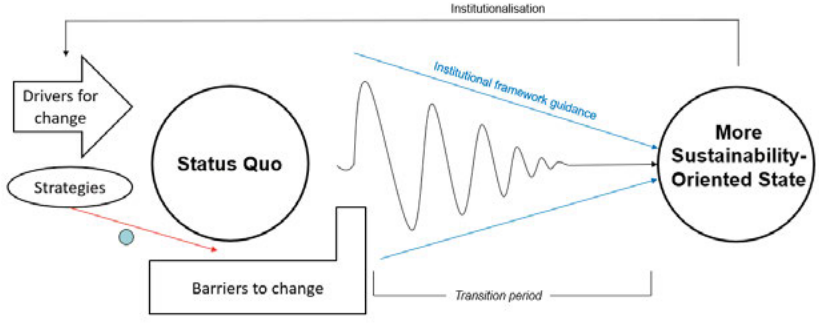


Figure 3. Orchestrating organisational changes towards sustainability. Source: adapted from (Lozano, 2012).

Most research in OCM for sustainability has focused on the implementation of sustainability in organisations, e.g. the drivers for and overcoming resistance to change (Lozano, 2012; Stouten et al., 2018), the role of leadership in driving changes (Ferdig, 2007; Thakhathi et al., 2019), the approaches of small and medium enterprises to change (Jaramillo et al., 2019; Wiesner et al., 2018), SBMs development (Rauter et al., 2017), and sustainability reporting efforts (Domingues et al., 2017; Lodhia et al., 2012). Research on this topic has mainly focused on understanding changes from an intra-organisational perspective (Millar et al., 2012), studying how organisations can govern their own systems to navigate their change processes towards sustainability (Doppelt, 2010). There is still limited research on organisational changes towards sustainability from a process perspective, in which organisations pursue sustainability through interactions in broader systems (such as SPP) (Lozano, 2022).

2.2.2 Organisational changes for sustainable public procurement

Research on OCM for SPP has focused on disseminating good practices to boost public organisations' efforts (Sönnichsen and Clement, 2020) and developing applicable knowledge to be considered whilst planning changes (Grandia *et al.*, 2015; Testa *et al.*, 2016). Previous studies have been providing insights into drivers for (Amann *et al.*, 2014; Grandia *et al.*, 2015), barriers to SPP (Brammer and Walker, 2011; Delmonico *et al.*, 2018; Shaikh and Channa, 2022), and strategies to overcome such barriers (Berg *et al.*, 2022; Leal Filho *et al.*, 2019; Testa *et al.*, 2016). However, there is a lack of a more complete understanding of change processes towards SPP. For example, there are limited efforts in organising the whole set of forces (i.e. drivers, barriers, and strategies) that can contribute to and hinder change efforts towards SPP. Research efforts in categorising and analysing the importance of forces are also limited.

There is also little to no literature discussing the impact of contexts on those change processes – except Nemec et al. (2023) analysing hindering factors to SPP in four post-socialist countries. Most research on OCM for SPP has focused on a single organisation (Eikelboom et al., 2018; Mendonça et al., 2021) or country (e.g. Berg et al., 2022; Delmonico et al., 2018; Oruezabala &

Rico, 2012; Shaikh & Channa, 2022). There is a call for comparing different contexts of changes towards SPP (Cheng et al., 2018; Sattari et al., 2022; Vejaratnam et al., 2020), contributing to resisting a generalisation trend in OCM (Hughes, 2022).

Comparative analyses in OCM for SPP can contribute to understanding different contexts and efforts, considering that SPP change approaches can differ among countries (Brammer and Walker, 2011; UNEP, 2023; World Bank, 2021b). For example, an approach more related to national policies (i.e. based on national structures and guidelines supporting regional and local implementation, as recommended by UNEP, 2021) has been adopted in countries such as China (Zhu et al., 2013) and Sweden (Daugbjerg, 2023; Sattari et al., 2022), whereas a more practical approach (i.e. based on practitioner efforts rather than on more structured policies) has been adopted in countries such as Brazil (Delmonico et al., 2018), Pakistan (Shaikh and Channa, 2022), and Russia (Shadrina et al., 2022). However, there has been limited research analysing different contexts in organisational changes for SPP and offering an understanding of how contexts can influence drivers for, barriers to SPP, and strategies to overcome such barriers.

The following subsubsections will present drivers for SPP, barriers to SPP, and strategies to overcome such barriers based on previous research, using the internal-connecting-external categorisation (presented in Subsection 2.2.1). This categorisation considers stakeholder influence and provides inputs for promoting systemic changes that consider internal and external stakeholders. This literature categorisation complements this thesis's theoretical framework by providing an overview of forces identified by previous research efforts, considering different contexts and organisations. This categorisation of previous literature also contributes to building an additional dataset to be compared with empirical data, following data analysis techniques discussed in Section 3.6 (e.g. the literature was reviewed throughout the data collection and analysis efforts and coded after the empirical data was analysed).

2.2.3.1 Drivers for sustainable public procurement

Drivers for implementing SPP include clear and supportive regulations (Brammer and Walker, 2011; Smith *et al.*, 2016), personnel's commitment to sustainability (Grandia *et al.*, 2015; Leal Filho *et al.*, 2019), policy and guidelines (Allen, 2021; van Berkel and Schotanus, 2021), and collaboration between stakeholders (Betiol et al., 2015; Oruezabala and Rico, 2012). Although several drivers for SPP have been proposed, they have not yet been categorised in the literature.

Figure 4 presents the main 24 drivers for SPP identified in the literature, grouped according to the internal-connecting-external categorisation.

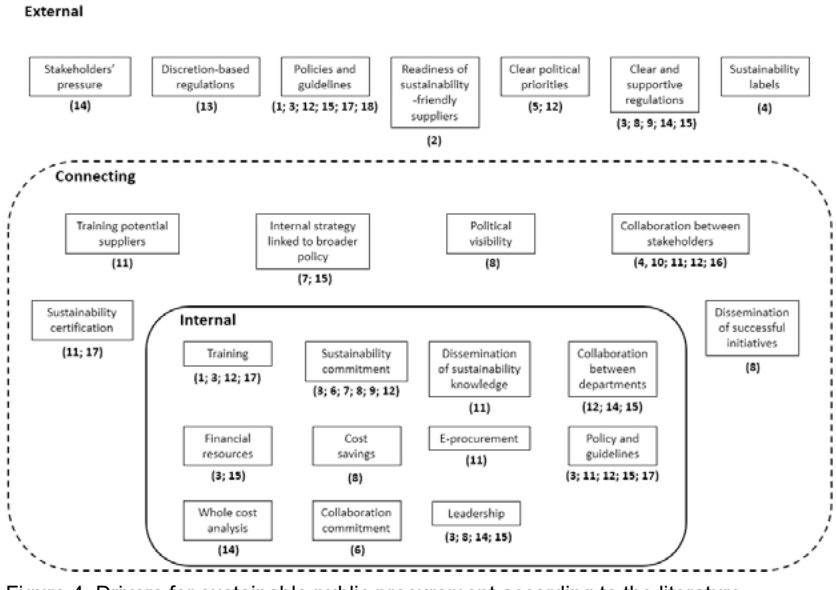


Figure 4. Drivers for sustainable public procurement according to the literature.

Sources: **1** (Allen, 2021); **2** (Amann *et al.*, 2014); **3** (Brammer and Walker, 2011); **4** (Betiol *et al.*, 2015); **5** (Daugbjerg, 2023); **6** (Eikelboom *et al.*, 2018); **7** (Grandia *et al.*, 2015); **8** (Leal Filho *et al.*, 2019); **9** (Meehan and Bryde, 2011); **10** (Oruezabala and Rico, 2012); **11** (Preuss, 2009); **12** (Sattari *et al.*, 2022); **13** (Shadrina *et al.*, 2022); **14** (Shaikh and Channa, 2022); **15** (Smith *et al.*, 2016); **16** (Sparrevik *et al.*, 2018); **17** (Testa *et al.*, 2016); **18** (van Berkel and Schotanus, 2021).

2.2.3.2 Barriers to sustainable public procurement and strategies to overcome them

Barriers to SPP encompass lack of supportive regulations (Günther and Scheibe, 2006; Oyewobi and Jimoh, 2022; Shaikh and Channa, 2022), lack of financial resources (Brammer and Walker, 2011; Leal Filho *et al.*, 2019), lack of sustainability-friendly suppliers (Mendonça *et al.*, 2021; Oruezabala and Rico, 2012), lack of training (Berg *et al.*, 2022; Betiol *et al.*, 2015), and lack of collaboration between stakeholders (Delmonico et al., 2018; Preuss and Walker, 2011). Strategies to overcome barriers to SPP include assessment and reporting routine (Leal Filho *et al.*, 2019; Meehan and Bryde, 2011), empowerment of procurement officers (Eikelboom *et al.*, 2018), mandatory regulations (Zhu *et al.*, 2013), and collaboration between stakeholders (Berg et al., 2022; Preuss, 2009).

Despite growing attention to discussing barriers to SPP and strategies to overcome them (Berg et al., 2022; Brammer and Walker, 2011; Delmonico et al., 2018; Leal Filho et al., 2019), there are few efforts to categorise these forces. For example, barriers were categorised into themes (e.g. economic uncertainty, legal) (c.f. Delmonico et al., 2018; Vejaratnam et al., 2020) and psychological factors (e.g. individual, organisational, group adaptation) (see Preuss & Walker, 2011). However, these categorisations fail to organise these

forces through a perspective that integrates the organisation's internal and external systems. Research efforts on analysing barriers to SPP according to their order of importance are also limited and consider a reduced number of barriers (e.g. 16, see Couto & Ribeiro, 2016); previous research has highlighted lack of financial resources (Brammer and Walker, 2011) and lack of training (Couto and Ribeiro, 2016; Shaikh and Channa, 2022; Vejaratnam et al., 2020) as the most important barriers. There is an absence of literature providing rankings on the importance of strategies to overcome barriers to SPP.

Figure 5 presents the main 30 barriers to SPP identified in the literature, grouped according to the internal-connecting-external categorisation. Figure 6 presents the 18 main strategies to overcome such barriers according to the literature, also categorised into internal-connecting-external.

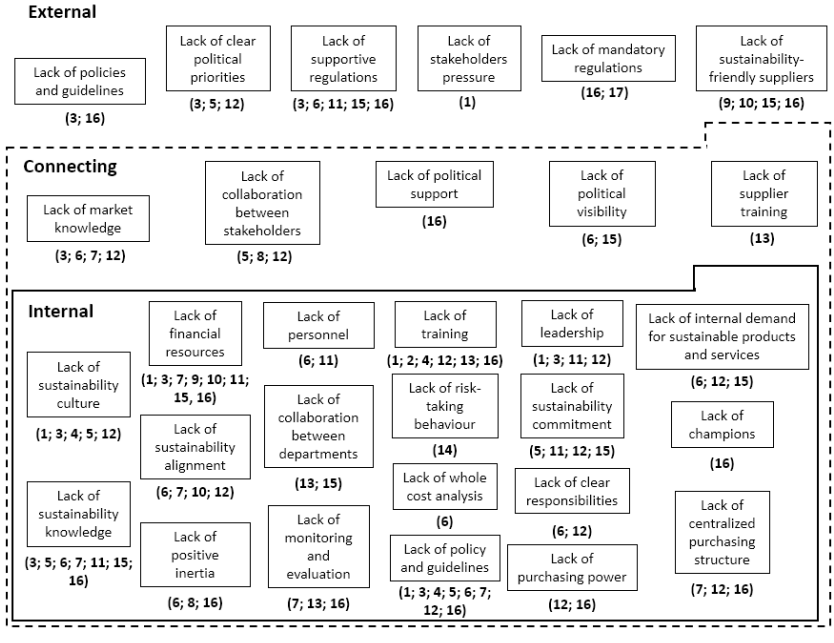


Figure 5. Barriers to sustainable public procurement according to the literature.

Sources: **1** (Berg *et al.*, 2022); **2** (Betiol *et al.*, 2015); **3** (Brammer and Walker, 2011); **4** (Couto and Ribeiro, 2016); **5** (Delmonico *et al.*, 2018); **6** (Günther and Scheibe, 2006); **7** (Leal Filho *et al.*, 2019); **8** (Meehan and Bryde, 2011); **9** (Mendonça *et al.*, 2021); **10** (Oruezabala and Rico, 2012); **11** (Oyewobi and Jimoh, 2022); **12** (Preuss and Walker, 2011); **13** (Sattari *et al.*, 2022); **14** (Shadrina *et al.*, 2022); **15** (Shaikh and Channa, 2022); **16** (Vejaratnam *et al.*, 2020); **17** (Zhu *et al.*, 2013).

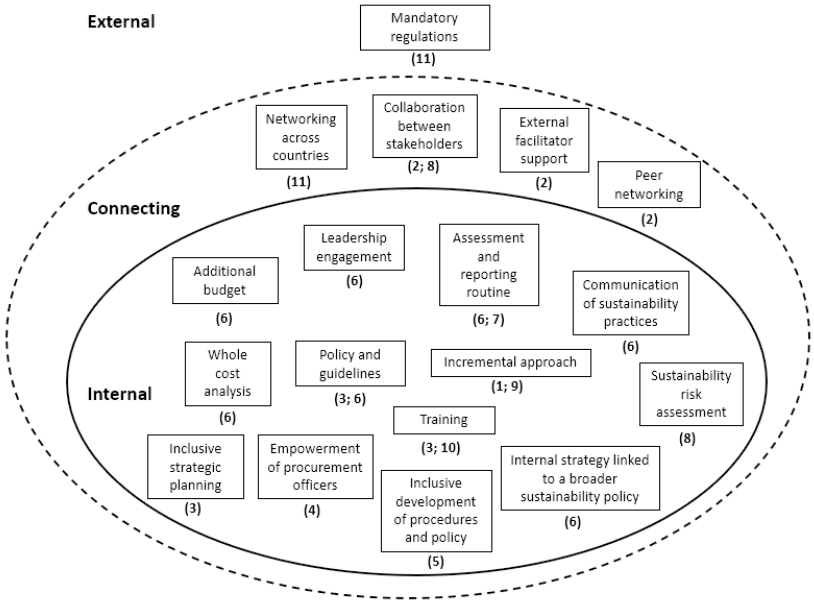


Figure 6. Strategies to overcome barriers to sustainable public procurement according to the literature.

Sources: **1** (Allen, 2021); **2** (Berg *et al.*, 2022); **3** (Betiol *et al.*, 2015); **4** (Eikelboom *et al.*, 2018); **5** (Grandia *et al.*, 2015); **6** (Leal Filho *et al.*, 2019); **7** (Meehan and Bryde, 2011); **8** (Preuss, 2009); **9** (Smith *et al.*, 2016); **10** (Testa *et al.*, 2016); **11** (Zhu *et al.*, 2013).

2.3 Collaboration between stakeholders

Stakeholders are individuals, groups, and organisations who can affect or are affected by an organisation's value creation (Bryson, 2004; Freeman, 1984; Parmar *et al.*, 2010). According to Freeman (1984), stakeholders can be divided into internal (e.g. employees, managers) and external (e.g. customers, suppliers, governments), the former contributing to building interactions between an organisation and its external stakeholders.

Research on interactions between stakeholders in SPP has offered an understanding of how establishing collaboration can contribute to incorporating sustainability issues into PP (Holma et al., 2020; Lingegård et al., 2021; Witjes and Lozano, 2016). Collaboration between stakeholders has been proposed as a driver for organisational changes towards SPP (Betiol et al., 2015; Oruezabala and Rico, 2012) and a strategy to overcome barriers to SPP (Berg et al., 2022; Preuss, 2009) – lack of collaboration between stakeholders was also described as a barrier to SPP (Delmonico et al., 2018; Preuss and Walker, 2011).

Although previous research has contributed to understanding collaboration in SPP through different theoretical lenses, such as multilevel policy implementation (Lingegård *et al.*, 2021), stakeholder tensions (Johnson and

Klassen, 2022), triadic structures and meaningful involvement (Holma *et al.*, 2020), and translation theory and travel of ideas (Sattari *et al.*, 2022), there is still a research gap in applying collaboration theory to the SPP context. Collaboration theory entails concepts and principles related to intense joint actions between stakeholders (Gajda, 2004; Jamal and Getz, 1995; Wood and Gray, 1991) and can contribute to unveiling the elements that compose collaborative SPP initiatives by being process- and system-oriented (Gray, 1985).

Subsection 2.3.1 will present collaboration theory and Subsection 2.3.2 will discuss collaboration in SPP.

2.3.1 Collaboration theory

Collaboration between stakeholders is a process aimed at proposing joint responses to common problems (Gray, 1985; Vangen and Huxham, 2003) by creating new solutions and understandings through interactions with a high level of involvement (Denise, 1999; Lozano, 2007a). The growing complexity of problems faced by organisations (Dyllick and Hockerts, 2002; Gray and Wood, 1991; O'Toole, 1995) has been demanding solutions based on knowledge built from different perspectives (Lozano, 2022; McNamara, 2012; Webb, 1991), which requires intense joint actions between stakeholders (Gray and Wood, 1991; Lozano, 2008b).

Joint actions between stakeholders can be based on at least three levels of involvement (Axelrod, 1984; Gray, 1985; Webb, 1991): 1) coordination, 2) cooperation, and 3) collaboration. Coordination stands for joint actions promoted by formal or informal authorities (Webb, 1991), which orchestrate efforts from different stakeholders towards a common goal (Bedwell et al., 2012; Denise, 1999). Cooperation denotes joint actions based on common interests (McNamara, 2012), in which each stakeholder has their own goals but foresees benefits in working together (Axelrod, 1984). Collaboration means joint actions motivated by shared goals and interests and collective responsibility (Bedwell et al., 2012; McNamara, 2012), using divergent insights and spontaneity (Denise, 1999; Lozano, 2007a) to build intense and interdependent interactions (Gray, 1985). The three levels of involvement are grounded on communication, by which stakeholders can transfer information and agree on shifting from acting individually to acting together (Denise, 1999; Lozano, 2007a). Figure 7 illustrates the levels of involvement in joint actions between stakeholders, with collaboration at the highest level.

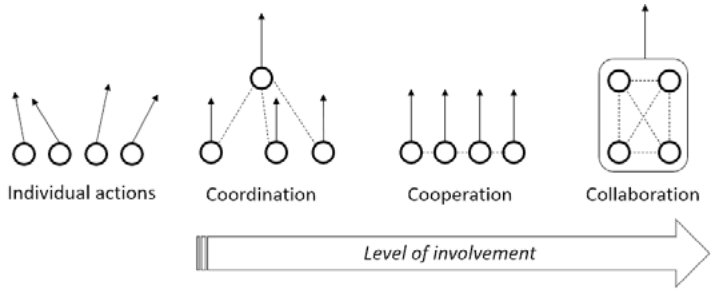


Figure 7. Levels of involvement in joint actions between stakeholders.

Gajda (2004) proposed five principles of collaboration between stakeholders. Firstly, collaboration is imperative to understand and address complex issues. Secondly, collaboration has been labelled by different names (e.g. joining forces, partnerships, strategic alliances) but comprises a complex process of working together to develop relationships, through a multitude of arrangements. Thirdly, collaboration is a journey entailing different levels of integration between stakeholders, from sharing information to unifying structures. Fourthly, collaboration depends on personal relations and emotional connections, requiring trust development and conflict acknowledgements. Lastly, collaboration develops in stages, such as 1) assembling and forming, 2) planning and defining roles and strategies, 3) implementing, and 4) assessing and improving.

Collaboration development requires transdisciplinarity and long-term thinking (Lozano, 2007a), which can contribute to setting problems, establishing clear directions, and providing structures to support and sustain joint actions (Gray, 1985). Successful collaboration initiatives can build feedback loops that pave the way for future joint actions (Bedwell *et al.*, 2012). However, collaboration efforts can be hindered by several factors (Lozano *et al.*, 2021), including lack of needs alignment (Lozano, 2007a), lack of trust (Webb, 1991), and lack of personnel collaboration commitment (Huxham, 1993).

Collaboration theory has been applied to several contexts in management research (Gray and Wood, 1991), such as public management (Huxham, 1993; Webb, 1991), supply chain management (Soosay and Hyland, 2015), tourism (Jamal and Getz, 1995), and corporate sustainability (Lozano *et al.*, 2021; Vurro *et al.*, 2023).

Collaboration has the potential to contribute to sustainability efforts by supporting interdependent decisions that can benefit the whole system (Lozano, 2007a), co-developing sustainability solutions (Johnson and Klassen, 2022), and improving the environmental and social performance of organisations (Niesten *et al.*, 2017). According to Elia et al. (2020), collaborative arrangements for sustainability entail four main elements: focus (what), collaborators (who), mechanisms (how), and benefits (why) – orchestrated through a governance structure. Lozano et al. (2021) adds a fifth element to this topic: collaboration for sustainability has benefits but also drawbacks (why not), which should be balanced to reach optimal collaboration

– collaboration costs impose limits to its growth. Research on collaboration in sustainability includes studies on corporate sustainability (Fobbe and Hilletofth, 2021; Lozano *et al.*, 2021), circular economy (Barreiro-Gen and Lozano, 2020; Köhler *et al.*, 2022), and SPP (Holma *et al.*, 2020; Johnson and Klassen, 2022).

2.3.2 Collaboration in sustainable public procurement

Collaboration between stakeholders in SPP can promote the development of more SBMs and supply chains (Witjes and Lozano, 2016) and foster the co-creation of sustainability-oriented products and services (Lingegård *et al.*, 2021). Collaboration can also improve the knowledge flow through meaningful interactions between stakeholders (Holma *et al.*, 2020), leveraging the potentiality of developing sustainability-friendly solutions (Johnson and Klassen, 2022).

Research has been providing frameworks for understanding collaboration between stakeholders in SPP, including a circular approach to reduce raw materials' use and waste generation (Witjes and Lozano, 2016), a proposal for involving multiple stakeholders with the support of facilitators (Lozano *et al.*, 2020), and a model for early involvement of stakeholders (Holma *et al.*, 2020). However, there is still a research gap in providing a more complete approach to this topic, considering who collaborates (collaboration channels), how to collaborate (collaboration mechanisms), why collaborate (benefits), why not collaborate (drawbacks), and who can help to establish and maintain collaboration (facilitators).

The following subsubsections will present collaboration channels, mechanisms, benefits, drawbacks, and facilitators, considering previous research efforts and aiming at complementing this thesis's theoretical framework. This previous research presentation will also provide an additional dataset to be used as a comparison source when analysing empirical data, following this thesis's research design (see Subsection 3.4.2).

2.3.2.1 Who collaborates with whom (i.e. collaboration channels)

Several stakeholders can participate in collaborative SPP initiatives (Oliveira *et al.*, 2020). Governmental stakeholders collaborating in SPP include public buyers (Erridge and Greer, 2002; Preuss and Walker, 2011; Sattari *et al.*, 2022) and policy agencies (Berg *et al.*, 2022; Daugbjerg, 2023). Market stakeholders include suppliers (Sonnino, 2009; Witjes and Lozano, 2016), potential suppliers (Johnson and Klassen, 2022; Lingegård *et al.*, 2021), and consultancies (Bortz *et al.*, 2024; Testa *et al.*, 2012). Society stakeholders include academia (Alhola and Nissinen, 2018; Sparrevik *et al.*, 2018), citizens (i.e. end-users) (Alhola and Nissinen, 2018; Sonnino, 2009), the natural environment (Lozano *et al.*, 2020), and non-governmental organisations (NGOs) (Kristensen *et al.*, 2021).

The collaboration channels in SPP proposed by the literature can be defined as intragovernmental (c.f. Erridge & Greer, 2002; Kristensen et al., 2021), government + market (see Uttam & Roos, 2015; Witjes & Lozano, 2016), government + society (c.f. Daugbjerg, 2023; Sonnino, 2009), and government + market + society (see Alhola & Nissinen, 2018; Lozano et al., 2020).

2.3.2.2 How to collaborate (i.e. collaboration mechanisms)

The procurement process can be strengthened through collaboration between stakeholders (Holma *et al.*, 2020; Lozano *et al.*, 2020). The main mechanisms for collaboration proposed by SPP literature are presented in Table 1 (8 in total), organised into the procurement phases.

Table 1. Mechanisms for collaboration according to the sustainable public procurement literature.

Procurement phase	Mechanism	Source
Needs and strategy definition	Direct contact between stakeholders aimed at encouraging SPP implementation	(Behravesh *et al.*, 2022)
	Fairs promoted by the industry, where information can be disseminated and stakeholders can connect	(Daugbjerg, 2023)
	Task force per procurement category	(Edquist and Zabala-Iturriagagoitia, 2012)
Procurement planning	Facility presentation for the potential suppliers to familiarise themselves with the buyer organisation and structure	(Holma *et al.*, 2020)
	Public consultation for needs clarification	(Edquist and Zabala-Iturriagagoitia, 2012; Holma *et al.*, 2020)
	Stakeholder dialogue	(Alhola and Nissinen, 2018; Edquist and Zabala-Iturriagagoitia, 2012; Holma *et al.*, 2020; Kristensen *et al.*, 2021; Lozano *et al.*, 2020; Sparrevik *et al.*, 2018).
Supplier selection	Information session after advertising the tender	(Holma *et al.*, 2020; Lozano *et al.*, 2020; Uttam and Roos, 2015)
Contract execution	Permanent table for regular meetings between buyer-supplier	(Sonnino, 2009)

2.3.2.3 Why collaborate (i.e. benefits of collaboration)

Research on collaboration in SPP has proposed several benefits for building meaningful interactions in this context, including developing innovative and sustainable solutions (Alhola and Nissinen, 2018; Edquist and Zabala-Iturriagagoitia, 2012; Holma *et al.*, 2020) and mutual learning (Daugbjerg, 2023; Kristensen *et al.*, 2021; Witjes and Lozano, 2016). Table 2 collates the main benefits of collaboration between stakeholders in SPP according to the literature (24 in total).

Table 2. Benefits of collaboration between stakeholders in sustainable public procurement according to the literature.

Benefits	Source
Aggregating demands	(Erridge and Greer, 2002)
Building capacity	(Berg *et al.*, 2022; Erridge and Greer, 2002)
Building trust	(Erridge and Greer, 2002; Sparrevik *et al.*, 2018)
Developing innovative and sustainable solutions	(Alhola and Nissinen, 2018; Edquist and Zabala-Iturriagagoitia, 2012; Holma *et al.*, 2020; Johnson and Klassen, 2022; Lingegård *et al.*, 2021; Uttam and Roos, 2015)
Developing procurement strategies	(Erridge and Greer, 2002)
Developing requirements (e.g. sustainability criteria and specifications)	(Alhola and Nissinen, 2018; Bortz *et al.*, 2023; Bratt *et al.*, 2013; Lozano *et al.*, 2020; Sattari *et al.*, 2022; Sonnino, 2009; Testa *et al.*, 2012; Witjes and Lozano, 2016)
Disseminating best practices	(Berg *et al.*, 2022; Lingegård *et al.*, 2021)
Driving long-term change	(Lingegård *et al.*, 2021)
Enhancing information flow	(Erridge and Greer, 2002)
Fostering conflict resolution	(Erridge and Greer, 2002; Witjes and Lozano, 2016)
Fostering fair competition	(Holma *et al.*, 2020)
Improving accountability	(Berg *et al.*, 2022)
Increasing legitimacy	(Berg *et al.*, 2022)
Mutual learning	(Daugbjerg, 2023; Edquist and Zabala-Iturriagagoitia, 2012; Erridge and Greer, 2002; Johnson and Klassen, 2022; Kristensen *et al.*, 2021; Lingegård *et al.*, 2021; Lozano *et al.*, 2020; Witjes and Lozano, 2016)
Mutual understanding of needs	(Alhola and Nissinen, 2018; Holma *et al.*, 2020; Lozano *et al.*, 2020; Uttam and Roos, 2015)
Providing a safe environment for pre-commercial prototyping	(Johnson and Klassen, 2022)
Providing experience for further collaborations	(Witjes and Lozano, 2016)
Reducing efforts duplication	(Erridge and Greer, 2002)
Reducing transaction costs	(Erridge and Greer, 2002)
Savings	(Preuss and Walker, 2011)
Sharing contract monitoring	(Daugbjerg, 2023; Erridge and Greer, 2002; Sonnino, 2009)

Sharing ownership of solutions	(Erridge and Greer, 2002; Sonnino, 2009; Sparrevik *et al.*, 2018; Witjes and Lozano, 2016)
Supporting decision-making	(Berg *et al.*, 2022)
Understanding the market readiness	(Alhola and Nissinen, 2018; Holma *et al.*, 2020)

2.3.2.4 Why not collaborate (i.e. drawbacks of collaboration)

The drawbacks of collaborating in SPP were also discussed by the literature, including coordination costs (Lozano *et al.*, 2020) and increasing work (Berg *et al.*, 2022; Sparrevik *et al.*, 2018). The main drawbacks of collaboration between stakeholders proposed by SPP research (10 in total) are presented in Table 3.

Table 3. Drawbacks of collaboration between stakeholders in sustainable public procurement according to the literature.

Drawback	Source
Accountability erosion	(Bortz *et al.*, 2024)
Coordination costs	(Lozano *et al.*, 2020)
Developing bias due to limited stakeholder participation	(Bratt *et al.*, 2013)
Developing supplier dependency	(Erridge and Greer, 2002)
Disclosing information that could cause a loss of commercial advantage	(Uttam and Roos, 2015)
Exposing stakeholders to inefficiency and corruption risks	(Edquist and Zabala-Iturriagagoitia, 2012)
Impeding new actors' entry	(Erridge and Greer, 2002)
Increasing work	(Berg *et al.*, 2022; Holma *et al.*, 2020; Sparrevik *et al.*, 2018)
Increasing procurement time	(Lozano *et al.*, 2020)
Using information against stakeholders	(Erridge and Greer, 2002)

2.3.2.5 Facilitators of collaboration

There is limited research on the role of facilitators in developing collaboration in SPP. Facilitators of collaboration in SPP are individuals (Alhola and Nissinen, 2018; Lozano *et al.*, 2020) and organisations (Berg *et al.*, 2022; Daugbjerg, 2023) aimed at driving collaboration forward by overcoming its

challenges (Lozano *et al.*, 2020). Individuals acting as facilitators can be project members working outside the procurement team (Lozano *et al.*, 2020) – which can be related to internal stakeholders that help link the organisation with external stakeholders (as proposed by Freeman, 1984) – and consultants contributing to understanding the market (Alhola and Nissinen, 2018). Organisations can perform a facilitator role by translating, transferring, and creating knowledge between stakeholders – as observed by Berg et al. (2022) in a public agency, Daugbjerg (2023) in an industry organisation, and Johnson and Klassen (2022) in a city council.

2.4 Managing organisational changes for collaboration in sustainable public procurement

Collaboration between stakeholders has been proposed as a driver for organisational changes towards implementing SPP (Betiol *et al.*, 2015; Preuss, 2009; Sattari *et al.*, 2022). However, although there is previous research proposing that implementing collaboration in SPP initiatives also requires organisational changes (Erridge and Greer, 2002), there is limited research providing an understanding of how organisations can change towards implementing collaborative SPP initiatives. For example, the literature has not yet organised and categorised the set of forces (i.e. drivers, barriers, and strategies) that can influence changes for collaboration in SPP.

The internal-connecting-external categorisation can be adopted to better organise forces influencing organisational changes for implementing collaboration in SPP, considering previous literature, similarly with the categorisation of forces acting in changes for implementing SPP (discussed in Subsubsection 2.2.3.1). This categorisation effort will also contribute to improving this thesis's theoretical framework and building an additional dataset to be compared with findings from empirical data (see Subsection 3.6).

2.4.1 Drivers for collaboration

Research on collaboration in SPP has been proposing drivers that contribute to implementing collaboration in this context: for instance, clear roles (Bortz *et al.*, 2024; Johnson and Klassen, 2022), champions (Lingegård *et al.*, 2021; Lozano *et al.*, 2020), and demand-offering of needs (i.e. buyer demanding and supplier offering needs fulfilment that may include products, services, and functional requirements) (Lozano *et al.*, 2020; Uttam and Roos, 2015).

The main drivers for collaboration in SPP, according to the literature (10 in total), are presented in Table 4, categorised into internal, connecting, and external.

Table 4. Drivers for collaboration between stakeholders in sustainable public procurement according to the literature.

Level	Driver	Sources
Connecting	Clear roles	(Bortz *et al.*, 2024; Johnson and Klassen, 2022)
	Innovation hub	(Johnson and Klassen, 2022)
Internal	Champions	(Lingegård *et al.*, 2021; Lozano *et al.*, 2020)
	Collaboration-friendly policy	(Johnson and Klassen, 2022)
	Demand-offering of needs	(Lozano *et al.*, 2020; Uttam and Roos, 2015)
	Discretion	(Sparrevik *et al.*, 2018)
	Leadership	(Lingegård *et al.*, 2021)
	Long planning phase	(Alhola and Nissinen, 2018; Sparrevik *et al.*, 2018)
	Socio-cultural specifications	(Alhola and Nissinen, 2018)
	Stakeholder mapping	(Alhola and Nissinen, 2018; Holma *et al.*, 2020; Johnson and Klassen, 2022)

2.4.2 Barriers to collaboration

Barriers hindering collaboration efforts in SPP include lack of collaboration commitment (Erridge and Greer, 2002), lack of supportive regulations (Edquist and Zabala-Iturriagagoitia, 2012; Lingegård *et al.*, 2021), and lack of trust (Erridge and Greer, 2002; Sonnino, 2009).

The main barriers to collaboration in SPP according to the literature (21 in total) are presented in Table 5, categorised into internal, connecting, and external.

Table 5. Barriers to collaboration between stakeholders in sustainable public procurement according to the literature.

Level	Barrier	Sources
External	Lack of competitive market	(Preuss and Walker, 2011)
	Lack of supportive regulations	(Edquist and Zabala-Iturriagagoitia, 2012; Erridge and Greer, 2002; Lingegård *et al.*, 2021)
Connecting	Lack of clear roles	(Bratt *et al.*, 2013; Holma *et al.*, 2020)
	Lack of early involvement	(Holma *et al.*, 2020)
	Lack of expectations alignment	(Sattari *et al.*, 2022)
	Lack of informal arrangements	(Erridge and Greer, 2002)

	Lack of initiative hub	(Preuss and Walker, 2011)
	Lack of interaction transparency	(Bratt *et al.*, 2013)
	Lack of participative-building internal policies	(Erridge and Greer, 2002)
	Lack of sustainability alignment	(Bratt *et al.*, 2013)
	Lack of trust	(Erridge and Greer, 2002; Sonnino, 2009)
Internal	Lack of category knowledge	(Edquist and Zabala-Iturriagagoitia, 2012; Lingegård *et al.*, 2021)
	Lack of collaboration commitment	(Erridge and Greer, 2002)
	Lack of common goals	(Erridge and Greer, 2002; Preuss and Walker, 2011)
	Lack of employee retention	(Erridge and Greer, 2002)
	Lack of long-term thinking	(Bratt *et al.*, 2013)
	Lack of positive inertia	(Erridge and Greer, 2002)
	Lack of risk-taking behaviour	(Erridge and Greer, 2002; Holma *et al.*, 2020)
	Lack of strategic positioning of procurement	(Erridge and Greer, 2002)
	Lack of time	(Holma *et al.*, 2020; Lingegård *et al.*, 2021)
	Lack of training	(Erridge and Greer, 2002; Holma *et al.*, 2020; Preuss and Walker, 2011)

2.4.3 Strategies to overcome barriers to collaboration

Strategies to overcome such barriers include participative-building policies and training (see Erridge & Greer, 2002). The main strategies to overcome barriers to collaboration in SPP according to the literature (5 in total) are presented in Table 6, categorised into internal, connecting, and external.

Table 6. Strategies to overcome barriers to collaboration in sustainable public procurement according to the literature.

Level	Strategy	Sources
Connecting	Participative-building internal policies	(Erridge and Greer, 2002)
Internal	Discretion	(Erridge and Greer, 2002)
	Professionalisation of procurement staff	(Erridge and Greer, 2002)
	Strategic positioning of procurement	(Erridge and Greer, 2002)
	Training	(Erridge and Greer, 2002)

2.5 Theoretical framework summary

This chapter presented this thesis's theoretical framework, built to support data collection and analysis and contribute to the development of a systemic understanding of collaboration between stakeholders in SPP. This framework was based on literature discussing sustainability, ST, SPP, OCM for sustainability, and collaboration theory.

ST has been used in social sciences to better understand complex phenomena, such as developing sustainable societies. SD efforts include SPP initiatives, aimed at connecting governments and businesses through sustainability-friendly commercial relationships. SPP complexity has been addressed with the support of organisational changes, considering that legal and policy efforts on promoting SPP require subsequent implementation efforts by public organisations (as public buyers).

OCM for sustainability contributes to understanding and promoting planned and orchestrated changes that can contribute to developing sustainable organisations. Drivers for, barriers to change, and strategies to overcome barriers can be considered whilst planning changes for sustainability, such as in the SPP context. Several drivers, barriers, and strategies related to organisational changes for SPP have been discussed in the literature, although categorisation and ranking efforts regarding such forces are still limited. An internal-connecting-external categorisation, based on previous research on OCM for sustainability that considers stakeholder influence, was proposed in this theoretical framework to organise drivers, barriers, and strategies from the literature.

Collaboration between stakeholders has been proposed as a driver for implementing SPP, contributing to co-developing sustainable solutions and providing mutual learning. Previous research has not yet provided a systemic understanding of the elements that compose collaborative SPP initiatives, such as collaboration channels, mechanisms, benefits, drawbacks, and facilitators. This theoretical framework used collaboration theory to organise previous literature on these elements.

Efforts on implementing collaboration in SPP were also discussed in previous research, although in a limited way when considering organisational changes for such implementation. OCM for sustainability can also be used for understanding drivers for, barriers to implementing collaboration in SPP, and strategies to overcome such barriers. This theoretical framework synthesises previous literature on such forces, organising them into internal, connecting, and external.

This thesis uses this theoretical framework to support theory-building efforts and strengthen data collection and analysis, such as in contributing to interpreting SPP discourses and acting as a comparison source for empirical data.

3. Methodology

This chapter presents this doctoral thesis's methodology, including the research process, philosophical assumptions, and research design. The research strategies used in the papers included in this thesis are also introduced, including sections explaining the data collection and analysis processes. The efforts towards improving this thesis' research quality – such as discussing data quality and research ethics – are also presented. Lastly, this chapter also discusses this thesis's limitations, considering its research aim and design.

Methodology stands for the theory of how research should be undertaken (Saunders *et al.*, 2019), including research methods (i.e. how to gather and analyse data) and philosophical assumptions that influence the research design and process (Corbin and Strauss, 2008; Saunders *et al.*, 2019).

3.1 Research process

The research process is composed of a series of linked stages aiming at organising the research efforts from choosing a research topic to proposing and answering research questions (Saunders *et al.*, 2019). This doctoral thesis was supported by a research process aimed to provide a more complete understanding of collaboration between stakeholders in SPP, including five research papers developed during a four-year doctoral work: Papers I, II, III, IV, and V.

The research process supporting this doctoral thesis can be organised into four studies: 1) understanding the SPP system (Paper I), 2) exploring forces influencing SPP implementation (Papers II and III), 3) analysing collaboration between stakeholders in SPP (Paper IV), and 4) providing insights on how to implement collaboration in SPP (Paper V). The first two studies were dedicated to understanding the influence of collaboration between stakeholders in SPP by exploring the SPP system and implementation – before delving into collaboration characteristics. The last two studies were designed to provide an in-depth understanding of collaboration between stakeholders in SPP, including the elements related to and efforts towards implementing collaboration in SPP.

The first study (late 2021 to mid-2022, Paper I) aimed at analysing the elements that compose initiatives related to incorporating sustainability into PP whilst providing an understanding of the role of interactions between stakeholders – and collaboration – in the SPP system. The second study (mid-2022 to late 2023, Papers II and III) was dedicated to providing a more complete understanding of how organisations can move towards implementing SPP, including drivers for, barriers to organisational changes, and strategies to overcome such barriers. In this study, the role of collaboration as a force contributing to SPP implementation was also analysed.

The third study (late 2023 to early 2024, Paper IV) was designed to delve into collaborative SPP initiatives and propose insights into the elements that comprise such initiatives, focusing on building a more complete understanding of the phenomenon. The fourth study (early 2024 to late 2024, Paper V) focused on analysing organisational changes for implementing collaboration in SPP, including forces (i.e. drivers, barriers, and strategies) influencing such changes.

The four studies and five papers contributed to answering the research questions on which this thesis was built – from main to partial contributions. Table 7 presents how each paper relates to the research questions.

Table 7. Research papers' contribution to answering this thesis's research questions.

Research question	Paper I	Paper II	Paper III	Paper IV	Paper V
RQ1: What is the role of collaboration between stakeholders in SPP?	X	X	X	X	
RQ2: How can collaboration contribute to developing SPP initiatives?		X	X	X	
RQ3: What are the best practices for establishing collaboration in SPP?				X	X
RQ4: What are the forces fostering and blocking collaboration in SPP?					X

3.2 Research philosophy

Research philosophies are sets of assumptions and beliefs that guide the development of scientific knowledge (Creswell, 2009; Saunders et al., 2019). There are different assumptions and beliefs on how reality can be understood (i.e. ontology or theory of being), how knowledge can be considered valid and communicated (i.e. epistemology or theory of knowledge), and how researchers' values and ethics impact knowledge-building (i.e. axiology) (Lowndes et al., 2018; Saunders et al., 2019). Philosophy of science is dedicated to organising these assumptions and beliefs into research philosophies accepted by scientific communities (i.e. academia) as guiding systems to develop knowledge (Saunders et al., 2019).

This doctoral thesis was developed considering two main research philosophies: interpretivism and pragmatism. Interpretivism supported Paper I, considering this paper's aim of reinterpreting SPP discourses – commonly developed separately – to propose a more complete interpretation (i.e. an approach interconnecting elements) of SPP.

Interpretivism is based on the idea that social phenomena are formed by interpretations of reality (Creswell, 2014; Lowndes et al., 2018), with

meanings that can be studied and reinterpreted (Saunders et al., 2019). Proposing a disconnection from natural sciences (e.g. reality is relative rather than objective) (Mills et al., 2006), interpretivism understands reality as a complex construct of social phenomena (Guba and Lincoln, 1994; Lowndes *et al.*, 2018), with each phenomenon being formed by different interpretations and inner realities (e.g. different cultural backgrounds, circumstances, and times) (Saunders et al., 2019). According to interpretivism's ontology, these multiple realities and interpretations can be assessed to create new interpretations and understandings (Schutt, 2012). This means that interpretivism's epistemology focuses on creating meanings and interpretations (Guba and Lincoln, 1994), building knowledge by exploring reality's complexity and contexts instead of trying to reduce it through law-like generalisations (Creswell, 2014; Saunders et al., 2019). Interpretivism contributions are linked to offering new understandings and interpretations of social phenomena and contexts (Saunders et al., 2019). The axiology adopted by interpretivism endorses the role of the researcher in interpreting data (Mills et al., 2006), including an active researcher's role in accessing each social context to understand different perspectives from the inside (Saunders et al., 2019).

Papers II to V were developed using pragmatism, aiming to explore and propose applicable knowledge on the studied phenomena (i.e. organisational changes for SPP in Papers II and III, collaboration in SPP in Paper IV, and organisational changes for collaboration in SPP in Paper V) considering practitioners' actions, contexts, experiences, and thoughts. Pragmatism is the research philosophy that questions the debates between other philosophies and their assumptions (Morgan, 2014; Saunders et al., 2019), aiming at transforming reality more practically (Creswell, 2014). Pragmatism's ontology sees reality as an outcome of ideas (Saunders et al., 2019), emphasising human experience and contexts (Morgan, 2014). Its epistemology understands knowledge as an instrument of thoughts and actions (Saunders et al., 2019), created through interactions between self-reflective beings (Corbin and Strauss, 2008). Pragmatism contributions are linked to solving problems and proposing practical solutions that inform future actions (Saunders et al., 2019). Pragmatism's axiology acknowledges researchers' influence on knowledge-building (Morgan, 2014), considering that research questions are based on problems identified and chosen by researchers (Morgan, 2014), in an inquiry process influenced by researchers' values (Saunders et al., 2019).

Table 8 presents the research philosophies supporting this thesis, focusing on the philosophical assumptions that guided this thesis's papers.

Table 8. Philosophical assumptions adopted by the thesis.

Paper	Research philosophy	Philosophical assumptions
I	Interpretivism	*Ontological*: SPP is formed by different discourses (i.e. interpretation of realities) that can be reinterpreted to build a new and more complete understanding. *Epistemological*: a new understanding of SPP can be built by exploring SPP and its complexity, offering a more complete approach that considers and interconnects SPP elements. *Axiological*: the researcher's academic and professional backgrounds (e.g. political scientist, public manager, SPP professional) can contribute to providing a more complete interpretation of SPP.
II to V	Pragmatism	*Ontological*: the studied phenomena (i.e. organisational changes for SPP in Papers II and III, collaboration in SPP in Paper IV, and organisational changes for collaboration in SPP in Paper V) are outcomes of practitioners' actions and ideas, linked to their contexts. *Epistemological*: a more complete understanding of the phenomena can be built through interactions between the researcher and practitioners (e.g. interviews), aimed at exploring practitioners' experiences and thoughts and acknowledging the influence of contexts in their actions; a new understanding of the topics can be used as input for better implementation of the phenomena. *Axiological*: the researcher's values, influenced by his academic and professional backgrounds, influence knowledge building by guiding research problem identification and research question formulation.

3.3 Research design

Research philosophies contribute to building research designs, which are the methodological plans that define how the research gaps (i.e. a lack of knowledge identified by researchers) will be addressed (Creswell, 2009). The integrity of each research relies, initially, on the coherence between its research philosophy and its research design (Saunders et al., 2019). The main elements of a research design (see Saunders *et al*, 2019) are 1) approach to theory

development, 2) methodological choice, 3) research purpose, 4) research strategy, and 5) time horizon.

The methodological plans for developing the papers appended to this doctoral thesis included an inductive approach to theory development, a mono-method qualitative choice, an exploratory purpose, two main research strategies (archival research and grounded theory), and a cross-sectional time horizon.

Approaches to theory development are related to how research engages with theory testing or theory building (Lowndes *et al.*, 2018; Saunders *et al.*, 2019). According to Corbin and Strauss (2015), theories are frameworks explaining why things happen and can be divided into substantive, middle-range, and more general or formal. Substantive theories are based on concepts more specific to a research context, deriving from a study of certain groups and places. Middle-range theories are developed by applying substantive theories to different contexts, adding more concepts and abstraction to the theoretical framework. More general theories are less context-specific and arise from the development of middle-range theories, by adding more concepts and raising their level of abstraction. More general theories are more suitable for explaining a wider range of social phenomena.

The inductive approach adopted by this thesis's appended papers is suitable for deriving theories from data (Lowndes *et al.*, 2018; Saunders *et al.*, 2019), guiding research through a theory-building journey (Creswell, 2009). Underresearched social phenomena may require efforts to generate theories (Saunders *et al.*, 2019; Schutt, 2012), including defining key concepts (Lowndes *et al.*, 2018) and proposing frameworks for understanding actors, interactions, and processes within their contexts (Creswell, 2009). Theories can be further complemented and tested (Saunders *et al.*, 2019), contributing to the development of disciplines and study fields (Lowndes *et al.*, 2018; McNabb, 2015).

Paper I used an inductive approach to build theory, aiming at providing a more complete understanding of the SPP system. Papers II to V adopted an inductive approach to build and complement theories, considering literature gaps identified during the research journey – including insights from Paper I's analysis. For instance, lack of a more complete understanding of which forces influence organisational changes for SPP (Papers II and III), lack of research efforts exploring and integrating the elements comprising collaboration in SPP (Paper IV), and lack of a more complete understanding of organisational changes for implementing collaboration in SPP (Paper V).

The methodological choice of the papers appended to this thesis (mono-method qualitative) is based on qualitative methods, which are aimed at generating and exploring non-numerical data, such as words, images, and audio recordings (Corbin and Strauss, 2008; Creswell, 2014). According to Saunders *et al.* (2019), these methods and their data collection and analysis techniques can be used individually or combinedly. The appended papers were based on mono-methods, i.e. single methods and techniques. Paper I collected and analysed secondary data (discussed in Subsection 3.5.1) – SPP discourses. Papers

II-V analysed data collected through interviews (discussed in Subsection 3.5.2).

Research purposes or goals are related to the objectives of the research and link research problems to research questions (Creswell, 2014; Lowndes *et al.*, 2018). Exploratory research – adopted by the papers appended to this thesis – aims to provide a more complete understanding of a phenomenon (Corbin and Strauss, 2008; Yin, 2014) through a flexible and adaptable process that supports questions such as "what" and "how" (Saunders *et al.*, 2019).

Research strategies are specific plans on how the research questions will be answered (McNabb, 2015; Saunders *et al.*, 2019), contributing to delineating the range of possible methods to collect and analyse data (Creswell, 2014). For instance, research strategies adopted in management studies include (see Saunders *et al.*, 2019) experiment, survey, archival research, case study, ethnography, action research, grounded theory, and narrative inquiry. Paper I analysed SPP discourses using archival research as its research strategy – presented in Subsection 3.4.1. Papers II to V used grounded theory as their research strategy (presented in Subsection 3.4.2).

Different research strategies were considered while designing this research, such as a) conducting a systematic literature review (see Sauer and Seuring, 2023) to organise research findings on collaboration in SPP, b) supporting a collaborative SPP initiative in Sweden through action research (mixing data collected through observation with interviews with professionals from a public buyer and potential suppliers), c) exploring an SPP facilitator perspective through a single-case study (see Yin, 2014) (collecting data by interviewing professionals from the facilitator, public buyers, and suppliers), and d) testing findings on OCM for SPP by carrying out a survey with SPP practitioners in Brazil and Sweden (focusing on questionnaires and quantitative data). However, this research's questions and scope contributed to composing its design with archival research (interpreting SPP discourses to understand the role of collaboration) and GT (building theory on underexplored topics through an immersive data collection and analysis journey).

Research can be aimed at understanding phenomena at a particular time or a given period (Saunders *et al.*, 2019). The papers appended to this thesis adopted a cross-sectional time horizon, related to a particular time (see Lowndes *et al.*, 2018; Sapsford and Jupp, 2006).

Table 9 presents the methodological plans of the papers appended to this doctoral thesis.

Table 9. Methodological plans of the thesis's appended papers.

Paper	Approach to theory	Methodological choice	Research purpose	Research strategy	Time horizon
I	Inductive	Mono-method qualitative	Exploratory	Archival research	Cross-sectional
II to V				Grounded theory	

3.4 Research strategies

This section discusses the research strategies included in the methodological plans designed to guide the development of the papers appended to this doctoral thesis. Subsection 3.4.1 presents archival research, adopted in Paper I. Subsection 3.4.2 discusses grounded theory, the research strategy used in Papers II to V.

3.4.1 Archival research (Paper I)

Archival or documentary research is a research strategy aimed at analysing and offering interpretations of documentary materials (Saunders *et al.*, 2019), including recorded information (i.e. extension of human memory) such as letters, governmental documents, and academic journals (McNabb, 2015). Archival research comprises interpreting social meaning from documents (Lowndes *et al.*, 2018), considered secondary data (i.e. created for purposes other than research) (Sapsford and Jupp, 2006; Schutt, 2012). Social researchers can analyse documents to interpret discourses expressed in texts – entailing ideas, knowledge, and statements that are relevant at a particular time (Sapsford and Jupp, 2006) and influence reality construction and understanding (Fergus and Rowney, 2005).

Paper I adopted archival research as a research strategy to interpret SPP discourses (i.e. secondary data) and provide a more complete understanding of the SPP system. SPP has, generally, been explained by disconnected discourses on the elements and approaches. Such discourses can be better understood and interpreted through a more complete perspective that considers and interconnects different elements. Archival research has been used in a similar way to analyse sustainability discourses, for example, to understand and explain theories of the firm to provide a holistic perspective for company engagement in sustainability (see Lozano et al., 2015).

3.4.2 Grounded theory (Papers II-V)

Grounded theory (GT) is a research strategy based on an inductive approach (Saunders *et al.*, 2019) and designed to generate theory from a set of data (Corbin and Strauss, 1998; Glaser and Strauss, 1967). GT aims to analyse, interpret, and explain social phenomena (Corbin and Strauss, 1998; McNabb, 2015), considering the meanings that social actors provide to their experiences, interactions, and realities (Creswell, 2009; Williams, 2007). This research strategy supports the gathering and analysis of qualitative data (Glaser and Strauss, 1967; McNabb, 2015) and allows researchers to analyse the data whilst the data is collected (Corbin and Strauss, 1998; Saunders *et al.*, 2019). The coding or categorization of data is essential to this strategy (Corbin and Strauss, 1998; McNabb, 2015), as is the process of constant comparison between the data and the categories (Creswell, 2009; Glaser and Strauss, 1967; Saunders *et al.*, 2019) – presented in Section 3.6.

According to Corbin and Strauss (2015), theories entail interconnected concepts, categories (i.e. major concepts), and central categories (i.e. more abstract concepts), integrated to offer theoretical explanations. GT focuses on building

theory from concepts derived and integrated from data collected during the research process, requiring an open and flexible approach to data collection and analysis. GT can contribute to developing substantive, middle-range, and more general theories. GT can also promote theory improvement by offering new concepts and extending a theory previously developed – using this theory to provide an initial set of concepts but remaining open to questioning them.

GT was initially designed to start developing categories solely from the collected data (Saunders et al., 2019; Williams, 2007), ignoring previous literature to avoid "contamination" of concepts (Glaser and Strauss, 1967). However, researchers have been acknowledging that concepts and categories found in previous literature can contribute to identifying research gaps, enhancing comparisons, and increasing theoretical sensitivity (Creswell, 2009; Mills et al., 2006; Tummers and Karsten, 2012). Previous literature can also be used in GT to confirm research findings and extend the body of knowledge (Corbin and Strauss, 1998). This GT's informed approach requires reviewing the literature throughout the entire research process (Thornberg, 2012).

Papers II to V used GT to analyse organisational changes in the SPP context and collaboration between stakeholders in SPP, based on an inductive approach to provide depth to the topics and propose a more complete understanding of these social phenomena. The use of empirical data is part of these papers' research design – as is the informed approach aimed at enhancing the research by adding categories from previous literature into the data analysis process (as a comparison source) and theoretical discussion (confirming research findings and refining the body of knowledge).

The empirical data collected and analysed for Papers II to V was based on two different contexts, borrowing elements from comparative studies to enhance these papers' research design. Comparative studies in social sciences are aimed at improving our understanding of social phenomena by comparing data between times and places (Ragin, 2014; Schutt, 2012), considering each context's individuals, institutions, and systems (Lowndes *et al.*, 2018). Comparative studies can be cross-sectional (i.e. comparing data from one time period) and longitudinal or historical (i.e. comparing data from more than one time period) (Schutt, 2012). Comparative studies can also vary from comparing a large number of contexts or cases – suitable for using sophisticated statistical analysis of extensive sets of quantitative data – to a small one (e.g. two) (Lowndes *et al.*, 2018).

Comparative studies entailing a small number of contexts can be used to build theories by analysing similarities and differences (Ragin, 2014), approaching contexts more holistically and considering an interactive process of comparing theory with data (Lowndes *et al.*, 2018). Such studies can also provide cross-context conclusions comprising similarities and differences in characteristics influencing a social phenomenon in multiple contexts (Schutt, 2012; Yin, 2014).

Papers II to V use elements of comparative studies to strengthen research efforts (mixing methods, such as proposed by Yin, 2014), aiming to provide a more complete understanding of organisational changes and collaboration in SPP. These papers offer a cross-sectional analysis of data collected from two

different countries, focusing on identifying similarities and differences in the studied social phenomena. This analysis contributes to providing a more complete approach to the studied topics, including a study of context influence on implementing SPP and collaboration in SPP. This analysis based on different contexts also contributes to answering a call for developing comparative studies to provide a more complete understanding of SPP implementation (Cheng *et al.*, 2018; Sattari *et al.*, 2022; Vejaratnam *et al.*, 2020).

The contexts analysed in Papers II to V were a developing (Brazil) and a developed (Sweden) country that have been fostering SPP initiatives (UNEP, 2017; World Bank, 2021b). Both countries have been using their government purchasing power to promote SD, having official records of successful SPP initiatives and a high government purchasing power (as presented in Subsection 2.1.2). However, these countries have distinct legal frameworks and bureaucratic cultures – which leverage the potential to understand organisational changes and collaboration in SPP in different contexts. For instance, Brazil and Sweden have different approaches to changes towards SPP. Brazil has been promoting SPP initiatives through an approach based on practical efforts (i.e. focusing on initiatives from practitioners) (Delmonico *et al.*, 2018), whilst Sweden has been supporting SPP initiatives through a policy-based approach (i.e. more related to national policies and structures) (Daugbjerg, 2023; Sattari *et al.*, 2022). For example, Brazilian SPP has adopted supporting and mandatory regulations (e.g. in a national school feeding program, local governments must spend at least 30 per cent of national-transferred resources on family farm products) monitored by control agencies (i.e. valuing coercive power) but depends mainly on practitioner efforts due to structural gaps at national and local levels (e.g. lack of national encouragement and monitoring by policy agencies, lack of strategic positioning of procurement in public organisations) (Mazzucato *et al.*, 2024; Oliveira *et al.*, 2020). Swedish SPP has adopted non-mandatory regulations (e.g. in a national organic food programme, local governments can adopt their own goals and strategies) supported by national goals, policies, and structures, but depends mainly on local willingness to set and implement their own policies (i.e. valuing discretionary power) (Bryngemark *et al.*, 2023; Daugbjerg, 2023; Hunka *et al.*, 2023).

3.5 Data collection

This section discusses the data collection techniques used in the papers appended to this doctoral thesis. Several research techniques and procedures are available for data collection (Sapsford and Jupp, 2006), varying according to each research design and method (Creswell, 2014; Lowndes *et al.*, 2018). For example, qualitative research includes data-gathering techniques such as observation (participant, structured, and internet-mediated), interviews (structured, semi-structured, and in-depth), and questionnaires (self-completed and researcher-completed) (Saunders *et al.*, 2019). Each technique requires decisions to be made by the researchers (e.g. which type of interview, how to engage the data sources) and can present limitations that have to be

acknowledged to improve the quality of a research design (Creswell, 2014; Saunders *et al.*, 2019).

The techniques discussed in this section comprise secondary data (Paper I) and semi-structured interviews (Papers II to V), which are linked to the methodological plans for these papers, presented in section 3.3. Paper I used secondary data – discussed in Subsection 3.5.1 – entailing a collection of SPP discourses that were further analysed using hermeneutics. Papers II to V adopted semi-structured interviews as a data collection technique – Subsection 3.5.2 will discuss semi-structured interviews and present the interview process aimed at collecting the data analysed in Papers II to V.

3.5.1 Secondary data (Paper I)

Secondary data can include raw data and published summaries (i.e. compiled data) collected for other purposes but available for further analyses and interpretations (Creswell, 2014; Lowndes *et al.*, 2018; Saunders *et al.*, 2019). According to Saunders *et al.* (2019), secondary data can be organised into surveys (e.g. census), documents (e.g. texts, audio), and multiple-source compilations. The multiple-source compilation includes continuously updated (e.g. crime records), longitudinal (e.g. government and international organisations publications), and snapshot (e.g. academic papers) datasets.

The process of gathering secondary data for Paper I included multiple-source compilations of academic papers and publications from governments and international organisations. A literature review contributed to collecting academic papers for Paper I. Literature reviews are commonly aimed at exploring the research context, positioning research within an open dialogue in the academic field, and providing a theoretical framework for theory testing and development (Creswell, 2014; Saunders *et al.*, 2019). Literature reviews in hermeneutic research – flexible in data collection techniques (Hay, 2011) – are immersive journeys focused on pondering, questioning, and thinking (Smythe and Spence, 2012). In conceptual papers such as Paper I, reviewing the literature is a technique to access data (SPP discourses in this case) but not the research objective (as proposed by Jaakkola, 2020).

The literature review carried out to collect SPP discourses for Paper I included keywords such as “sustainability”, “PP”, “government procurement”, “SPP”, “green PP”, and “social PP”, considering their combinations. The online databases used for scientific paper search included Scopus, Web of Science, and Google Scholar.

Publications from governments and international organisations were also collected for building Paper I’s dataset. For instance, SPP guidelines published by international organisations (e.g. UNEP, World Bank) – accessed through an Internet search – were considered in the SPP discourse collection.

3.5.2 Semi-structured interviews (Papers II-V)

Interviews are focused on gathering information on beliefs, meanings, personal experiences, and understandings (Lowndes *et al.*, 2018; Yin, 2014) through an insightful conversation between researchers (asking questions) and interviewees (who interact and give meaning to their social realities) (Saunders *et al.*,

2019). The interviewer's role requires listening skills to understand links between interviewees' contexts, emotions, and terminology whilst inferring meanings intended by their words (Yin, 2014). Interviews are based on confidentiality (i.e. guaranteed anonymity), reflexivity (i.e. researcher and interviewees construct data together), and sensitivity (i.e. researcher immersion in data) (Corbin and Strauss, 2008).

According to Saunders *et al.* (2019), interviews can be organised into structured, unstructured, and semi-structured types. Structured interviews use researcher-completed, fixed questionnaires to collect standardised data suitable for quantitative analyses. Unstructured or in-depth interviews are informal conversations aimed at exploring a research topic by asking non-planned questions. Semi-structured interviews use a predetermined set of questions and themes to guide the conversation, following an emergent process that allows for the exploration of research questions with flexibility – complementing questions to clarify and delve into each topic (c.f. Corbin & Strauss, 2015).

Papers II to V used semi-structured interviews to collect data regarding SPP, organisational changes for SPP, collaboration between stakeholders in SPP, and organisational changes towards implementing collaboration in SPP. The interview themes and questions were designed considering previous literature, theoretical frameworks, the researcher's experience on the topics, and discussion with the researcher's tutors (as proposed by Saunders *et al.*, 2019).

Table 10 presents the summary of the interview themes and questions, including the initial expectations regarding each theme. The interview questions are presented in detail in Appendices A (in English) and B (in Portuguese).

Table 10. Summary of interview questions, themes, and initial expectations.

Questions	Theme	Initial expectations
1-7	Identification	Start the interview, understand the interviewee's context
8-12	Sustainability and PP	Identify the interviewee's understanding of key concepts, such as sustainability and SPP
13-19	OCM and SPP	Discuss organisational changes towards implementing SPP, including forces influencing such changes and approaches to change
20-32	Collaboration in SPP	Identify the interviewee's understanding of how interactions between stakeholders can contribute to SPP and discuss organisational changes towards implementing collaborative SPP initiatives

3.5.2.1 Interview application

Twenty-two semi-structured interviews were carried out with SPP professionals to collect data for Papers II-V: 12 from Brazil and 10 from Sweden.

Theoretical sampling was used to select potential interviewees, considering an initial idea of where to sample (e.g. SPP practitioners from Brazil and Sweden), choosing participants as they were needed, and expanding the sample following GT's data analysis development (see Saunders et al., 2019). The sample expansion was also complemented through a volunteer basis – snowball sampling (see Sapsford and Jupp, 2006; Schutt, 2012), with some participants identifying further potential interviewees to facilitate access.

The selection criteria of interviewees were that they: 1) have been SPP professionals; 2) have had recent experience working in public organisations; and 3) have contributed to the implementation of SPP. The researcher's professional and academic networks were used to connect with potential interviewees. The interviewees were positioned in different public organisations: 9 Brazilian and 6 Swedish ones working as public buyers. Table 11 shows the interviewees' professional profiles.

Table 11. Professional profiles of interviewed Brazilian and Swedish practitioners.

Code	Position	Organisation level	Years working with public procurement	Interview date and duration
BR1	Audit officer	National	24	Oct 2022, 1h47
BR2	Procurement manager	National	16	Oct 2022, 49m
BR3	Procurement officer	National	9	Oct 2022, 2h05
BR4	Strategic advisor	National	20	Nov 2022, 1h22
BR5	Chief Procurement Officer	National	17	Dec 2022, 1h05
BR6	Audit manager	National	7	Dec 2022, 52m
BR7	Procurement manager	National	12	Dec 2022, 49m
BR8	Strategic advisor	National	22	Dec 2022, 1h01
BR9	Chief Procurement Officer	Regional	15	Dec 2022, 1h24
BR10	Procurement officer	National	15	Feb 2023, 1h35
BR11	Strategic advisor	National	11	May 2023, 1h 24
BR12	Chief Procurement Officer	National	16	May 2023, 1h12
SE1	Procurement officer	Local	20	Jan 2023, 47m
SE2	Sustainability officer	National	7	Feb 2023, 56m
SE3	Chief Sustainability Officer	National	4	Feb 2023, 50m
SE4	Chief Executive Officer	National	10	Feb 2023, 46m
SE5	Project manager	Regional	14	Feb 2023, 1h35
SE6	Sustainability manager	Local	5	Feb 2023, 46m

SE7	Project manager	Regional	11	Feb 2023, 49m
SE8	Chief Procurement Officer	Local	25	Feb 2023, 1h12
SE9	Chief Procurement Officer	Regional	5	Mar 2023, 45m
SE10	Sustainability manager	Regional	7	May 2023, 51m

Four semi-structured interviews with external practitioners working at organisations acting as SPP facilitators were additionally conducted to increase the validity of the data and its analysis by triangulation (as described by Saunders et al., 2019). Table 12 presents the professional profiles of these interviewees. The data collected through these additional interviews was used in Paper IV's data analysis, which explored collaboration in SPP through perspectives from public buyers and SPP facilitators.

Table 12. Professional profiles of external practitioners interviewed.

Code	Position	Country	Organisation role	Years working with public procurement	Interview date and duration
EX1	Researcher	Germany	Research institute	1	Oct 2022, 1h19
EX2	Director	Germany	NGO	5	Jan 2023, 1h26
EX3	Manager	Sweden	NGO	1	Feb 2023, 0h57
EX4	Director	Germany	Parliamentary group	9	Mar 2023, 1h09

The 26 interviews were carried out from October 2022 to May 2023 and most of them (23) were done digitally, via Zoom – three were done in person. The interviews ranged between 45 minutes and 2 hours, comprising 29 hours and 33 minutes in total. The interviews were conducted in different languages: Portuguese for the Brazilian interviewees and English for the German and Swedish ones. The interviews were transcribed in their original languages, aiming at reducing language nuances whilst managing and analysing the collected data. The data collection was finished after reaching conceptual saturation (see Corbin & Strauss, 2008; Saunders et al., 2019), when categories and central categories were clearly understood and the latest data provided by the interviewees did not reveal new insights.

The interview transcriptions comprised 384 pages, 2.227 paragraphs, and 213.457 words. After the transcriptions, the interviewees were identified by a code to guarantee anonymity: the Brazilian professionals with a BR followed by a number (e.g.: BR1, BR2), the external practitioners with EX followed by a number (e.g. EX1, EX2), and the Swedish professionals with SE followed by a number. The collected data was categorised with the support of NVivo 1.0

(QSR, 2020). The complete data categorisation resulted in 376 codes, based on 1.990 text references.

3.6 Data analysis

This section presents the data analysis methods used in the papers appended to this doctoral thesis, based on qualitative data. Qualitative data analysis is aimed at organising and interpreting data through a holistic and intuitive process that considers the interconnections between each data piece and the whole set of data (Corbin and Strauss, 2008; Saunders *et al.*, 2019).

Paper I's data analysis was based on the hermeneutics method, concerned with reconstructing meanings and beliefs that shape social phenomena (Hay, 2011). Social phenomena are highly complex and have a multi-dimensional nature, formed by individual experiences and their historical contexts, allowing different interpretations (Gibbons, 2006). The social sciences can analyse, explain, and contextualise problems through a process of interpretation and reinterpretation of social phenomena (Heller, 1989).

Hermeneutics provides an inductive approach to interpreting social phenomena by analysing texts and their internal consistency (Baškarada and Koronios, 2018). Hermeneutics is based on the inquiry into the conditions of human understanding (Harrington, 2001; Heidegger, 1996; Leyh, 1988), guiding researchers to a meta-interpretation work (Dreyfus, 1980), i.e. an interpretation of inner realities formed by individual experiences, their historical contexts, and their own interpretations (Dilthey, 1972). This meta-interpretation is essential to provide a critical understanding of a social phenomenon, especially when facing competing discourses motivated by conflicting understandings (Dreyfus, 1980).

The dialogical relationship between researcher and data in hermeneutics supports a circular process of data analysis that comprises approaching, reading, reflecting, re-reading, and writing (Moules, 2002; Smythe and Spence, 2012). This hermeneutic circle supports a more complete interpretation of meanings given to a phenomenon by building knowledge through understanding each part via the whole and the whole via its parts (Schleiermacher, 1998) – interpretation and analysis are synonymous in hermeneutics (Moules, 2002).

The data collected for Papers II-V was analysed using GT's constant comparative method. According to Corbin and Strauss (1998), this method comprises the following stages: 1) coding data by creating and identifying conceptual headings (including in-vivo codes by using terms from the data), constantly comparing them for similarities and differences; 2) searching for relationships between the codes to build categories and subcategories, continuously comparing them with the data; 3) identifying and integrating central categories and their properties; and 4) developing a grounded theory from the categories and their relationships. In those papers, the concepts and categories found in the literature review contributed to identifying research gaps, enhancing comparisons, and increasing theoretical sensitivity (as proposed by Mills et al., 2006; Tummers and Karsten, 2012), being used as an additional data source and coded after the conceptual headings emerged from the data.

Table 13 presents examples of how the stages of GT's constant comparative analysis were carried out during Papers II-V development. The code structure and categorisation used for Papers II-V (comprising 332 codes) are presented in Appendix C.

Table 13. Examples of grounded theory's constant comparative analysis stages.

Paper	GT constant comparative analysis stages			
	Coding	**Building categories**	**Integrating central categories**	**Developing a grounded theory**
II	Identifying each driver for SPP	Organising drivers into internal, connecting, and external	Integrating drivers for SPP and organising them according to their contexts	Proposing a new understanding of drivers for SPP
III	Identifying each barrier to SPP and strategy to overcome them	Organising barriers and strategies into internal, connecting, and external	Integrating barriers and strategies and organising them according to their contexts	Offering a new understanding of barriers to SPP and strategies to overcome them
IV	Identifying each collaboration benefit and drawback	Organising collaboration benefits and drawbacks into the procurement phases	Consolidating collaboration benefits and drawbacks	Building a framework for understanding collaboration between stakeholders in SPP
V	Identifying each force influencing organisational changes for collaboration in SPP	Organising forces into internal, connecting-protagonist, connecting-supporting, and external	Integrating forces influencing organisational changes for collaboration in SPP	Proposing a new understanding of organisational changes for collaboration in SPP

The data was analysed in their original languages (Portuguese and English). Only a minimum translation was done during the data analysis (following the lessons by Corbin and Strauss, 2008). For instance, the data in Portuguese was coded in English, following the language of the papers.

3.7 Research quality

This section discusses the efforts towards improving the quality of this doctoral thesis's research, including a presentation of data quality considering the methods used in its appended papers and a discussion regarding research ethics and scientific integrity.

Research quality can be related to researchers' efforts to ensure the validity and reliability of research findings (Creswell, 2014; Saunders et al., 2019; Yin, 2014). Validity entails checking the accuracy of the findings (Creswell, 2014). Validity can be divided into internal (more related to explanatory studies and causal relationships) and external (concerned with generalisation of findings to other contexts) (Saunders et al., 2019; Yin, 2014). Strategies to strengthen validity entail triangulation (i.e. using additional data sources), rich description of the findings, clarifying bias (e.g. presenting limitations of the methods), and spending prolonged time in the research field (Creswell, 2014).

Reliability comprises the findings' consistency (Creswell, 2014). According to Saunders et al. (2019), threats to reliability include participant error (i.e. factors affecting interviewees' performance), participant bias (i.e. factors inducing false answers), researcher error (i.e. factors influencing researchers' interpretation), and researcher bias (i.e. factors leading to response selectivity).

Research ethics and scientific integrity are aimed at improving the quality of research (Saunders et al., 2019) by following common standards, values, and practices directed to prevent research misconduct (Kretser et al., 2019) – anticipating ethical issues that may arise during the research process (Creswell, 2014).

3.7.1 Data quality

The research design of Paper I has data quality issues associated with using hermeneutics as a data analysis method. For example, each SPP discourse is presented in a non-exhaustive way due to formal limitations (e.g. size of a scientific journal paper). Reliability in this paper may be affected by researcher error and bias given the nature of the method, which endorses the influence of researchers' values (as discussed by Heller, 1989), perceptions (see Moules, 2002), and experience (proposed by Dilthey, 1972). Validity in this paper, especially external validity, or generalisation, is limited by the context where the SPP discourses were developed and the context of the paper.

According to Moules (2002), research quality in hermeneutics also requires credibility (e.g. sharing interpretations with other readers), dependability (e.g. documenting the analysis to demonstrate how an interpretation was reached), and transferability (e.g. readers understanding the applicability of an interpretation to another context). In Paper I, credibility was ensured by intensely discussing interpretations between the three researchers who developed the paper. Dependability was granted by structuring the paper with sections of findings (i.e. SPP discourse analysis) that subsidised the interpretation – in an evolving writing style culminating with a proposal of a holistic framework of SPP. Transferability was verified by an academic journal's double-blinded peer-review process that this paper passed through before its publication and citations from other researchers. Transferability was also granted by an international award for quality research on PP received by Paper I, the TED Ambassador Award 2023, promoted by the Publications Office of the European Union.

The research design used in Papers II-V included interviews as a data collection technique, which also has data quality issues that threaten the reliability and validity of the research findings (see Saunders et al., 2019). For example,

the interviewees' selection criteria (e.g. SPP practitioners, external practitioners working in SPP facilitators) may be a threat related to participation bias. The engagement of the interviewer with sustainability and SPP may create researcher bias. The use of a second language (English) for interviewing German and Swedish professionals may have led to cultural and language nuances that affected interactions with the interviewees and interpretation of the data. The number of interviews limits the external validity of the findings.

Papers II-V's methods also include using previous research to support data analysis in GT (e.g., to identify research gaps, enhance categorization, and improve theoretical sensitivity), which also has pitfalls that may threaten data quality. In this context, the pitfalls that were acknowledged and treated to strengthen this research's quality include overlooking certain issues pointed out by previous studies, selecting data according to literature findings, and hindering new insights (see Tummers and Karsten, 2012). In this research, strategies to overcome such threats entailed reviewing the literature throughout the entire research process (e.g. during the data collection and analysis) and categorising this literature after the data.

3.7.2 Research ethics

Good research practices can improve research ethics and scientific integrity (Saunders *et al.*, 2019), including promoting a culture of integrity, ensuring training in research design, and respecting the confidentiality of the data when required (All European Academies, 2017). Kretser et al. (2019) present two principles for scientific integrity and nine best practices to improve research quality. The principles are: 1) foster a culture of integrity and 2) policy interests should not interfere with scientific integrity. The best practices include 1) training in research methods; 2) training in ethics and conduct; 3) reproducibility and transparency; 4) open science; 5) communication and outreach skills; 6) strengthening the peer review process; 7) publishing findings even if they were unanticipated (i.e. contrary to research hypothesis); 8) transparent processes of correction and retraction of academic papers; 9) reward high standards of integrity.

Research ethics and scientific integrity are relevant parts of the research carried out for this doctoral thesis. The author worked towards incorporating best research practices (such as those proposed by Kretser et al., 2019), aiming to ensure the appropriate standards of ethical behaviour and improve this research's quality. For example, the ethical basis of this doctoral thesis included standards on scientific integrity promoted by the University of Gävle and the Sustainability and Procurement in International, European, and National Systems (SAPIENS) Network, a European Union-funded research project to which the author's doctoral work was linked.

The author engaged in doctoral studies related to research philosophies, methods, and ethics to ensure the appropriate use of research protocols. For instance, courses on such topics included research methods and philosophy at the University of Gävle and research ethics and open science at the SAPIENS Network – hosted by the University of Turin, Italy. SAPIENS Network also provided opportunities to develop and present research plans at project

meetings (at least one per semester). These meetings allowed the author to check the methodological and ethical aspects of his doctoral research. These project meetings also provided additional feedback (including from external experts) that was used to improve his research plan.

Four research papers appended to this doctoral thesis include data from interviews. Best practices for planning and collecting data using this technique (c.f. Saunders et al., 2019) were used. For example, before data collection, the interviewees consented to the interview recording for transcription purposes. However, most of the interviewees did not allow data sharing. Some interviewees mentioned political reasons in their home country for not authorising data sharing. Nonetheless, the data was anonymised, is available at the University of Gävle secured database, and was checked by the author's research group, improving the data validation.

The authorship of the papers appended to this doctoral thesis was discussed by each research team. Only the members of the team who contributed to the paper development were included in the publications.

The selection of journals for submitting the papers appended to this doctoral thesis was done using several indexes (e.g. impact factor) and a discussion between the authors. A list of potential journals was built according to each paper's objectives, aiming at publishing them in peer-reviewed and recognised (i.e. non-predatory) journals.

Open science was a mandatory requirement of the research project linked to this doctoral thesis, following the SAPIENS Network guidelines. This research project and the University of Gävle offered funding for full open science, which allowed the selection of the most transparent publishing option (e.g. gold open access). The author was also engaged in social media for research purposes, amplifying his research dissemination – research outreach and communication courses and duties were included in the SAPIENS Network activities.

3.8 Research limitations

This thesis has limitations related to its research aims and design, presenting gaps that can be addressed by future research efforts. This thesis adopts an exploratory purpose, aiming to provide a more complete understanding of collaboration between stakeholders in SPP through a flexible process. This thesis's findings should not be used to address questions based on different research purposes, such as who has been implementing collaborative SPP initiatives (i.e. descriptive research), why collaboration in SPP contributes to developing sustainability requirements (i.e. explanatory research), and how specific collaborative SPP initiatives have been undertaken (i.e. evaluative research).

The interviews used to collect data for this thesis included questions regarding sustainability understanding, SPP implementation, collaboration in SPP, and implementing collaboration in SPP – entailing discussions on drivers for, barriers to change, and strategies to overcome barriers. However, the questions and interview development were not aimed at discussing direct relationships between strategies and barriers (e.g. which strategies are aimed at overcoming

a specific barrier), which limits the depth of this topic's analysis. The interviews were applied to SPP practitioners mainly working in public organisations, i.e. buyers – another limitation of this research is related to the absence of suppliers' perspectives on the topics, which could contribute to a more complete understanding of the studied phenomena.

This thesis uses GT as a research strategy to support an inductive approach to building knowledge on collaboration in SPP. The GT approach adopted by this thesis was not aimed at building more general theories but helped to develop middle-range theories on OCM for sustainability and collaboration theory – by applying the knowledge previously built on these theories to a new context (i.e. SPP). These middle-ranged theories – including the theoretical contributions provided by this thesis – may be tested and eventually applied to other contexts for further development – although there are limits to the generalisation of this research's findings (as pointed out in Subsection 3.7.1). The pragmatism and interpretivism adopted as research philosophies in this thesis' efforts support the idea that the theoretical contributions of this research should be tested and applied to other contexts – knowledge is socially constructed and intertwined with actions, requiring different research efforts on the same topic, including analysing different contexts, to provide better understandings of social phenomena (c.f. Saunders *et al.*, 2019).

4. Summary of appended papers

This doctoral thesis analyses the influence of collaboration between stakeholders in SPP, focusing on how organisations can change towards implementing collaborative initiatives. The five papers appended to this thesis are summarised in this chapter. The first paper discusses a systemic approach to SPP, aiming to understand how interactions between stakeholders support the incorporation of sustainability into PP. The second paper delves into organisational changes for SPP through an analysis of drivers for implementing SPP, unveiling the role of collaboration between stakeholders in driving such changes. The third paper focuses on barriers hindering efforts towards organisational changes for SPP and strategies to overcome such barriers, considering the role of collaboration. The fourth paper analyses the elements that comprise collaborative SPP initiatives, providing a systemic understanding of collaboration in the SPP context. The fifth paper discusses organisational changes towards implementing collaboration in SPP, focusing on forces contributing to or hindering such changes.

4.1 Paper I: Developing a harmonic sustainable public procurement framework

4.1.1 Research background

Discourses and practical efforts aiming at understanding and implementing SPP have been focusing on the following elements: 1) demand-offering (e.g. Parikka-Alhola, 2008) – i.e. the transactional relationship through which buyers (demanding) and suppliers (offering) exchange products and services for an agreed sum of money (Baily *et al.*, 2015); 2) procurement specifications of products and services (see Thai, 2009); 3) stakeholder interactions (such as Alhola et al., 2017); and 4) disciplinary approaches (for example Ferk & Ferk, 2018; Günther & Scheibe, 2006; Sanchez-Graells, 2018). However, most efforts have taken a compartmentalised approach, i.e. addressing only one of the elements, with some limited exceptions such as demand-offering and some disciplinary approaches (for example Wesseling & Edquist, 2018) and specifications and stakeholder interactions (see Witjes & Lozano, 2016).

Paper I aimed to develop a framework that considers the interconnections between the elements comprising SPP initiatives, proposing a holistic understanding entailing demand-offering, procurement specifications, stakeholder interactions, disciplinary approaches, and sustainability.

4.1.2 Contribution to the thesis

Paper I used hermeneutics to provide an explanation of SPP discourses to propose the Harmonic SPP framework (Figure 8). The Harmonic SPP framework

integrates the following system elements: 1) demand-offering (including products, services, functions, and needs); 2) procurement specifications (with technical, non-technical, and socio-cultural ones); 3) stakeholder interactions (buyers, suppliers, and other stakeholders); 4) disciplinary approaches (technical, public policy, legal, economic, management, marketing, behavioural, societal, and institutional); and 5) sustainability (economic, environmental, social, and time dimensions). The framework is based on the interrelations within and between system elements to fully capture the complexity of SPP. The Harmonic SPP framework provides a better understanding of SPP complexity by explicitly considering the system elements and their interactions.

Paper I contributes to this thesis by showing that interactions between stakeholders are a key element of SPP initiatives, leveraging the potential of developing collaboration to better implement SPP – by harmonising SPP elements. Harmonising the interrelations between the demand-offering, procurement specifications, stakeholder interactions, disciplinary approaches, and the four sustainability dimensions is *sine qua non* in achieving SPP.

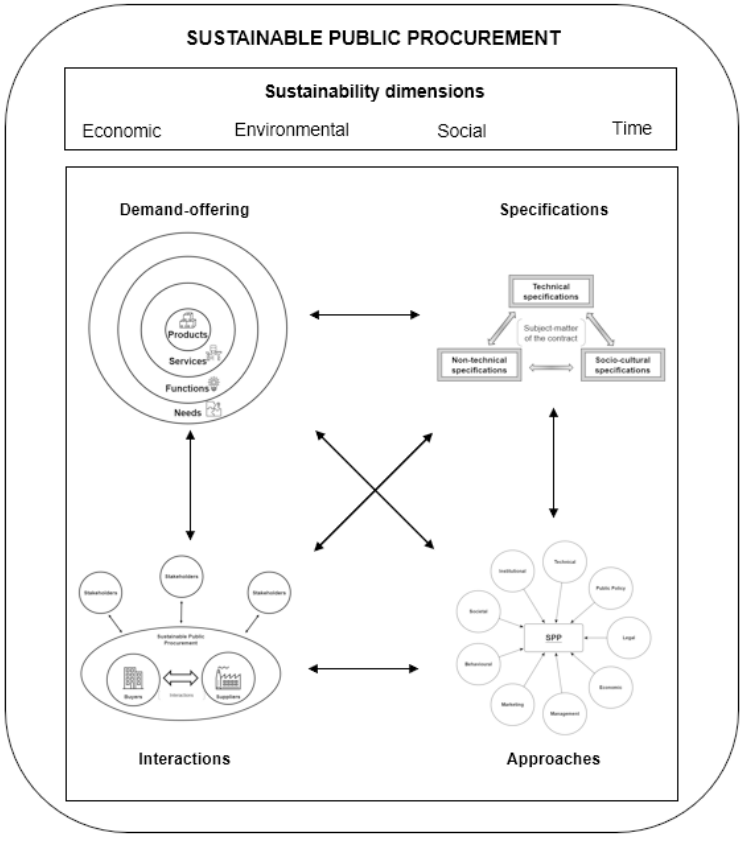


Figure 8. The Harmonic Sustainable Public Procurement framework.

4.2 Paper II: Analysing the drivers for sustainable public procurement

4.2.1 Research background

The managerial approach to implementing SPP (identified by Paper I) entails practical and research efforts based on OCM. Although there is a growing body of literature on OCM for SPP (Cheng *et al.*, 2018; Sönnichsen and Clement, 2020), most studies exploring drivers for organisational change focus on a single context – for example, a single country (Berg *et al.*, 2022; Testa *et al.*, 2016) or a single organisation (Eikelboom *et al.*, 2018; Mendonça *et al.*, 2021). There has been limited to no research regarding drivers for change towards SPP in different contexts (Cheng *et al.*, 2018; Sattari *et al.*, 2022; Vejaratnam *et al.*, 2020). Additionally, there has been limited research taking a more complete perspective on drivers for change processes towards SPP, i.e. considering organisations as whole systems interconnected in broader systems (Hjorth and Bagheri, 2006; Jackson, 2003; Ragsdell, 2000).

Paper II aimed to analyse the drivers for organisational changes towards implementing SPP from different contexts, based on semi-structured interviews with SPP practitioners from Brazil and Sweden.

4.2.2 Contribution to the thesis

Paper II provides a more complete understanding of drivers for SPP, expanding the compilation of drivers and providing a categorisation of internal, connecting, and external levels (based on Lozano's (2015) categorisation). This paper collates 46 drivers for SPP, including 24 drivers identified in the literature review (presented in Figure 4) and 22 new drivers mentioned by the interviewees.

The drivers for SPP mentioned by the interviewees were ranked according to the number of interviewees who mentioned them, divided into high-ranked, medium-ranked, and low-ranked, complementing the literature by providing a perspective on drivers' importance. The ranking of drivers for SPP proposes an understanding of which drivers are more important according to the interviewees, providing a path for prioritising research and practical initiatives in exploring each driver. Collaboration between stakeholders was the most important driver for SPP following this ranking.

This paper also shows that, whilst some of the drivers are common in Brazil and Sweden, other drivers for SPP are context-specific. This allowed to position the drivers in a continuum from those related to practical efforts, i.e. praxis-driven, as identified by the 15 drivers in the Brazilian context (in line with Delmonico et al., 2018) to those related to policy efforts, i.e. policy-driven, as identified by the 9 drivers in the Swedish context (in line with Daugbjerg, 2023; Sattari et al., 2022), with the 22 common drivers in the middle. Figure 9 shows the 46 drivers for SPP divided into internal, connecting, and external and set up in the continuum from praxis-driven to policy-driven contexts, with 24 drivers related to these approaches and 22 listed as common drivers. The common drivers include 4 drivers indicated in the SPP literature

but not mentioned by the interviewees. The drivers were organised into high-ranked, medium-ranked, and low-ranked. The drivers not mentioned by the interviewees were positioned in the low rank.

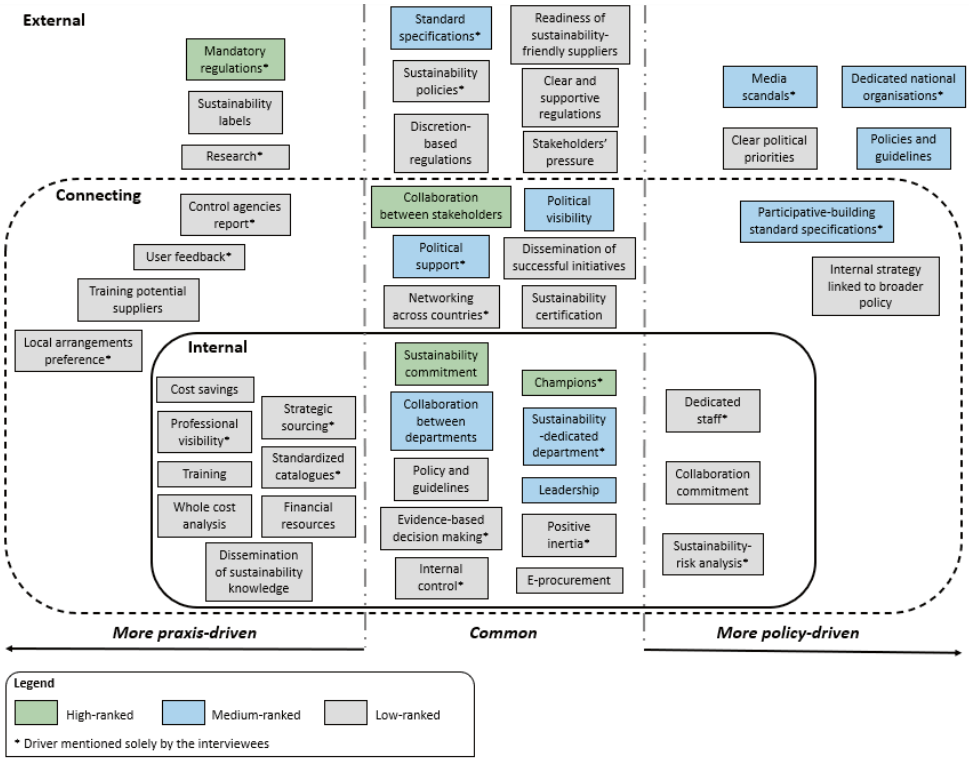


Figure 9. Drivers for sustainable public procurement in a continuum from praxis-driven to policy-driven contexts.

The analysis of drivers for sustainability within a process, such as SPP, considering different contexts, can improve the understanding of drivers for sustainability by showing that they can be expanded from an organisation context to a process one, be context-specific, and organised in a continuum. This research provides depth on this topic and shows a contextual influence on drivers for sustainability, complementing the previous efforts on categorising (see Lozano, 2015) and ranking (c.f. Lozano & von Haartman, 2018) drivers through focusing on understanding changes on a process level.

4.3 Paper III: Overcoming the barriers to sustainable public procurement

4.3.1 Research background

Research on OCM for SPP has been exploring barriers to organisational changes for SPP (Delmonico *et al.*, 2018; Preuss and Walker, 2011) and

strategies to overcome such barriers (e.g. Berg et al., 2022; Leal Filho et al., 2019). However, there is a lack of a more complete understanding of how organisations can promote efforts to overcome barriers to SPP considering the influence of internal, connecting, and external forces. For example, there are limited efforts in categorising the barriers that can hinder change efforts towards SPP and the strategies to overcome such barriers (e.g. limited to internal-external categories). Research efforts in analysing the importance of barriers to SPP and strategies to overcome them are also limited. There is also little to no literature discussing the influence of contexts on those forces – although there is a call for comparing different contexts of changes towards SPP (Cheng *et al.*, 2018; Sattari *et al.*, 2022; Vejaratnam *et al.*, 2020).

Paper III aimed to provide a more complete understanding of barriers to organisational changes towards SPP and strategies to overcome such barriers by analysing data from two distinct world regions (Brazil and Sweden). This research analysed data collected through semi-structured interviews with SPP practitioners from both countries.

4.3.2 Contribution to the thesis

Paper III contributes to a more complete understanding of barriers to SPP and strategies to overcome them by proposing a catalogue of these forces, categorising them into internal, connecting, and external (as proposed by Lozano et al., 2020). By organising the barriers into these three levels, this paper provides an understanding of the systemic influence of each hindering force in SPP-related organisations. This categorisation goes beyond the internal-external one and provides an awareness of the connecting barriers and strategies – forces that can link the internal to the external levels, offering a more complete approach to the topic.

This paper compiles 51 barriers to SPP, comprising 30 barriers previously identified in the literature (see Figure 5) and 21 new barriers mentioned solely by the interviewees, complementing the previous research efforts. A variety of strategies can be used to overcome such barriers, including the 31 strategies organised by this paper, of which 18 were previously identified in the literature (see Figure 6) and 13 strategies are new (i.e. mentioned solely by the interviewees), expanding the literature.

The barriers to SPP and strategies to overcome them mentioned by the interviewees were ranked considering the number of interviewees mentioning them. Both forces were organised in rankings with three levels: high-rank, medium-rank, and low-rank. These rankings can be used to understand the degree of attention given to those barriers and strategies by the interviewed practitioners, contributing to supporting future research and practical efforts. Lack of collaboration was identified as a barrier to SPP (low-ranked). Collaboration between stakeholders was the most important strategy to overcome barriers to SPP following the strategy ranking.

The findings of this paper also contribute to the literature by showing that barriers to SPP and strategies to overcome them are context-related (contrary to the findings of previous research, e.g. Shaikh & Channa, 2022). There are 17 barriers and 10 strategies linked to contexts based on practical efforts, as

identified in the Brazilian context (in line with Delmonico *et al.*, 2018), and 9 barriers and 7 strategies related to contexts based on policy efforts, as identified in the Swedish context (concurring with Daugbjerg, 2023; Sattari *et al.*, 2022). Common forces (i.e. identified in both contexts) include 25 barriers and 14 strategies. Lack of collaboration between stakeholders was categorised as a barrier in contexts based on practical efforts, which may be explained by reduced institutional and policy support to engage with stakeholders in such contexts when compared to contexts based on policy efforts.

The barriers to SPP and strategies to overcome them can be organised in a continuum from praxis-driven to policy-driven contexts, encompassing the common forces (related to both approaches, including the barriers and strategies mentioned in the SPP literature but not identified in the contexts analysed). There are 26 barriers and 17 strategies related to praxis-driven and policy-driven contexts, whilst 25 barriers and 14 strategies are common forces. Figure 10 and Figure 11 present the 51 barriers to SPP and 31 strategies to overcome them in continuums from praxis-driven to policy-driven contexts, categorised into internal, connecting, and external. The barriers and strategies were highlighted according to their position in the respective ranking, whilst forces not mentioned by the interviewees were positioned in the low rank.

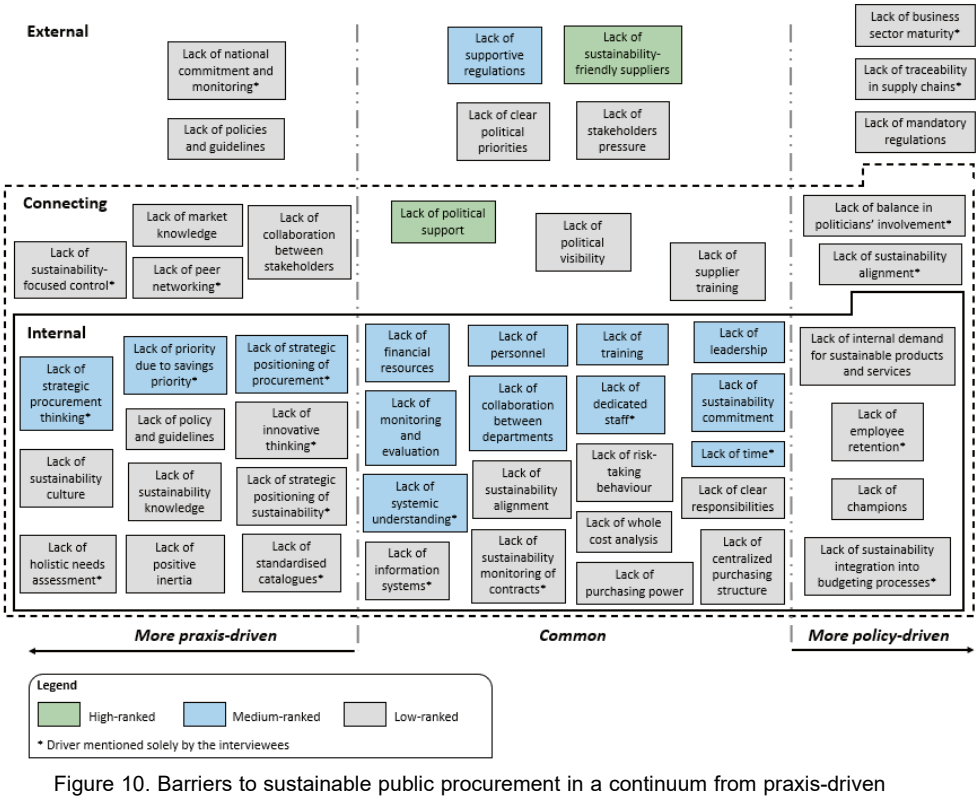


Figure 10. Barriers to sustainable public procurement in a continuum from praxis-driven to policy-driven contexts.

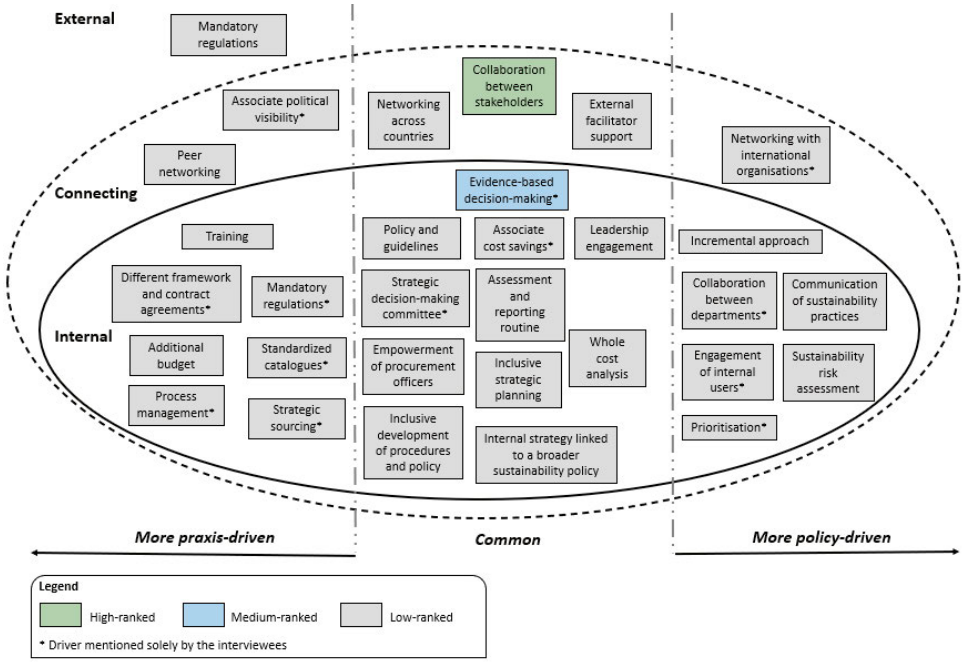


Figure 11. Strategies to overcome barriers to sustainable public procurement in a continuum from praxis-driven to policy-driven contexts.

This paper provides depth to the understanding of barriers to sustainability and strategies to overcome them by presenting the contextual impact of those forces and providing a continuum for organising them. This paper also contributes to answering a call for analysing the importance of barriers to sustainability and strategies to overcome them (see Lozano & von Haartman, 2018) and complements the previous efforts in categorising these forces (c.f. Jaramillo et al., 2019; Klein et al., 2022; Lozano et al., 2020). Research efforts in understanding changes within processes, such as SPP, are still limited – OCM for sustainability has mainly studied how organisations can act considering their own systems to move towards sustainability. This paper contributes to analysing how organisations can move towards sustainability through interactions in broader systems, expanding the body of literature on this topic.

4.4 Paper IV: Understanding collaboration between stakeholders in sustainable public procurement

4.4.1 Research background

Collaboration between stakeholders was highlighted in Papers II and III as a key driver for SPP and the most important strategy for overcoming barriers to SPP. However, the literature has not yet provided a systemic understanding of collaboration in SPP (Kabra *et al.*, 2023; Rejeb *et al.*, 2023), including elements such as collaboration channels (i.e. who collaborates with whom) (c.f.

Kristensen *et al.*, 2021), benefits and drawbacks (i.e. why and why not collaborate) (as proposed by Guarnieri and Gomes, 2019), and mechanisms (i.e. how to collaborate) (see Sönnichsen and Clement, 2020; Uttam and Roos, 2015). There is also a research gap in exploring who can contribute to establishing and maintaining collaboration in SPP (i.e. facilitators) (as proposed by Lozano *et al.*, 2020). Developing countries' perspectives on this topic are also underresearched – previous research on collaboration in SPP has also focused mainly on exploring developed countries' experiences, such as Finland (e.g. Alhola and Nissinen, 2018; Holma *et al.*, 2020), Sweden (see Sattari *et al.*, 2022; Uttam and Roos, 2015), and the United Kingdom (e.g. Erridge and Greer, 2002).

Paper IV aimed to provide depth on collaboration between stakeholders in SPP by analysing the collaboration channels, mechanisms, benefits, drawbacks, and facilitators. This paper was based on empirical data from two different world regions – including a developed country in Europe (Sweden) and a developing country in South America (Brazil), offering a more complete approach to the topic. The perspective of the external practitioners working at organisational facilitators in the SPP context was also considered in this paper.

4.4.2 Contribution to the thesis

Paper IV proposes a more complete understanding of collaboration in SPP by analysing and organising its elements, providing depth to collaboration channels (i.e. who collaborates with whom in SPP initiatives), collaboration mechanisms (i.e. how to collaborate in SPP), benefits (why collaborate in SPP), drawbacks (why not collaborate in SPP), and facilitators of collaboration (i.e. who can contribute to establishing and maintaining collaboration in SPP). This paper also sheds light on the similarities and differences between the collaboration elements in both contexts analysed (Brazil and Sweden).

Paper IV proposes a framework for Understanding Collaboration between Stakeholders in SPP – UnCoStake (including 4 channels, 15 mechanisms, 27 benefits, 10 drawbacks, and 2 types of facilitators). This framework, presented in Figure 12, can also support decision-making processes towards breaking the PP walls and establishing collaborative SPP initiatives.

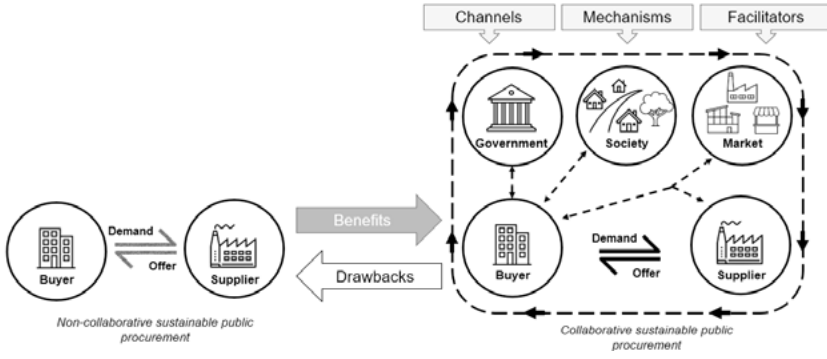


Figure 12. Framework for Understanding Collaboration between Stakeholders in Sustainable Public Procurement – UnCoStake.

The UnCoStake may be the first systematisation of collaboration between stakeholders in SPP, considering that previous efforts in understanding collaboration between stakeholders in SPP have been mostly dedicated to analysing specific elements of this process. The UnCoStake adds to collaboration theory by explicitly considering six elements of collaboration for sustainability: focus (what; in this case, SPP), channel (who), mechanisms (who), why (benefits), why not (drawbacks), and who can facilitate collaboration (expanding the frameworks proposed by Elia *et al.*, 2020; Lozano *et al.*, 2021). The UnCoStake adds to the SPP literature by providing a more complete approach to collaboration and contributes to collaboration for sustainability by applying its concepts to a specific process (i.e. SPP).

4.5 Paper V: Managing organisational changes for collaboration in sustainable public procurement

4.5.1 Research background

Collaboration between stakeholders in SPP has been driving organisational changes for SPP (see Paper II), overcoming barriers to implementing SPP (see Paper III), and offering 27 benefits to SPP initiatives (see Paper IV). However, implementing collaboration between stakeholders in SPP also requires organisational changes (Erridge and Greer, 2002), but there is limited research focusing on how organisations can change towards developing collaborative SPP initiatives. There is also little to no research providing an understanding of the intersections between organisational changes for SPP and changes for implementing collaboration – changes for SPP are usually understood as a single and monolithic process.

Paper V aimed to provide depth to the OCM approach for SPP by exploring forces influencing organisational changes towards collaboration – whilst analysing the interconnection between changes towards SPP and changes for collaboration, focusing on a more plural and systemic perspective. This paper analysed empirical data collected through semi-structured interviews with SPP practitioners from Brazil and Sweden.

4.5.2 Contribution to the thesis

Paper V offers a catalogue of forces (i.e. drivers, barriers, and strategies) influencing organisational changes for collaboration between stakeholders in SPP. This paper collates 23 drivers that contribute to moving organisations towards implementing collaboration in SPP, of which 10 drivers were previously identified in the literature (see Table 4) and 13 new drivers were mentioned by the interviewees but not identified in previous research – complementing the literature. A variety of barriers can hinder change efforts towards implementing collaboration in SPP, including the 40 barriers identified by this paper, of which 21 were previously identified in the literature (see Table 5) and 19 new barriers were mentioned by the interviewees but not identified in previous research, expanding the literature. Several strategies can contribute to overcoming barriers to implementing collaboration in SPP, such as the 24 strategies collated by this paper, comprising 5 strategies previously identified in the literature (see Table 6) and 19 new strategies (i.e. not identified in previous research), which complement research efforts.

Paper V proposes a new categorisation of change forces in the sustainability context, based on a more complete approach to the forces influencing organisational changes towards collaboration in SPP – from a changing organisation's perspective. This categorisation entails an internal level, i.e. more dependent on internal stakeholders (i.e. high organisational influence), and an external level, i.e. more dependent on external stakeholders (i.e. limited organisational influence). This categorisation also includes a connecting level that entails forces more dependent on both internal and external stakeholders and their systems (concurring with Lozano, 2015; Lozano et al., 2019). However, this new categorisation expands the literature by arranging connecting forces into two sets, according to which role an organisation can perform to link the internal to external systems: protagonist (i.e. more related to internal stakeholder initiatives and medium organisational influence) and supporting (i.e. more related to external stakeholder initiatives and low organisational influence). Figure 13 presents the Change Forces System – ChaFoS, a holistic approach to forces influencing organisational changes for sustainability that explicitly considers stakeholder dependence and organisational influence.

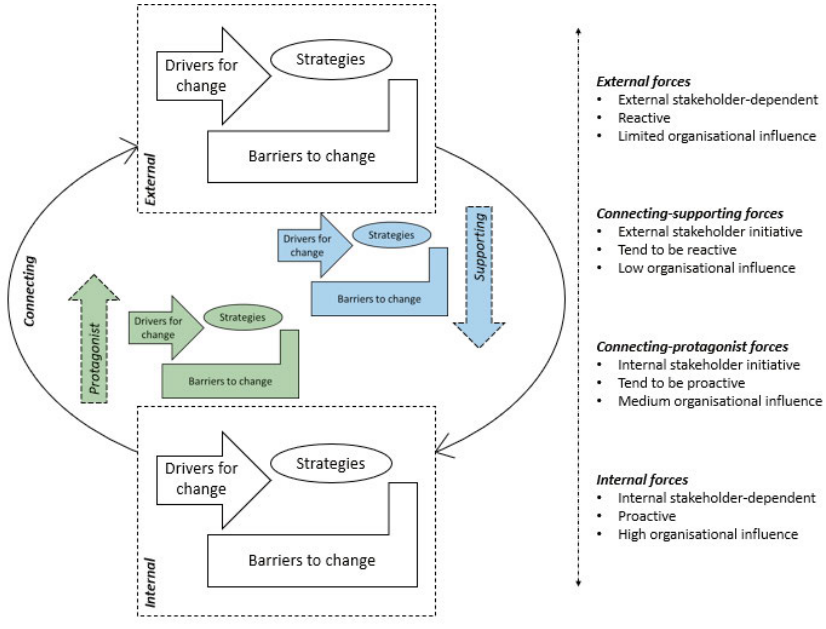


Figure 13. Change Forces System.

Paper V also contribute to OCM for sustainability by offering an understanding of simultaneous change processes contributing to moving organisations towards implementing SPP. Organisations can change for implementing SPP but, if they intend to use a key driver such as collaboration between stakeholders, an additional and specific change must be carried out (in agreement with Erridge & Greer, 2002). The change process for implementing collaboration in SPP can be understood as derived from the main change towards SPP, suggesting that other change processes may be needed to use additional forces from the main change. The multiple change processes derived from and linked to the main change can be defined as nested changes. The concept of nested changes expands the concept of multiple change processes (see Pettigrew et al., 2001) and adds to the OCM literature by considering that some change processes are multiple and composite instead of single and monolithic.

Figure 14 presents a framework for understanding nested organisational changes in the context of implementing SPP, based on a simplified version of the framework proposed by Lozano (2012) for orchestrating changes for organisational sustainability. This framework considers the different forces influencing the main and nested change processes and includes organisational changes for collaboration in SPP as a nested change.

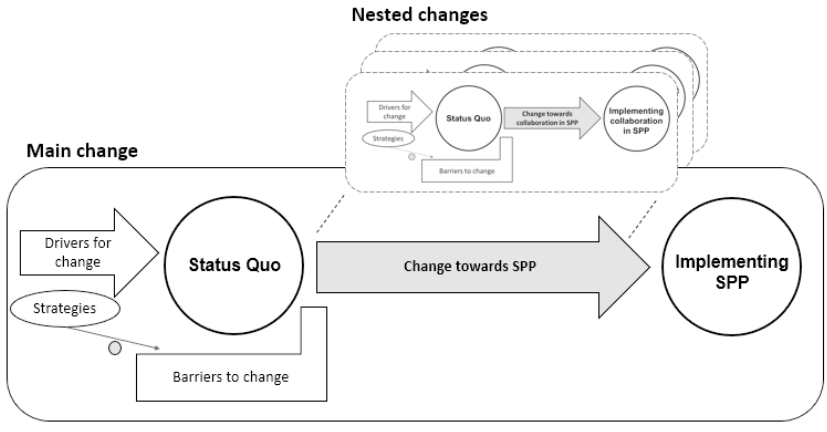


Figure 14. Nested organisational changes in the context of implementing sustainable public procurement.

Paper V also adds to the literature by expanding the dimensions of organisational capacity for change (c.f. Meyer & Stensaker, 2006), highlighting the need to build the capability to implement changes within main changes. In the context of nested changes (such as changes for SPP), the success of change processes can be increased by developing four dimensions of organisational capacity for change: capability to implement the main change, capability to maintain the operations, capability to implement nested changes, and capability to implement subsequent (i.e. future) changes.

5. Discussion

This chapter discusses the research questions for this doctoral thesis based on the contributions of Papers I-V. The following sections show analyses supported by findings on a holistic approach to SPP (Paper I), drivers for SPP (Paper II), barriers to SPP and strategies to overcome such barriers (Paper III), elements of collaboration between stakeholders in SPP (Paper IV), and managing organisational changes towards implementing collaboration in SPP (Paper V).

5.1 Research question 1: What is the role of collaboration between stakeholders in sustainable public procurement?

This doctoral thesis contributes to unveiling the role of collaboration between stakeholders in SPP by showing that interactions between stakeholders are a systemic element of SPP initiatives (see Paper I), leveraging the potential of developing collaboration to better implement SPP – by harmonising SPP elements. For instance, collaboration between stakeholders (see Paper IV) can contribute to building procurement specifications that incorporate sustainability issues, materialising sustainability solutions considering interrelations between demand-offering categories – supported by joint efforts between academia (e.g. scholars from different disciplinary backgrounds) and practitioners, aiming to develop transdisciplinary approaches that can foster more complete understandings and implementation of SPP.

Collaboration is a key to harmonising the systemic elements that compose SPP initiatives. The Harmonic SPP framework proposed by Paper I can be updated to consider the role of collaboration in SPP initiatives explicitly. The SPP Symphony framework, presented in Figure 15, positions interactions between stakeholders (leveraged through collaboration) at the centre of SPP initiatives, considering its conductor and harmonic role (as a maestro in an orchestra). Collaboration can be a maestro in conducting and harmonising the SPP symphony.

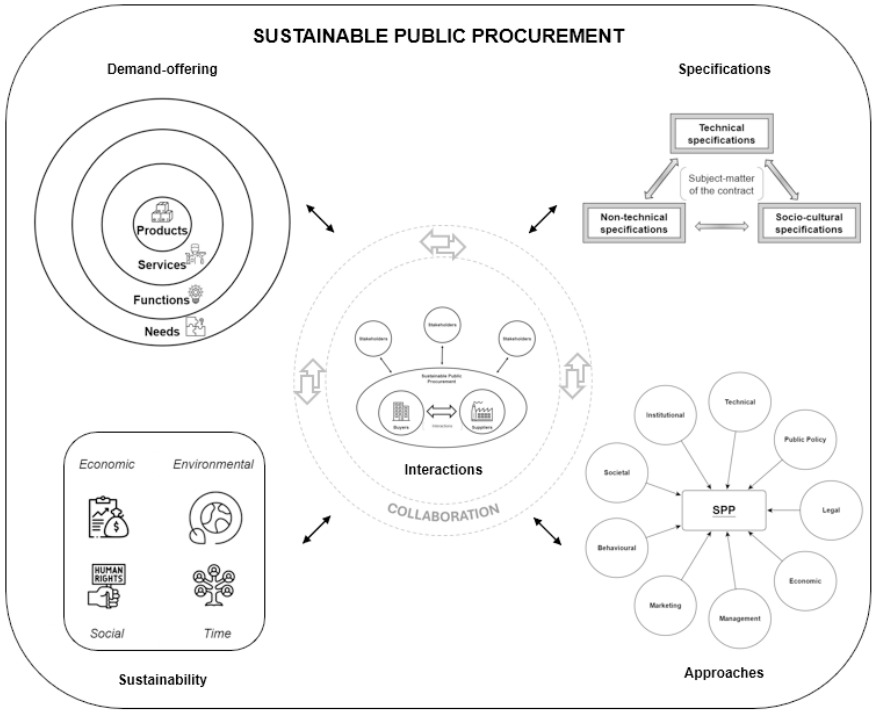


Figure 15. The Sustainable Public Procurement Symphony framework.

This doctoral thesis also highlights the harmonic role of collaboration between stakeholders by identifying its key influence in promoting organisational changes for implementing SPP. Collaboration has been the most important driver for such changes (see Paper II), influencing praxis-driven and policy-driven contexts. Collaboration has also been used as a strategy to overcome barriers to SPP in both contexts (see Paper III). Lack of collaboration was also identified as a barrier to SPP in praxis-driven contexts (see Paper III).

The key influence of collaboration between stakeholders in promoting organisational changes for SPP contributes to a more complete understanding of such change processes – by expanding the understanding of organisational changes for sustainability from an organisational context (e.g. buyers, suppliers) to a process one (i.e. SPP). Organisational changes for sustainability are aimed at moving organisations towards a more sustainability-oriented state. In the SPP context, such changes are also aimed at moving organisations to more sustainability-oriented states; however, the relational nature of SPP processes requires that buyers and suppliers move together towards a state of implementing SPP (e.g. demand-offerings shall not be materialised when there is a lack of buyer's demand or supplier's offer). For instance, the lack of sustainability-friendly suppliers was identified as the most important barrier to SPP from a buyer's perspective (see Paper III), hindering efforts in praxis-driven and policy-driven contexts.

Moving together – collaborating between them (i.e. buyers and suppliers) and with other stakeholders – can be a relevant element in understanding how organisations can move towards sustainability through interactions in broader systems (such as SPP). A representation of organisational changes for sustainability considering interactions between organisations in broader systems can be offered by this thesis, expanding Lozano's (2012) proposal on orchestrating changes towards sustainability (see Figure 3). The framework for harmonising organisational changes towards SPP (presented in Figure 16) considers the prominence of interactions between organisations (e.g. buyers and suppliers) in implementing SPP (see Paper I). In changes towards SPP, buyers and suppliers can move together from a *status quo* to a more sustainability-oriented state (in which both organisations become more sustainability-oriented), requiring meaningful interactions through collaboration. This framework entails the main forces fostering and hindering organisational changes towards SPP from a buyer's perspective: drivers (see Paper II), barriers (see Paper III), and strategies (see Paper III). This framework also considers the influence of contexts on drivers, barriers, and strategies (see Papers II and III) – contributing to a more complete understanding of organisational change processes towards sustainability.

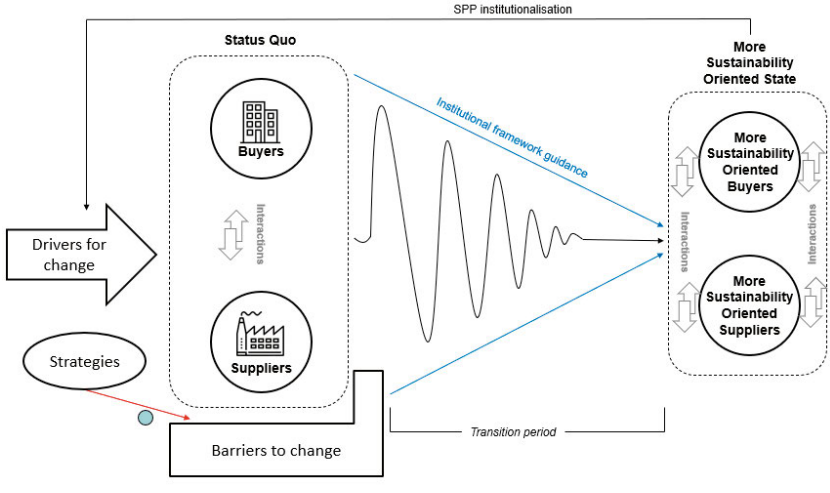


Figure 16. Harmonising organisational changes towards sustainable public procurement.

5.2 Research question 2: How can collaboration contribute to developing sustainable public procurement initiatives?

This doctoral thesis defines collaboration between stakeholders in SPP (see Paper IV) as a process aimed at proposing joint responses to the complexity of SPP initiatives by creating new solutions and understandings through interactions with a high level of involvement (concurring with Denise, 1999; Lozano, 2007). Collaboration between stakeholders in SPP focuses on improving the connection between demand (i.e. buyer's needs) and offer (i.e.

supplier's products and services) by joining knowledge and efforts from government, market, and society stakeholders (in agreement with Behravesh *et al.*, 2022).

This doctoral thesis provides insights into how collaboration between stakeholders has been contributing to implementing SPP. Papers II and III offer an understanding of collaboration as an important force (e.g. the most relevant driver and strategy to overcome barriers) in organisational changes, contributing to implementing SPP initiatives. SPP is a dynamic state (c.f. Lozano, 2007b; Martin *et al.*, 2005), requiring successive organisational changes and a continuous implementation process. Paper IV shows that collaboration between stakeholders can contribute to this permanent SPP implementation, offering benefits such as driving long-term changes and re-signifying procurement.

Paper IV also offers a catalogue of 27 benefits of collaboration that contribute to SPP initiatives, considering how each benefit has been more related to each context analysed (Brazil and Sweden). These benefits, presented in Figure 17, were organised into the procurement phases – considering that some benefits influence the whole procurement cycle and governance. The most important benefits of collaboration according to Paper IV are 1) mutual learning, 2) developing sustainability requirements, 3) disseminating best practices, 4) mutual understanding of needs, and 5) understanding market readiness.

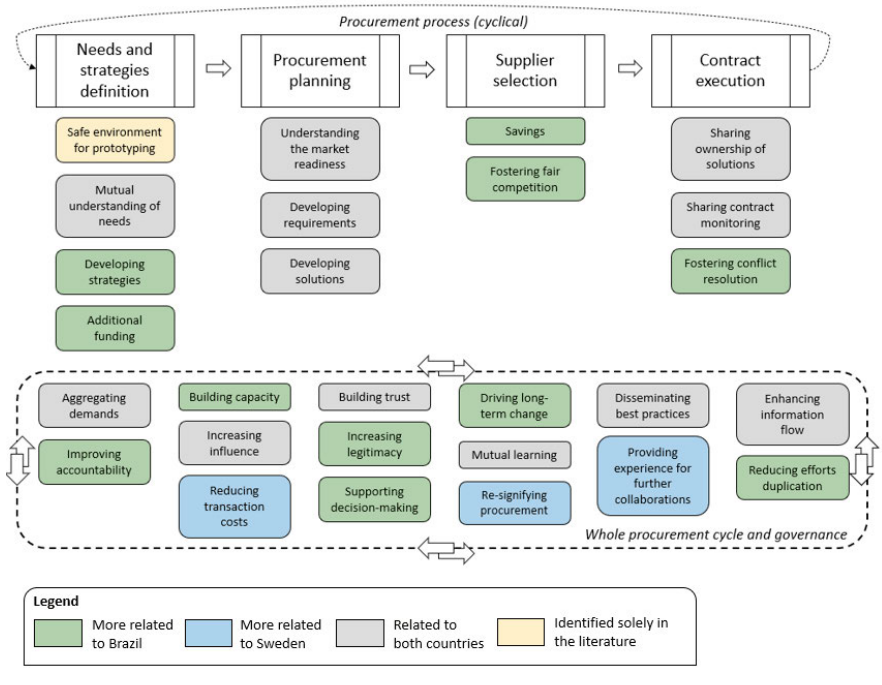


Figure 17. Benefits of collaboration between stakeholders in sustainable public procurement.

Paper IV proposed that the benefits of collaboration in SPP should be balanced with the drawbacks of collaboration, considering the costs and threats that may be imposed on the collaborators. The 10 drawbacks of collaboration were also organised into the procurement phases; however, only 1 drawback (developing bias due to limited stakeholder participation) can be related to a single procurement phase (procurement planning). The other 9 drawbacks are related to the whole procurement cycle and governance: 1) accountability erosion, 2) coordination costs, 3) developing supplier dependency, 4) disclosing information that could cause a loss of commercial advantage, 5) exposing stakeholders to inefficiency and corruption risks, 6) impeding new actors' entry, 7) increasing work, 8) increasing procurement time, and 9) using information against stakeholders. The most important drawbacks of collaboration in SPP according to Paper IV are 1) coordination costs and 2) developing bias due to limited stakeholder participation.

Paper IV proposes that several challenges of SPP can be tackled by using collaborative approaches – contributing to breaking the cultural and operational walls hindering joint actions between public organisations and other stakeholders (in agreement with Erridge and Greer, 2002). However, there are limits to which aspects of SPP can be addressed through collaboration and costs to be considered whilst structuring collaborative initiatives in SPP. Collaboration is not a silver bullet for addressing the complexity of SPP – costs and threats in collaborating should be considered by organisations, which must balance benefits and drawbacks to reach optimal collaboration (concurring with Lozano *et al.*, 2021).

5.3 Research question 3: What are the best practices for establishing collaboration in sustainable public procurement?

This doctoral thesis offers a catalogue of 15 mechanisms related to how collaboration in SPP can be established, organised into the procurement phases in which they occur (see Paper IV). The collaboration mechanisms include the 8 mechanisms identified by previous research (see Table 1) and 7 new collaboration mechanisms unveiled by Paper IV, expanding the literature. Figure 18 presents the collaboration mechanisms in SPP, highlighting how they relate to the contexts analysed (Brazil and Sweden). There are more similarities than differences between collaboration mechanisms in both contexts – although each mechanism can be implemented through different procedures (e.g. more attention to streaming, recording, and publishing stakeholder dialogues in Brazil), which can derive from each country's accountability developments and legal frameworks (c.f. Behravesh *et al.*, 2022).

Paper IV also highlights a collaboration mechanism that comprehends the four procurement phases: the centralised arena. Although centralised arrangements (e.g. buyer collaboration) for improving procurement practices have been studied by SPP research (see Erridge & Greer, 2002), this thesis offers a new perspective on the topic by highlighting the potential of these arrangements to work as collaboration arenas. Collaboration arenas in SPP can

be defined as formal or informal arrangements that promote structured interactions between stakeholders, aimed at improving governance resources and joint procurement processes whilst developing individual commitment and organisational capacity for collaboration.

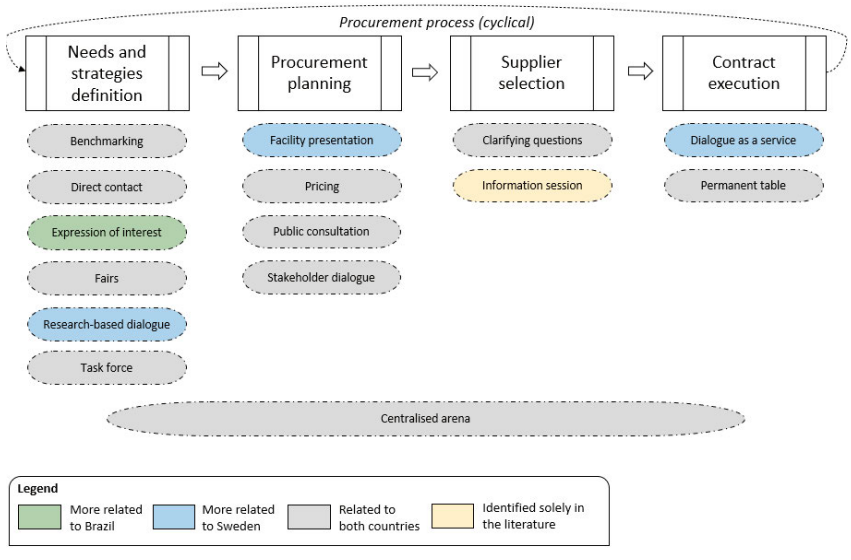


Figure 18. Mechanisms for collaboration between stakeholders in sustainable public procurement.

This thesis also provides depth to the role of facilitators in contributing to establishing and maintaining collaboration between stakeholders in SPP. Paper IV presents two types of facilitators of collaboration: individual and organisational. Individual facilitators (in agreement with Alhola & Nissinen, 2018; Lozano et al., 2020) can be employees working to build connections between stakeholders, mobilise internal resources to collaborative initiatives, and stimulate stakeholder participation. Organisations can also work as facilitators of collaboration (concurring with Berg et al., 2022; Daugbjerg, 2023; Johnson and Klassen, 2022), contributing by balancing stakeholders' interests, building collaboration mechanisms, developing collaboration commitment, engaging other stakeholders, and reducing information asymmetry. Organisational and individual facilitators were identified in both contexts analysed (Brazil and Sweden).

This thesis also offers an understanding of when organisations could establish collaboration between stakeholders in SPP (see Paper IV). Organisations engaged with SPP can decide when to establish collaboration by analysing the benefits and drawbacks of collaboration in their contexts, evaluating their expertise in using established collaboration mechanisms and the need to develop new ways of collaborating, and assessing the availability of individual and organisational facilitators.

The influence of contexts in establishing collaboration in SPP was also explored by this thesis, offering a more complete approach to the topic by analysing data from different world regions (Brazil and Sweden). Although these countries have different legal frameworks and bureaucratic cultures, Paper IV suggests the idea that collaboration in SPP can occur in different contexts. Despite minor discrepancies (e.g. more attention to a specific collaboration mechanism in one country), both contexts analysed were aligned towards the use of collaborative approaches to implement SPP. This alignment indicates that collaboration initiatives may not be determined by context but by individual commitment and organisational capacity to navigate into the legal and bureaucratic pathways to SPP. The non-deterministic role of contexts in establishing and developing collaboration between stakeholders in SPP was also confirmed by the perspective of external practitioners working at organisational facilitators.

This thesis also contributes to providing an understanding of how organisations can establish collaboration in SPP by offering a more complete approach to organisational changes for implementing collaboration. The catalogue of forces influencing such changes provided by Paper V can foster organisations' efforts towards establishing collaborative SPP initiatives, considering that a specific change process must be undertaken by such organisations.

Paper V also sheds light on the multiple and composite characteristics of organisational change processes for implementing SPP – organisations can change for implementing SPP but, if they intend to establish and use a key driver such as collaboration between stakeholders, an additional and specific (nested) change must be carried out. The understanding of multiple changes in SPP is strengthened by a comparison between the forces influencing changes for SPP (presented in Papers II and III) and the forces influencing changes for implementing collaboration in SPP (collated by Paper V). These changes have only one driver (collaboration commitment), one barrier (lack of sustainability alignment), and one strategy (associate political visibility) in common, suggesting that different forces have been influencing these change processes and reinforcing the understanding of their distinction. This thesis's findings on nested changes in the SPP context reinforce the idea that collaboration is not a silver bullet for addressing SPP complexity – nor is it self-applicable.

5.4 Research question 4: What are the forces fostering and blocking collaboration in sustainable public procurement?

Implementing collaboration between stakeholders in SPP initiatives requires organisational changes, which are influenced by forces contributing to (i.e. drivers and strategies to overcome barriers) and hindering (i.e. barriers) change processes. This doctoral thesis identifies and categorises these forces (i.e. drivers for, barriers to collaboration in SPP, and strategies to overcome such barriers, see Paper V) to provide a more complete understanding of changes towards implementing collaboration in SPP.

Paper V offers a catalogue of 23 drivers, 40 barriers, and 24 strategies. According to this paper, the most important drivers for collaboration in SPP are collaboration commitment and multidisciplinary teams. The most relevant

barriers to collaboration in SPP are lack of willingness from external stakeholders, lack of personnel, and lack of time. The most important strategy to overcome barriers to implementing collaboration in SPP is providing incentives for stakeholder participation, as actively informing potential collaborators regarding the benefits of engaging in collaborative SPP initiatives, such as a better understanding of procurement processes, mutual learning, and sharing needs.

Paper V organised the drivers for, barriers to collaboration in SPP, and strategies to overcome such barriers considering stakeholder dependence and organisational influence (according to the ChaFoS, see Figure 13) – into internal, connecting-protagonist, connecting-supporting, and external (expanding the internal-connecting-external categorisation proposed by Lozano, 2015; Lozano et al., 2019). A holistic approach to forces influencing organisational changes towards collaboration in SPP is needed, explicitly considering stakeholder dependence and organisational influence.

This thesis also contributes to understanding the forces fostering and hindering organisational changes towards collaboration in SPP by presenting their malleability. Multiple interconnections between barriers and strategies to overcome them were identified by Paper V – linking different forces, at different levels, in a malleable way. These inter-level connections suggest that barriers to collaboration in SPP may not be overcome only by using specific or same-level strategies but also by developing a set of strategies that can holistically tackle hindering forces.

6. Conclusions

The purpose of this doctoral thesis was to provide a more complete understanding of how organisations can contribute to sustainability efforts through SPP by exploring the role of collaboration between stakeholders and considering organisational changes required to move such organisations towards implementing SPP and collaborative initiatives. This purpose was fulfilled by answering the RQs for this thesis with the support of the five appended papers. The following sections present this thesis's key findings, theoretical contribution, and practical implications. The last section discusses further research topics that may follow this thesis.

6.1 Key findings

This thesis's key findings are summarised in Table 14, organised according to the questions proposed by this research:

Table 14. Summary of key findings according to the research questions.

Research question	Key findings
RQ1: What is the role of collaboration between stakeholders in SPP?	- Stakeholder interactions are an important element of the SPP system, leveraging the potential for developing collaborative initiatives - Collaboration contributes to harmonising the systemic elements that compose SPP initiatives (e.g. developing procurement specifications, exploring demand-offering possibilities, joining disciplinary approaches) and organisational changes towards implementing SPP - Collaboration has been driving organisational changes for SPP and overcoming barriers to such changes - Lack of collaboration is a barrier to implementing SPP in praxis-driven contexts, which may be explained by reduced institutional and policy support to engage with stakeholders in such contexts when compared to policy-driven ones
RQ2: How can collaboration contribute to developing SPP initiatives?	- Collaboration is the most important driver for organisational changes towards implementing SPP, influencing praxis-driven and policy-driven contexts

	- Collaboration promotes 27 benefits for SPP initiatives, influencing the whole procurement cycle and governance - Benefits of collaboration in SPP include mutual learning, developing sustainability requirements, disseminating best practices, mutual understanding of needs, and understanding market readiness - Organisations must balance benefits and drawbacks to reach optimal collaboration in SPP initiatives - There are limits to which aspects of SPP can be addressed through collaboration and costs to be considered - Collaboration can contribute to breaking the PP walls but is not a silver bullet for addressing the complexity of SPP
RQ3: What are the best practices for establishing collaboration in SPP?	- Collaboration can be established through 15 mechanisms, influencing the whole procurement cycle - Centralised arenas can develop collaborative approaches aimed at improving governance resources and joint procurement processes whilst developing individual commitment and organisational capacity for collaboration - Individual and organisational facilitators can contribute to establishing and maintaining collaboration in SPP - Collaboration initiatives may not be determined by context but by individual commitment and organisational capacity to navigate into the legal and bureaucratic pathways to SPP - Organisations engaged with SPP can decide when to collaborate by analysing the benefits and drawbacks, evaluating their expertise in developing collaboration mechanisms, and assessing the availability of facilitators - Organisations can change for implementing SPP but, if they intend to use a key driver such as collaboration between stakeholders, an additional and specific change must be carried out
RQ4: What are the forces fostering and blocking collaboration in SPP?	- Forces influencing organisational changes for collaboration in SPP include 23 drivers, 40 barriers, and 24 strategies to overcome such barriers, organised into internal, connecting-protagonist, connecting-supporting, and external levels

	- A holistic approach to forces influencing organisational changes towards collaboration in SPP is needed, explicitly considering stakeholder dependence and organisational influence - Forces influencing changes for collaboration in SPP can be malleable (i.e. interconnected through different levels); for example, barriers to collaboration in SPP may not be overcome only by using specific or same-level strategies but also by developing a set of strategies that can holistically tackle hindering forces

6.2 Theoretical contributions

This thesis offers theoretical contributions mainly related to the SPP system (by applying an ST approach to SPP), OCM for sustainability (developing OCM for SPP and collaboration in SPP), and collaboration between stakeholders (by applying collaboration theory to the SPP context).

6.2.1 Sustainable public procurement system

This thesis proposes a holistic framework, developed through hermeneutics, to interconnect SPP system elements and explain the complexity of SPP more clearly. The Harmonic SPP framework, updated in the SPP Symphony framework, contributes to SPP research by explicitly presenting the main elements that compose SPP initiatives, providing a systemic framework for positioning further research and offering pathways to develop studies on harmonising SPP elements.

6.2.2 Improving organisational changes for sustainability

This thesis contributes to OCM for sustainability by offering a catalogue of drivers for, barriers to SPP, and strategies to overcome such barriers. These forces were categorised by their level (internal, connecting, or external), considering organisational systems and their interconnections, and ranked. The key role of collaboration in harmonising organisational changes for SPP was also proposed by this thesis, contributing to the literature.

This thesis shows that forces influencing organisational changes for SPP can be context-specific and organised in continuums from praxis-driven to policy-driven. The continuums can help to better understand the forces that can influence changes towards SPP and the specificity of particular contexts. This thesis also complements the efforts on understanding forces influencing changes for sustainability by analysing a process (i.e. SPP), instead of an organisation, in different contexts and presenting the importance of contexts on such forces. Moving towards sustainability in processes is affected by the contexts, which must be considered when planning changes.

This thesis also provides a catalogue of forces influencing organisational changes towards collaboration in SPP. These forces were categorised into internal, connecting-protagonist, connecting-supporting, and external – the SPP Change Forces System, a new categorisation that understands organisations as whole systems interconnected in broad systems (in this case, the SPP system) and explicitly considers stakeholder dependence and organisational influence. The set of forces organised according to this categorisation can be used to support further research aimed at understanding collaborative SPP initiatives through an OCM perspective.

The analysis of how the barriers to collaboration in SPP and strategies to overcome such barriers are interrelated, considering the different categories, contributes to understanding the malleability of such forces. This malleability suggests that barriers hindering change efforts towards collaboration in SPP may require developing a set of strategies that can holistically tackle such forces.

This thesis also adds to the literature by offering an understanding of how organisational changes for collaboration in SPP are interconnected with changes for implementing SPP, the former deriving from the latter. Both changes have been driven mainly by different forces, reinforcing the understanding of these changes as separate but linked processes. This research offers the concept of nested changes to better understand these simultaneous change processes, considering change processes for SPP as multiple and composite instead of single and monolithic. This thesis also contributes to OCM for sustainability research by improving the dimensions of organisational capacities for changes, proposing the capability to implement nested changes as a dimension that can increase the success of some change processes.

6.2.3 Developing collaboration theory

This thesis contributes to collaboration theory by offering a systematisation of the collaboration elements in the SPP context. The framework for Understanding Collaboration between Stakeholders in SPP – UnCoStake was proposed, organising the collaboration elements and providing a more complete understanding of decision-making processes towards collaborative SPP initiatives.

This thesis also adds to the literature by offering definitions of key concepts regarding collaboration between stakeholders in SPP. Collaboration between stakeholders is a process aimed at proposing joint responses to the SPP complexity and can be implemented in different contexts. Collaboration can break the PP walls and contribute to sustainability efforts, but it is not a silver bullet for addressing the complexity of SPP.

6.3 Practical implications

This thesis offers practical tools to better understand SPP initiatives and to promote collaboration in SPP, contributing to practitioners' efforts. These tools can be adopted by practitioners and incorporated into SPP guidelines, fostering a better SPP implementation.

6.3.1 Harmonising sustainable public procurement initiatives

The Harmonic SPP and SPP Symphony frameworks proposed by this thesis provide a more complete perspective on SPP, contributing to SPP understanding and implementation. Practitioners can apply the frameworks to their contexts to better understand how the SPP elements can be harmonised for implementing SPP initiatives. An analogy of the Harmonic SPP and SPP Symphony frameworks is a composer's creativity, who creates a beautiful symphony (SPP in this case) by harmonising notes (e.g. the elements in the demand-offering, or the stakeholder interactions) and chords (i.e. the demand-offering, specifications, stakeholder interactions, disciplinary approaches, and sustainability dimensions) through a melody of complex interactions. In this harmonic way, supported by collaborative approaches, SPP can better contribute to making societies more sustainable.

6.3.2 Implementing sustainable public procurement

This thesis also has managerial implications for SPP implementation through an organisational change approach. For example, organisations planning changes towards SPP need to consider the whole set of drivers, barriers, and strategies collated by this thesis – manoeuvring internal, connecting, and external influences. The rankings of drivers, barriers, and strategies could be used for prioritising efforts. Different stakeholders must be considered whilst planning such changes, considering the opportunity of moving together towards implementing SPP. Organisations should make efforts to understand their contexts of change towards SPP and focus on developing a more complete change approach considering the forces from praxis-driven and policy-driven contexts. A holistic approach, considering internal, connecting, and external influences that navigate through the praxis and policy spectrums, is needed to enact more successful changes towards SPP.

6.3.3 Collaborating in sustainable public procurement

This thesis offers a systematisation of the elements that comprise collaborative SPP initiatives, contributing to practitioners' efforts in understanding and implementing collaboration. The 4 collaboration channels proposed by this thesis can be used to understand which collaborators have been engaged in collaborative initiatives and potential collaborators to be engaged – aiming to expand the set of stakeholders contributing to SPP initiatives. The 15 collaboration mechanisms can be developed to establish collaborative initiatives. The 27 benefits and 10 drawbacks can be balanced, according to each context, to support decision-making towards implementing SPP using collaboration. The 2 types of facilitators can contribute to a better understanding of the additional support available for practitioners in their efforts to establish collaborative SPP initiatives. The UnCoStake can be used to guide management decisions on implementing collaboration in SPP, considering that the implementation of collaborative SPP initiatives may be determined by individual commitment and organisational capacity to navigate the legal and bureaucratic pathways to SPP. The UnCoStake can also be used to assess and follow up initiatives,

providing a pathway to improve collaborative arrangements, such as balancing benefits and drawbacks to reach optimal collaboration.

This thesis also contributes to practitioners' efforts to implement collaboration in SPP by offering a catalogue of forces influencing organisational changes towards collaboration. This catalogue can be used to plan organisational changes for collaboration in SPP, considering a more complete approach to how organisations and their stakeholders influence such forces. The malleability of such forces, as proposed by this thesis, can contribute to practical efforts by fostering systemic development of sets of forces instead of focusing on specific ones – such as building different level strategies to overcome single-level barriers.

This thesis also proposes a better implementation of collaboration in SPP – and SPP – by offering an understanding of the simultaneous change processes in complex contexts such as SPP, in which organisational changes can be multiple and composite. The nested changes concept can help practitioners implement changes in the SPP context and develop capacities for change in their organisations, avoiding simplifications that consider changes for SPP as single and monolithic.

6.4 Further research

This doctoral thesis should be followed up by several studies aimed at building and testing theories regarding the SPP system, organisational changes in the SPP context, and collaboration in SPP. For example, further research should explore the implementation of the Harmonic SPP and SPP Symphony frameworks, disciplinary approaches' interactions, and organisational impacts of adopting a holistic view of SPP.

Further research should also explore the new drivers, barriers, and strategies identified by this thesis to understand their contributions to implementing SPP through an OCM approach. The importance of the drivers, barriers, and strategies could be analysed by a quantitative study, assessing the validity of the whole set of forces in different contexts and updating the rankings presented by this research. Quantitative studies could also focus on the interconnections between barriers to SPP and strategies to overcome such barriers in different contexts. The suppliers' perspective on organisational changes for implementing SPP should also be explored by qualitative studies, using inductive approaches to build theory considering their contexts. The framework for harmonising organisational changes towards SPP could be explored by joining buyers' and suppliers' perspectives. More research is needed to understand the praxis-driven and policy-driven change approaches in the SPP context. More studies analysing different SPP contexts could help to develop knowledge on the systemic elements that compose the different SPP change approaches. The contextual influence on forces influencing changes for sustainability could be analysed in other processes.

Further research should explore collaboration in SPP through the market and society stakeholders' perspectives to provide depth on the topic and identify eventual collaboration led by non-governmental stakeholders. The

influence of legal frameworks and bureaucratic cultures in collaboration mechanisms, considering developing and developed countries, could be explored by case studies aimed at providing depth on such mechanisms. Legal comparative research (including developing and developed countries) could also be conducted to shed light on the different procedures used to implement collaboration mechanisms, considering changes in the SPP process according to each procedure. Collaboration drawbacks (e.g. coordination costs, developing supplier dependency) could be analysed through case studies considering how organisations are working to minimise them. The interconnections between the elements of collaboration in SPP could be explored by surveys. Comparative studies can be carried out to provide depth on the non-deterministic role of contexts in developing collaboration between stakeholders in SPP. More research efforts on developing countries' approaches to implementing SPP and collaboration are also needed, such as using inductive designs aimed at building theories considering their contexts (e.g. in-depth single case studies, action research, grounded theory) and deductive designs (e.g. surveys) testing and expanding more general frameworks (such as the UnCoStake). The UnCoStake can be used for supporting case studies that may focus on how organisations can build individual commitment and organisational capacity to implement collaborative SPP initiatives.

This thesis should also be followed up with quantitative studies aimed at ranking and offering a more complete analysis of the interconnections between the forces influencing organisational changes for collaboration between stakeholders in SPP (e.g. barriers vs. strategies). More depth should be provided on the specific drivers, barriers, and strategies in the collaboration context organised by this thesis. Additional studies on forces influencing organisational changes for SPP should be carried out to analyse which forces require nested changes. Other contexts of organisational changes towards sustainability should be analysed through a nested changes approach aiming at building knowledge on multiple and simultaneous change processes.

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Appendices

A. Semi-structured interview (English version)

Interview Questions

Identification
1) Name:
2) Email:
3) Gender:
4) Position in the organisation:
5) How long have you been in that position?
6) How long have you been in your organisation?
7) How long have you been working with public procurement?
Sustainability and PP
8) What is your understanding of sustainability?
9) What does the inclusion of sustainability issues in public procurement (PP) mean to you?
10) Why is it important to include sustainability issues in your PP processes?
11) Do you include sustainability issues only in products or only in services in your PP?
12) What types of sustainability specifications do you include in the procurement process?
Organisational Change Management and SPP
13) How is sustainable public procurement (SPP) being implemented by your organisation?
14) Where in the PP process were sustainability issues first implemented?
15) Who took the initiative to start implementing SPP in your organisation?
16) How has SPP been implemented throughout the process?
17) What have been the drivers for implementing SPP?
18) What are the main challenges in implementing SPP?
19) How has your organisation overcome such challenges?
Collaboration in SPP
20a) Does your organisation interact with stakeholders while implementing SPP? If yes, how? 20b) And with whom?
Collaboration in SPP (in case of having interactions)
21) Which stakeholders have the most influence to foster SPP?
22) Who usually takes the initiative to start interacting with stakeholders in a SPP process?
23) How does your organisation engage with stakeholders in SPP?
24) What are the advantages of interacting with stakeholders in SPP?
25) What are the disadvantages of interacting with stakeholders in SPP?
26) What has been driving the interactions with stakeholders in SPP?

27) What have been the main challenges in interacting with stakeholders in SPP?
28) How has your organisation overcome such challenges?
29) Which stakeholders do you think are the most engaged in SPP?
30) Which stakeholders should your organisation engage for improving your SPP processes?
31) What would you change in your organisation to improve the interactions with stakeholders in SPP?
32) If you had unlimited resources and influence, how could you improve the implementation of SPP?

Collaboration in SPP (in case of not having interactions)

21b) Which stakeholders should have the most influence to foster SPP?
22b) Which advantages would you foresee in interacting with stakeholders in SPP?
23b) Which disadvantages would you foresee in interacting with stakeholders in SPP?
24b) Who should start the interactions in SPP?
25b) Which stakeholders should your organisation interact with to improve the implementation of sustainability in your PP processes?
26b) Who should facilitate the interactions in SPP with your organisation?
27b) How should this facilitator contribute to these interactions in SPP?
28b) What should drive the interactions with stakeholders in SPP?
29b) What should be the main challenges in interacting with stakeholders in SPP?
30b) How should your organisation plan to overcome those challenges?
31b) If you had unlimited resources and influence, how could you improve the implementation of SPP?

B. Semi-structured interview (Portuguese version)

Questões para Entrevista

Identificação

1) Nome:
2) Email:
3) Gênero:
4) Posição em sua organização:
5) Há quanto tempo você está nessa posição?
6) Há quanto tempo você está em sua organização?
7) Há quanto tempo você está trabalhando com compras públicas?

Sustentabilidade e compras públicas

8) O que você entende por sustentabilidade?
9) O que significa para você a inclusão de questões sustentáveis nas compras públicas?
10) Por que é importante incluir questões sustentáveis em seus processos de compras públicas?
11) Você inclui questões sustentáveis em suas compras públicas somente em produtos ou somente em serviços?
12) Quais tipos de especificações sustentáveis você inclui em seu processo de compras?

Gestão da mudança organizacional e compras públicas sustentáveis

13) Como as compras públicas sustentáveis estão sendo implementadas por sua organização?

14) Onde, no processo de compras públicas, as questões sustentáveis foram implementadas primeiro?

15) Quem tomou a iniciativa de comecar a implementação das compras públicas sustentáveis em sua organizacão?

16) Como a compras públicas sustentáveis tem sido implementada por todo o processo?

17) Quais têm sido os estímulos/indutores para a implementacão das compras públicas sustentáveis?

18) Quais são os principais desafios na implementação das compras públicas sustentáveis?

19) Como sua organização tem superado esses desafios?

Colaboração e compras públicas sustentáveis

20a) Sua organização interage com partes interessadas (ex.: outras pessoas ou organizações) enquanto implementa compras públicas sustentáveis? Se sim, com como?

20b) E com quem?

Colaboração e compras públicas sustentáveis (caso haja interacões)

21) Quais partes interessadas têm maior influência para fomentar as compras públicas sustentáveis?

22) Quem usualmente toma a iniciativa para comecar a interação com partes interessadas em um processo de compras públicas sustentáveis?

23) Como a sua organização se conecta/envolve com partes interessadas nas CPS?

24) Quais são as vantagens da interação com partes interessadas nas compras públicas sustentáveis?

25) Quais são as desvantagens da interação com partes interessadas nas compras públicas sustentáveis?

26) O que tem induzido/estimulado as interações com partes interessadas nas compras públicas sustentáveis?

27) Quais têm sido os principais desafios na interação com partes interessadas nas compras públicas sustentáveis?

28) Como sua organização tem superado esses desafios?

29) Quais partes interessadas você acha que são as mais envolvidas nas compras públicas sustentáveis?

30) Com quais partes interessadas sua organização deveria se envolver para aprimorar seus processos de compras públicas sustentáveis?

31) O que você mudaria em sua organização para aprimorar as interações com partes interessadas nas compras públicas sustentáveis?

32) Se você tivesse recursos e influência ilimitados, como você poderia aprimorar a implementação das compras públicas sustentáveis?

Colaboração e compras públicas sustentáveis (caso não haja interações)

21b) Quais partes interessadas deveriam ter maior influência para fomentar as compras públicas sustentáveis?

22b) Quais vantagens você enxergaria na interação com partes interessadas nas compras públicas sustentáveis?

23b) Quais desvantagens você enxergaria na interação com partes interessadas nas compras públicas sustentáveis?

24b) Quem deveria iniciar as interações nas compras públicas sustentáveis?

25b) Com quais partes interessadas a sua organização deveria interagir para aprimorar a implementação da sustentabilidade em seus processos de compras públicas sustentáveis?

26b) Quem deveria atuar como facilitador das interações com sua organizacão nas compras públicas sustentáveis?

27b) Como esse facilitador deveria contribuir para essas interações nas compras públicas sustentáveis?

28b) O que deveria atuar como estímulo/indutor das interações com partes interessadas nas compras públicas sustentáveis?

29b) Quais seriam os principais desafios nas interações com partes interessadas nas compras públicas sustentáveis?

30b) Como a sua organização deveria planejar para superar esses desafios?

31b) Se você tivesse recursos e influência ilimitados, como você poderia aprimorar a implementação das compras públicas sustentáveis?

C. Code structure and categorisation (NVivo)

Central category	Category	Code	Sources	References
Driver for SPP				
	Internal			
		Champions	9	13
		Collaboration between departments	7	14
		Collaboration commitment	2	2
		Cost savings	4	6
		Dedicated staff	4	7
		Dissemination of sustainability knowledge	1	1
		Evidence-based decision making	3	3
		Financial resources	1	1
		Internal control	3	6
		Leadership	5	7
		Policy and guidelines	6	8
		Positive inertia	3	3
		Professional visibility	3	5
		Standardized catalogues	2	4
		Strategic sourcing	3	5
		Sustainability commitment	14	24
		Sustainability risk-analysis	2	5
		Sustainability-dedicated department	6	7
		Training	2	3
		Whole cost analysis	1	1
	Connecting			
		Collaboration between stakeholders	16	29
		Control agencies report	3	3
		Dissemination of successful initiatives	4	4
		Internal strategy linked to broader policy	1	1
		Local arrangements preference	1	1
		Networking across countries	2	2
		Participative-building standard specifications	5	10

		Political support	6	7
		Political visibility	6	11
		Training potential suppliers	1	1
		User feedback	1	2
	External			
		Clear and supportive regulations	2	2
		Clear political priorities	4	6
		Dedicated national organisations	5	5
		Mandatory regulations	9	16
		Media scandals	5	6
		Policy and guidelines	3	4
		Readiness of sustainability-friendly suppliers	3	3
		Research	1	1
		Standard specifications	5	7
		Sustainability labels	3	3
		Sustainability policies	2	2
Barrier to SPP				
	Internal			
		Lack of champions	1	1
		Lack of clear responsibilities	4	4
		Lack of collaboration between departments	5	10
		Lack of dedicated staff	5	6
		Lack of employee retention	2	2
		Lack of financial resources	8	15
		Lack of holistic needs assessment	2	3
		Lack of information systems	2	2
		Lack of innovative thinking	4	6
		Lack of internal demand for sustainable products and services	3	4
		Lack of leadership	7	8
		Lack of monitoring and evaluation	5	7
		Lack of personnel	8	11
		Lack of policy and guidelines	3	4
		Lack of positive inertia	2	2
		Lack of priority due to savings priority	6	7
		Lack of risk-taking behaviour	5	5
		Lack of standardised catalogues	1	1
		Lack of strategic positioning of procurement	7	10
		Lack of strategic positioning of sustainability	3	4
		Lack of strategic procurement thinking	5	5
		Lack of sustainability alignment	4	4
		Lack of sustainability commitment	5	5
		Lack of sustainability culture	3	4
		Lack of sustainability integration into budgeting processes	2	2
		Lack of sustainability knowledge	3	4
		Lack of sustainability monitoring of contracts	2	3
		Lack of systemic understanding	5	5
		Lack of time	5	6
		Lack of training	8	16
	Connecting			
		Lack of balance in politicians' involvement	1	1

		Lack of collaboration between stakeholders	4	8
		Lack of market knowledge	4	4
		Lack of peer networking	1	1
		Lack of political support	10	11
		Lack of sustainability alignment	1	1
		Lack of sustainability-focused control	3	3
	External			
		Lack of business sector maturity	1	1
		Lack of mandatory regulations	1	1
		Lack of national commitment and monitoring	4	4
		Lack of policies and guidelines	2	2
		Lack of supportive regulations	5	6
		Lack of sustainability-friendly suppliers	10	14
		Lack of traceability in supply chains	1	1
Strategy in SPP				
	Internal			
		Additional budget	1	1
		Associate cost savings	3	4
		Collaboration between departments	2	3
		Communication of sustainability practices	1	1
		Different framework and contract agreements	2	3
		Engagement of internal users	1	2
		Evidence-based decision-making	5	5
		Incremental approach	7	8
		Leadership engagement	2	2
		Mandatory regulations	1	1
		Peer networking	2	2
		Policy and guidelines	3	3
		Prioritisation	1	2
		Process management	1	1
		Standardized catalogues	1	1
		Strategic decision-making committee	2	2
		Strategic sourcing	1	1
		Sustainability risk assessment	1	1
		Training	3	3
	Connecting			
		Associate political visibility	3	3
		Collaboration between stakeholders	10	14
		Networking across countries	2	2
		Networking with international organisations	1	1
	External			
		Mandatory regulations	2	2
Contextual influence				
	Change approaches			
		Policy-driven	9	11
		Praxis-driven	12	23
Collaborator				
	Government			
		Buyers	10	14
		Buyers from other countries	2	2

		Control agencies	2	3
		International organisations	1	1
		Judicial system	1	1
		Politicians	5	6
		Policy agencies	4	5
	Market			
		Consultancy	3	4
		Industry organisations	8	10
		Potential suppliers	13	23
		Private buyers	3	3
		Suppliers	5	5
		Trade unions	4	6
	Society			
		Academia	7	9
		Citizens	2	2
		International NGOs	3	5
		NGOs	8	15
Potential collaborator				
	Government			
		Buyers	3	3
		Control agencies	4	4
		International organisations	3	3
		Politicians	3	3
		Policy agencies	4	7
	Market			
		Consultancy	1	2
		Industry organisations	2	2
		Potential suppliers	4	4
		Private buyers	1	1
		Shareholders	1	1
		Standards organisations	1	1
		Startups	2	2
		Suppliers	1	2
	Society			
		Academia	5	6
		Citizens	3	3
		NGOs	3	5
Collaboration mechanism				
	Needs and strategies definition			
		Benchmarking	7	8
		Direct contact	12	17
		Expression of interest	1	1
		Fairs	8	9
		Research-based dialogue	1	1
		Task force	3	3
	Procurement planning			
		Facility presentation	1	1
		Pricing and market research	3	3
		Public consultation	8	9
		Stakeholder dialogue	5	9

	Supplier selection			
		Clarifying questions	2	2
	Contract execution			
		Dialogue as a service	1	1
		Permanent table	3	3
	All phases			
		Centralised arena	8	11
Collaboration benefit				
	Needs and strategies definition			
		Additional funding	1	1
		Developing procurement strategies	1	1
		Mutual understanding of needs	8	13
	Procurement planning			
		Developing solutions	7	15
		Developing requirements	12	14
		Understanding the market readiness	8	8
	Supplier selection			
		Fostering fair competition	4	5
		Savings	3	3
	Contract execution			
		Fostering conflict resolution	1	1
		Sharing contract monitoring	6	9
		Sharing ownership of solutions	3	3
	All phases			
		Aggregating demands	5	6
		Building capacity	4	4
		Building trust	3	3
		Disseminating best practices	11	12
		Driving long-term change	3	4
		Enhancing information flow	4	6
		Improving accountability	2	2
		Increasing influence	7	9
		Increasing legitimacy	3	5
		Mutual learning	17	29
		Providing experience for further collaborations	3	3
		Reducing efforts duplication	4	4
		Reducing transaction costs	3	3
		Re-signifying procurement	1	1
		Supporting decision-making	2	3
Collaboration drawback				
	Procurement planning			
		Developing bias due to limited stakeholder participation	10	12
	All phases			

		Accountability erosion	1	1
		Coordination costs	12	15
		Developing supplier dependency	1	1
		Exposing stakeholders to inefficiency and corruption risks	7	8
		Increasing procurement time	5	5
		Increasing work	8	10
		Using information against stakeholders	1	1
Collaboration facilitator				
	Facilitator			
		Individual	4	5
		Organisational	12	38
Driver for collaboration				
	Internal			
		Balanced workload	1	1
		Champions	3	5
		Collaboration commitment	8	11
		Collaboration-friendly policy	1	1
		Dedicated staff	2	4
		Innovation culture	1	1
		Leadership	3	3
		Long planning phase	1	1
		Multidisciplinary team	4	5
		Positive inertia	3	3
		Professional visibility	1	1
		Stakeholder mapping	2	2
	Connecting-Protagonist			
		Political visibility	2	2
	Connecting-Supporting			
		Control agencies' support	1	1
		Political support	1	1
		Shared procurement system	1	1
	External			
		Clear and supportive regulations	2	2
		Market willingness	1	1
Barrier to collaboration				
	Internal			
		Lack of attractiveness	3	3
		Lack of category knowledge	2	2
		Lack of clear responsibilities	2	2
		Lack of collaboration commitment	6	7
		Lack of cost-benefit balance	1	1
		Lack of dedicated staff	4	6
		Lack of guidelines	3	3
		Lack of initiative	2	2
		Lack of knowledge management	1	1
		Lack of personal chemistry	1	1
		Lack of personnel	7	9
		Lack of risk-taking behaviour	2	2
		Lack of stakeholder mapping	6	6

		Lack of strategic positioning of procurement	4	5
		Lack of supplier relationship policy	3	4
		Lack of sustainability knowledge	1	1
		Lack of time	7	9
		Lack of training	2	3
	Connecting-Protagonist			
		Lack of expectations alignment	2	2
		Lack of political visibility	1	1
		Lack of sustainability alignment	2	3
		Lack of timing	2	2
		Lack of trust	5	8
	Connecting-Supporting			
		Lack of political support	2	2
	External			
		Lack of end-user association	1	3
		Lack of policies and guidelines	1	3
		Lack of procurement knowledge	1	1
		Lack of supportive regulations	2	4
		Lack of willingness	10	12
Strategy in collaboration				
	Internal	Annual procurement plan	1	1
		Balanced workload	1	1
		Collaboration between departments	2	2
		Engagement through leaders	1	2
		Incremental approach	1	1
		Integrity plan	2	2
		Multidisciplinary team	1	1
		Process management	1	1
		Socio-cultural specifications	1	1
		Training	2	2
	Connecting-Protagonist			
		Active networking	2	2
		Associate political visibility	1	1
		Formalised interactions	1	1
		Incentives for participation	3	3
		Long-term approach	2	2
		Peer pressure	1	1
		Supplier training	1	1
		Suppliers summit	2	5
		Triangulation	1	1
	Connecting-Supporting			
		Control agencies' support	1	2

Papers
Associated papers have been removed in the electronic version of this thesis.
For more details about the papers see:
http://urn.kb.se/resolve?urn=urn:nbn:se:hig:diva-46607